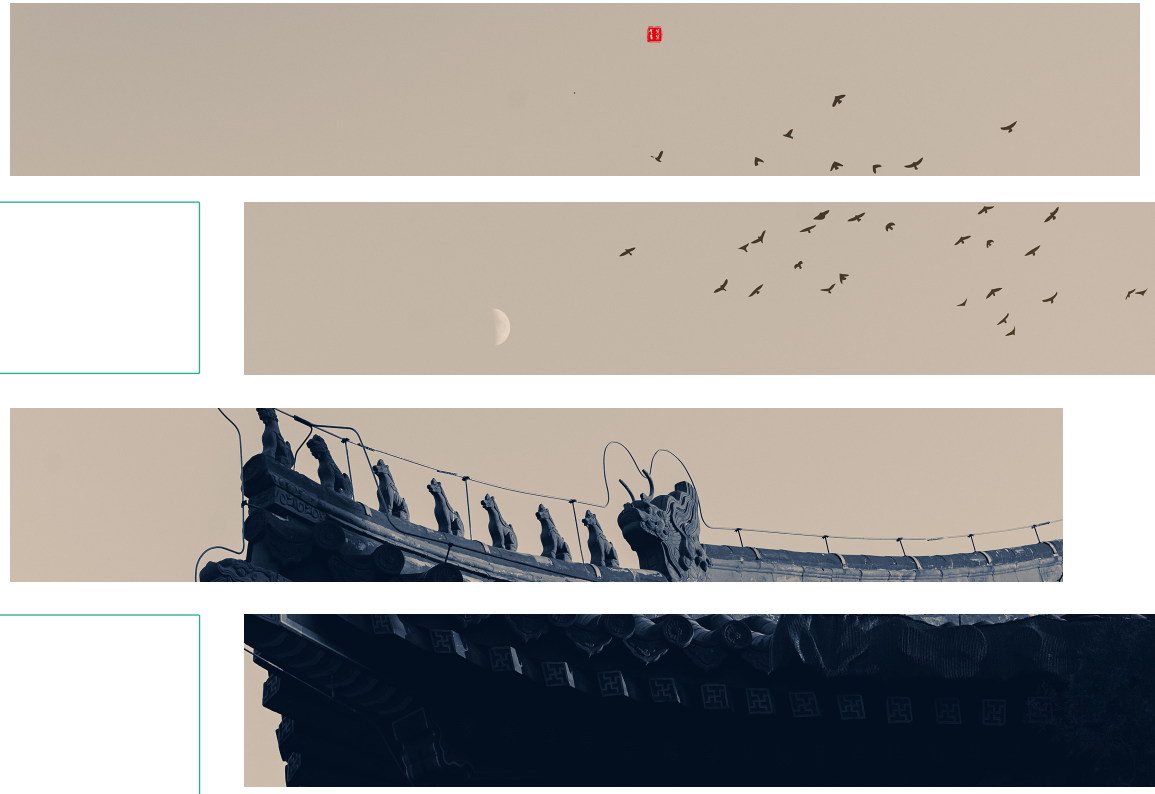
Cognitive Reasoning

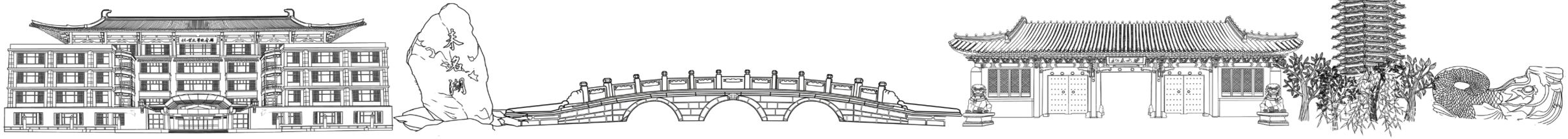
Fall 2026

Lecture 07

Causality

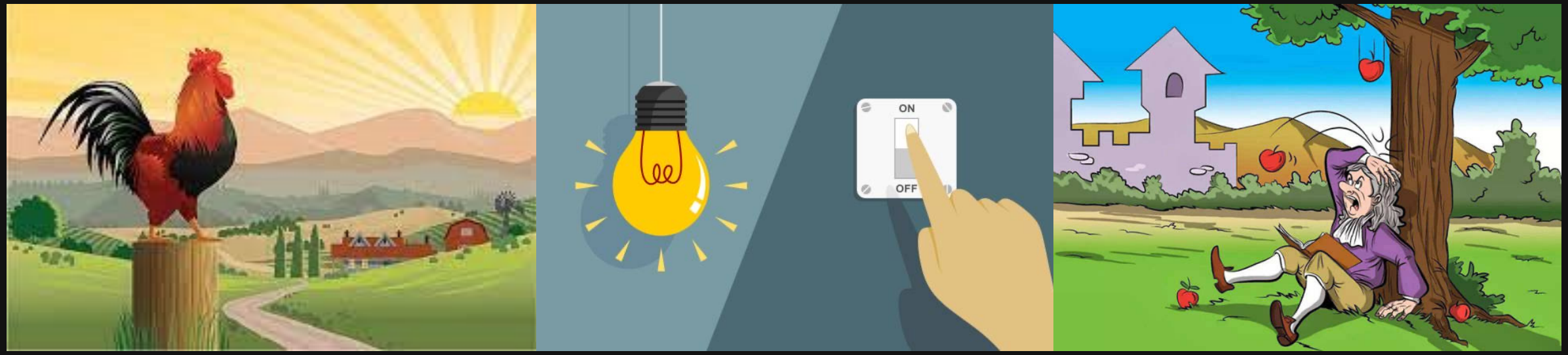


Instructor: Dr. Chi Zhang
School of Intelligence Science and Technology



咖啡因来自咖啡果。
咖啡果是因，
咖啡因是果。

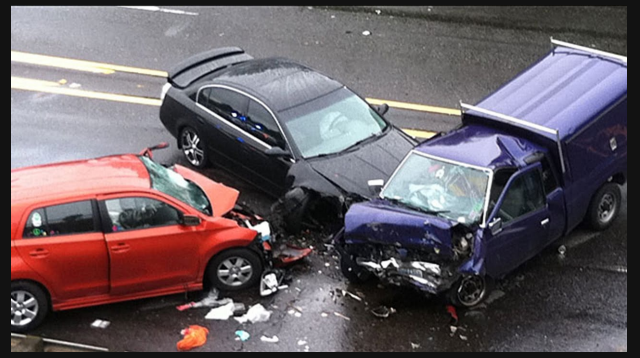
What comes to mind when you think of causality?



Causality is common in daily life.

Conjunctions

- 原来、因为、由于、这就是为什么、基于、理由是、原因是
- 因此、所以、是故、以致、以便、至乃、致使、于是
- 借由、借着
- 为了、为此、着眼于、向着
- 如果、如果不是、假设、若是、假使、倘若、要是、譬如、只要……就能……、在……的情况下、根据、否则
- because, since, so, as a result, therefore, then, furthermore, otherwise



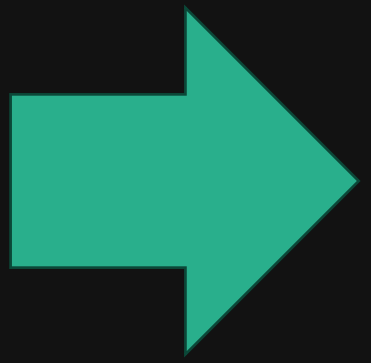
Causality is important for intelligence.

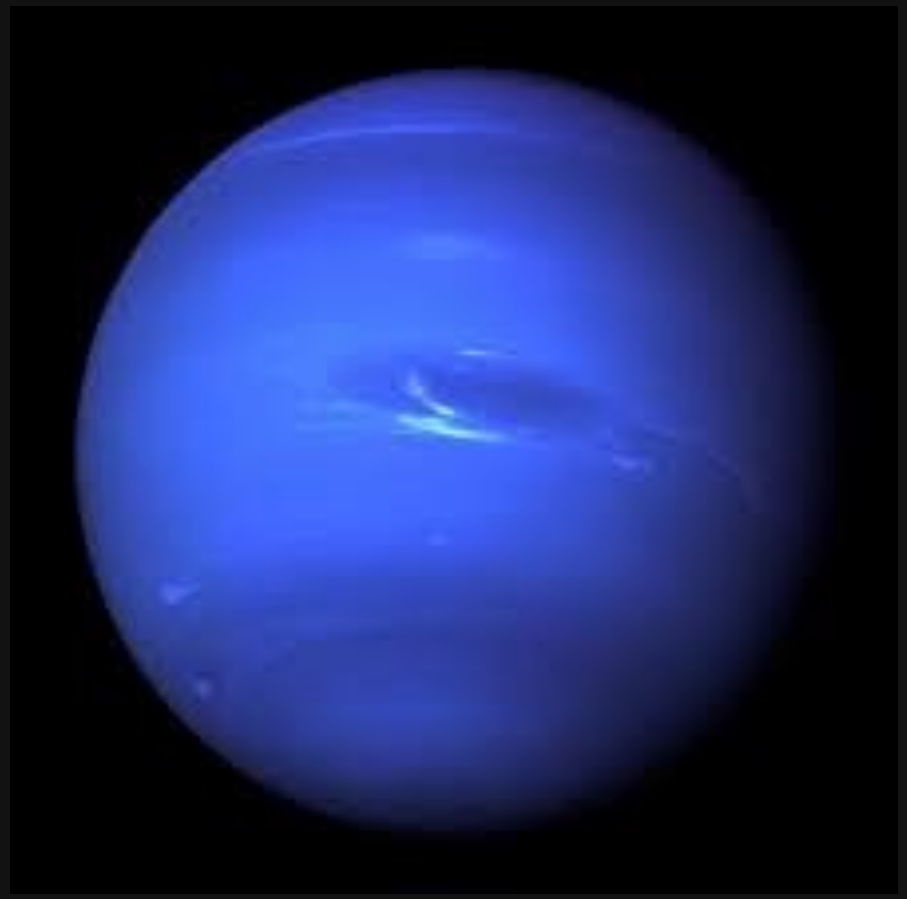
- Explanation

Causality is important for intelligence.

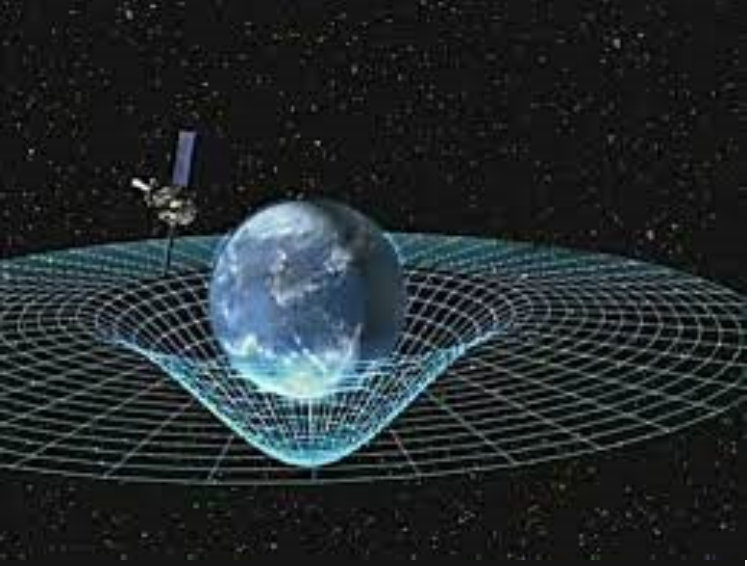
- Explanation
- Prediction

作业还没写完





The French astronomer Urbain Jean Joseph Le Verrier and the British astronomer John Couch Adams independently predicted the existence and position of the planet Neptune using only mathematics—a pivotal event in the history of astronomy. Neptune, whose existence was visually confirmed in 1846, was *the first planet to be discovered by mathematical rather than observational means*. The discovery of Neptune not only represents the greatest triumph for Newton's gravitational theory since the return of Halley's Comet in 1758, but it also marks the point at which mathematics and theory, rather than observation, began to take the lead in astronomical research. Discovered independently by two astronomers in working in England and in France, this discovery was not published in one or even two publications, but in a small group, thus making it somewhat more complicated than is usual in the study and collecting of documents in the history of science.



The Nobel Prize in Physics 2017

The Royal Swedish Academy of Sciences has decided to award the Nobel Prize in Physics 2017

with one half to

Rainer Weiss
LIGO/VIRGO Collaboration

and the other half jointly to

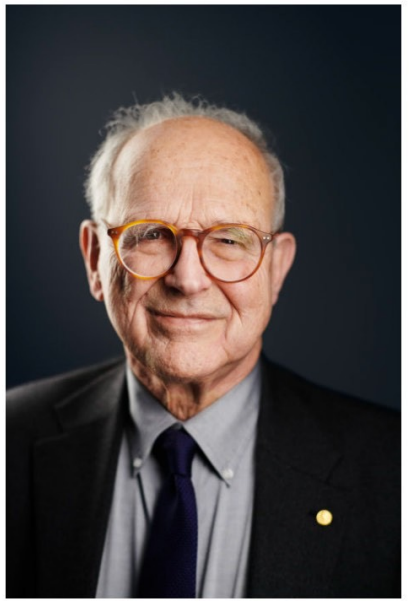
Barry C. Barish
LIGO/VIRGO Collaboration

and

Kip S. Thorne
LIGO/VIRGO Collaboration

"for decisive contributions to the LIGO detector and the observation of gravitational waves"

Gravitational waves finally captured



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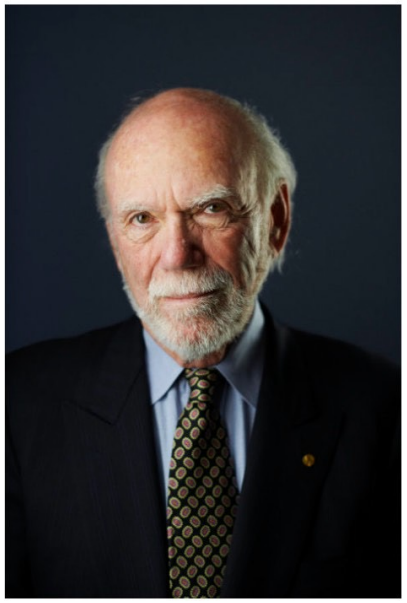
Rainer Weiss

Prize share: 1/2



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Barry C. Barish

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Kip S. Thorne

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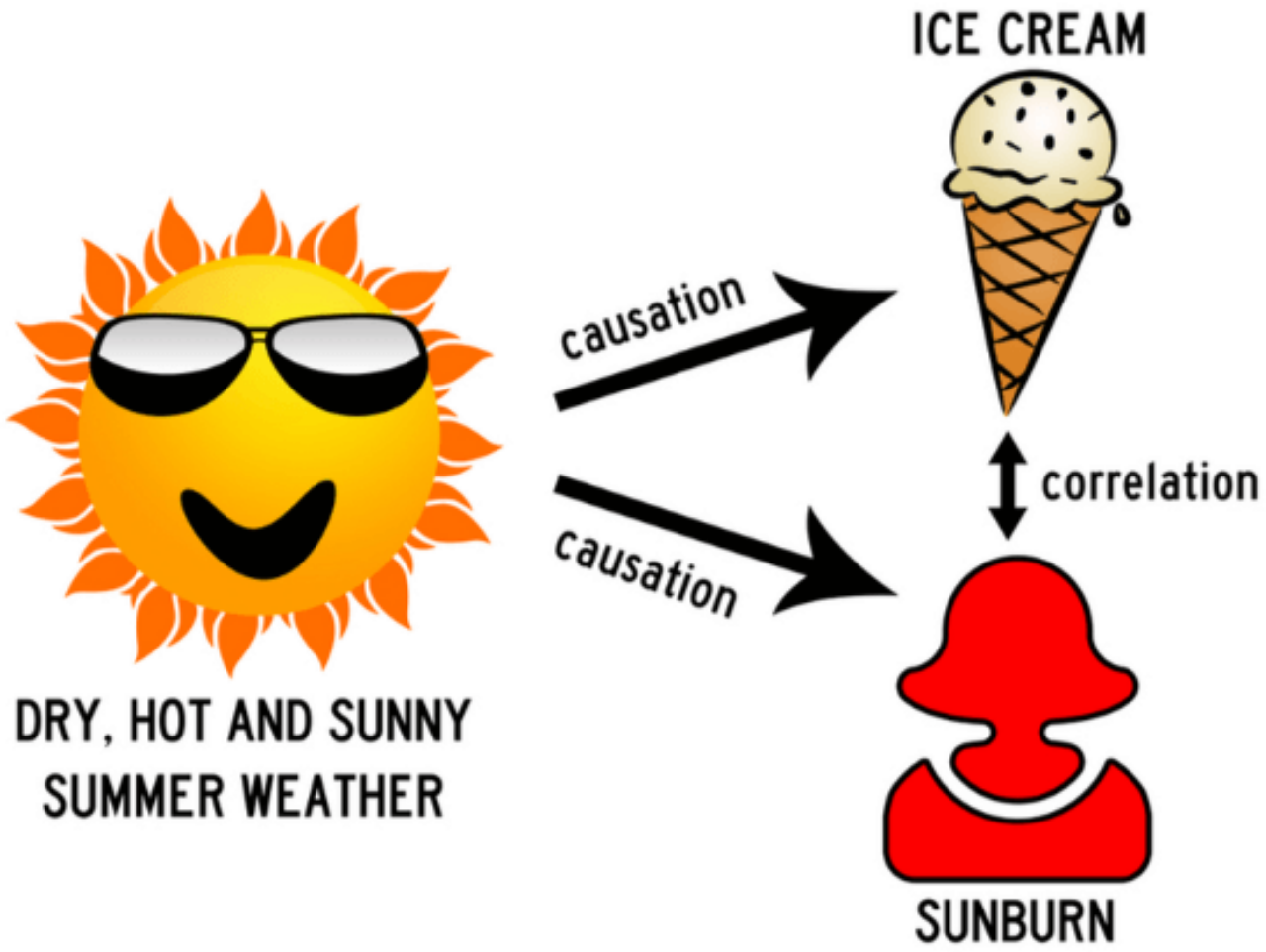
Causality is important for intelligence.

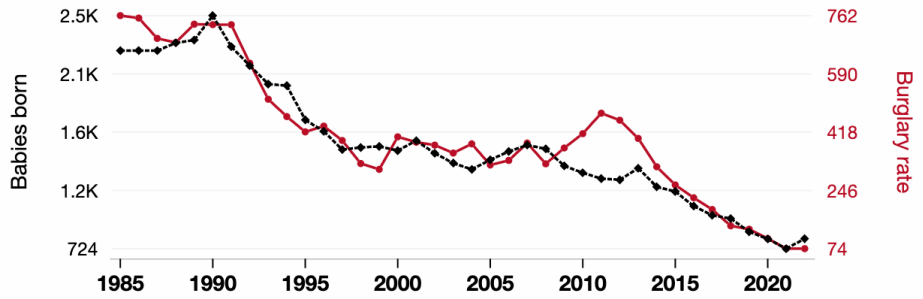
- Explanation
- Prediction
- Attribution


ICE CREAM
causation
correlation
causation
DRY, HOT AND SUNNY
SUMMER WEATHER
SUNBURN

Popularity of the first name Johnny

correlates with

Burglaries in New Hampshire



Babies of all sexes born in the US named Johnny · Source: US Social Security Administration

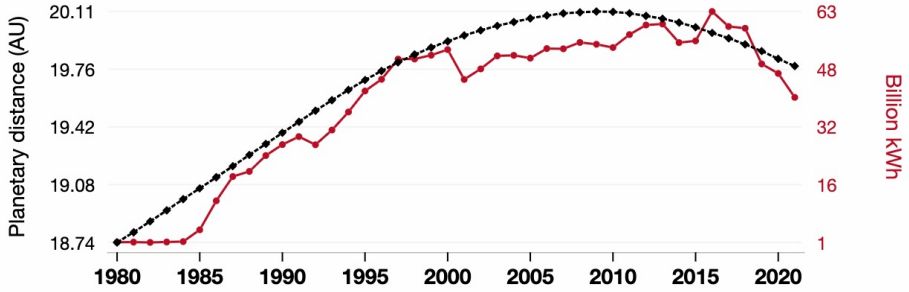
The burglary rate per 100,000 residents in New Hampshire · Source: FBI Criminal Justice Information Services

1985-2022, r=0.950, r²=0.902, p<0.01 · tylervigen.com/spurious/correlation/2496

The distance between Uranus and Earth

correlates with

Electricity generation in Paraguay



The average distance between Uranus and Earth as measured on the first day of each month · Source: Caclculated using Astropy

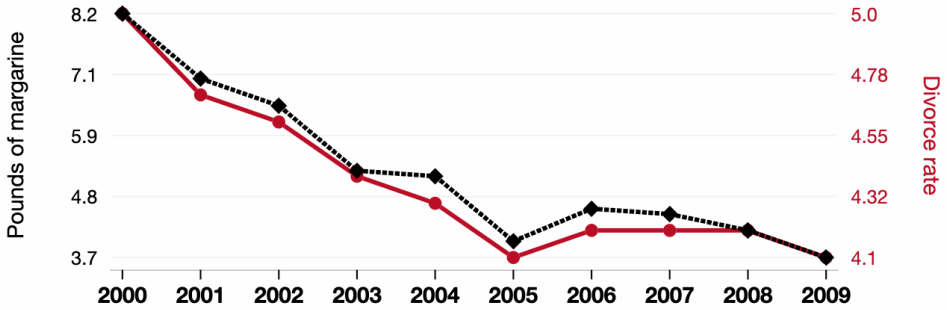
Total electricity generation in Paraguay in billion kWh · Source: Energy Information Administration

1980-2021, r=0.976, r²=0.952, p<0.01 · tylervigen.com/spurious/correlation/2727

Per capita consumption of margarine

correlates with

The divorce rate in Maine



Per capita consumption of margarine in the United States · Source: US Department of Agriculture

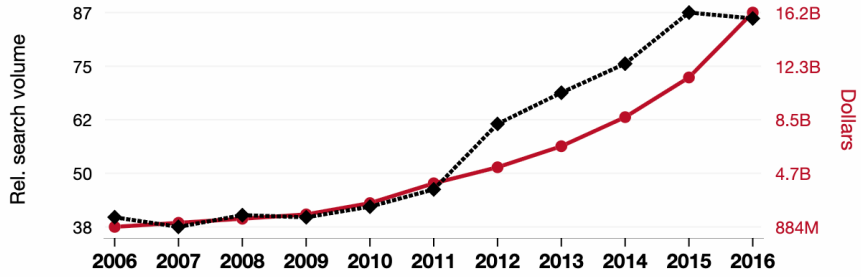
The divorce rate in Maine · Source: CDC National Vital Statistics

2000-2009, r=0.993, r²=0.985, p<0.01 · tylervigen.com/spurious/correlation/5920

Google searches for 'cute cats'

correlates with

Amazon's Annual Outbound Shipping Expenditure in Millions



Relative volume of Google searches for 'cute cats' (Worldwide, without quotes) · Source: Google Trends

Amazon's Annual Outbound Shipping Expenditure in Millions · Source: Statista

2006-2016, r=0.950, r²=0.902, p<0.01 · tylervigen.com/spurious/correlation/5231

https://www.tylervigen.com/spurious-correlations

冲浪选手卡里萨·摩尔无法穿幸运袜冲浪，但会给她的男朋友穿上

五次世锦赛冠军、美国冲浪选手卡莉莎·摩尔有一个独特的迷信习惯——不是在自己身上，而是在她的男朋友身上。这位2020年东京奥运会冠军坚信她男朋友穿幸运袜会给她的冲浪带来好运。“当我比赛时，我的男朋友必须穿上幸运袜。这很有趣。今年到目前为止，我输掉的所有比赛，他都没有穿上幸运袜，所以我的失利都是他的错！”摩尔在2014年接受《女性健康》杂志采访时分享道。

纳达尔：仪式之王

网球场上的一些迷信举动看起来最为有趣，使其成为一个每个细节和怪癖都可能成为选手在比赛日独特举动的运动。

最先让人想到的就是获得过22个大满贯冠军的拉法尔·纳达尔，这名西班牙选手在每场比赛前的45分钟整会进行冰冷的淋浴。他将这一仪式描述为进入一个新的、强大的精神空间的方式。在赛场上，他还会精心排列他的水瓶，使标签朝同一方向。他还遵循严格的行动顺序，例如不踩场地线，并在发球前进行特定的有关球衫、头发和脸部的一整套动作。

在2022年澳大利亚网球公开赛四分之一决赛对阵丹尼斯·沙波瓦洛夫时，纳达尔完成了146次他的衬衫拉扯、头发整理和脸部擦拭的例行动作。

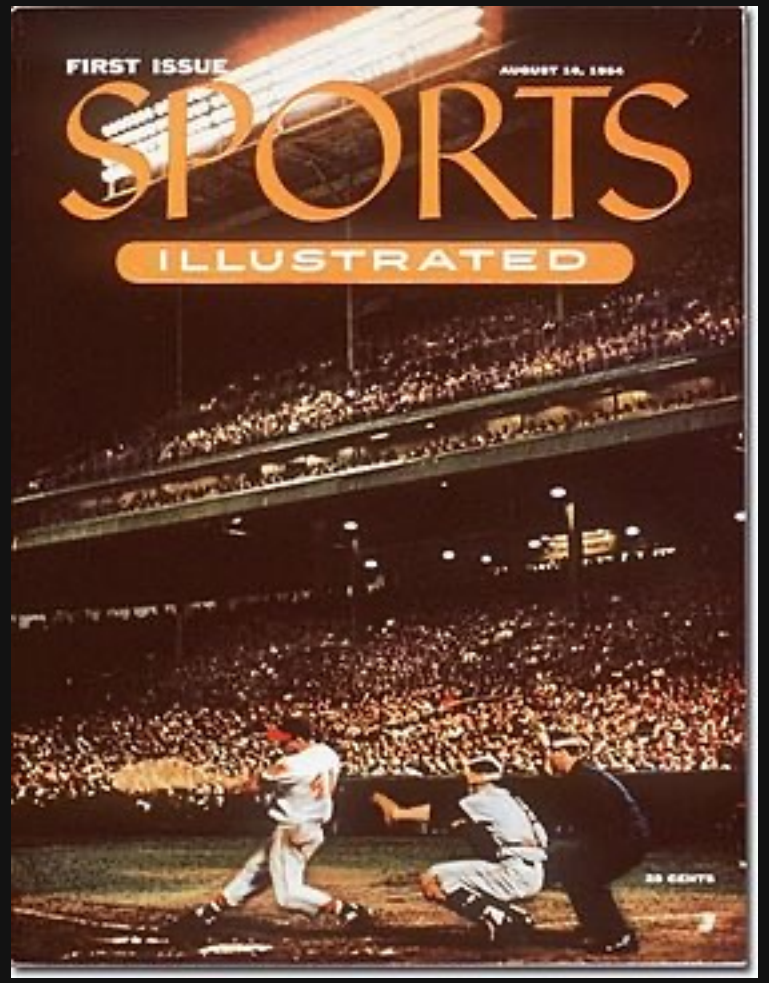
在他的回忆录《拉法》中，这位手握两块奥运金牌的网球运动员透露了一些自己的习惯。他说：“我必须把两个水瓶放在我椅子左边的脚前，一个整齐地排在另一个后面，成对角线地对准球场。”

纳达尔解释道：“网球是一项精神上极具攻击性的运动，任何可能受到外部干扰的东西，我总是能避免就尽量避免。”

大坂直美夺得迷信同样与水瓶有关

四次大满贯冠军大坂直美对她在球场上的水瓶也很讲究。和纳达尔一样，她的仪式围绕着如何精确排列水瓶展开，她要确保水平朝着同一方向并按特定顺序放置。

“我认为大多数运动员都有非常强的迷信，”大坂直美分享道，“可能是一样东西，可能是两样东西。对我来说，肯定是线条和标志。而且，我需要我的水瓶完全直放。我也不确定这么做时为什么。”这位网球明星解释道。



封面带来的诅咒 [编辑]

- 当美国职棒大联盟球员艾迪·马修斯成为第一期封面人物一个星期后，遭到伤患困扰，错失了七场比赛。“体育画报诅咒”因而诞生。
- 1957年11月18日奥克拉荷马大学赢得连续47场大学美式橄榄球比赛，是当时最长的连胜记录。封面大字标题写着“为什么奥克拉荷马是无法打败？”。不过在下一场比赛被圣母大学所击败。此外圣母大学在1953年是奥克拉荷马取得47连胜前最后一次击败他们的球队。
- 1970年12月14日，德州奥斯汀大学当时10胜0负以及有连胜30场的记录[2]，在棉花碗比赛出现9次漏接以11–24负于圣母大学。
- 2002年其中一期，封面标题写关于它们带来的诅咒，封面是一只黑猫[3]。圣路易斯公羊四分卫库尔特·瓦纳想与黑猫一起登上封面，但遭杂志拒绝。结果公羊队连胜两场，成为国联会的冠军。

VACCINES CAN CAUSE INJURY AND DEATH
FORCED VACCINES = MEDICAL TYRA
U.S. LAW CLASSIFIES VACCINES AS UNAVOIDABLY UNSAFE
VACCINE MANDATES VIOLATE BODILY AUTONOMY

You are expected to learn…

- Part 1: A historical tour

  How do pioneers understand causality from different perspectives?

- Part 2: Cognitive phenomena

  How people understand and perceive causation?

- Part 3: Computational models

  How to describe the causal cognition in computation?

- Part 4: Modern approaches

  What are researchers doing nowadays?

You are expected to think…

- How our understanding of causality changes across the history?
- How to position a literature on causality within the framework?
- Is cause and effect a phenomenon in nature, something that really exists, or is it a phenomenon of human cognition, something that exists in the human brain?
- How to approach causality in AI?

Outline

- Part 1: A historical tour

  How do pioneers understand causality from different perspectives?

- Part 2: Cognitive phenomena

  How people understand and perceive causation?

- Part 3: Computational models

  How to describe the causal cognition in computation?

- Part 4: Modern approaches

  What are researchers doing nowadays?

A History Roadmap

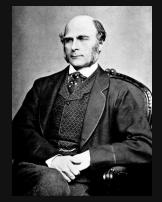
Philosophy

Aristotle (384–322 BC)
Four causes

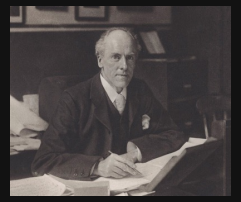
Hume (1711-1776)
Regularity

Kant (1724-1804)
Necessity

Statistics & Economics & Epidemiology



Francis Galton (1877)
Regression analysis



Karl Pearson (1904)
Correlation

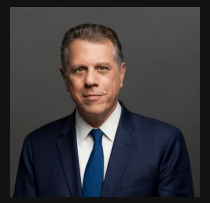
Sewall Wright (1921)
Path analysis

Donald Rubin (1980)
Rubin causal model

Judea Pearl (2000)
Pearl causal model

Psychology & Cognition

Albert Michotte (1963)
Causal perception



Michael Waldmann (1992)
Causal structure

New eras of causality research

Causal inference

Causal discovery

Causal learning

Causal reasoning

Causal judgment

Causal reinforcement learning

Causal representation learning

…

Look into philosophy

Aristotle (384–322 BC)
Four causes

Hume (1711-1776)
Regularity

Kant (1724-1804)
Necessity

What is causality?

Aristotle: four types of cause

There are four types of cause to understand a thing:

- Material cause: What is it made of? (Matter)
- Formal cause: What is its form or essence? (Shape, definition)
- Efficient cause: What brought it into being? (Agent of change)
- Final cause: What is its purpose or end? (Goal, function)

Aristotle: four types of cause

A famous marble statue

- Material cause: Marble block
- Formal cause: The form of David
- Efficient cause: The sculptor's hands and tools
- Final cause: Artistic expression, a tribute, or a masterpiece

Aristotle: four types of cause

- Historical Legacy:
  - The first systematic theory to study and classify the concept of "cause."
  - Initiated the West's systematic philosophical inquiry into "why" things are.
- Role in Aristotle's Philosophy:
  - A foundational core of his natural philosophy and metaphysics.
  - Emphasizes the Final Cause, embodying his teleological view: all things aim to fulfill their inherent purpose or potential.
- Influence on Posterity:
  - A cornerstone of medieval Scholasticism, with a lasting impact.
  - Challenged by the rise of modern science, particularly after Galileo and Newton.
  - Modern science shifted focus from the Final Cause to the Efficient Cause, explaining change through physical laws.

David Hume: Does causality really exist?

- David Hume, 1711-1776
  - Scottish philosopher, historian, economist, and essayist
  - Best known for his highly influential system of ***empiricism, philosophical scepticism and metaphysical naturalism***

- ***Core question:*** *How do we really know "A causes B"? Can we induce causality from sensory experience?*

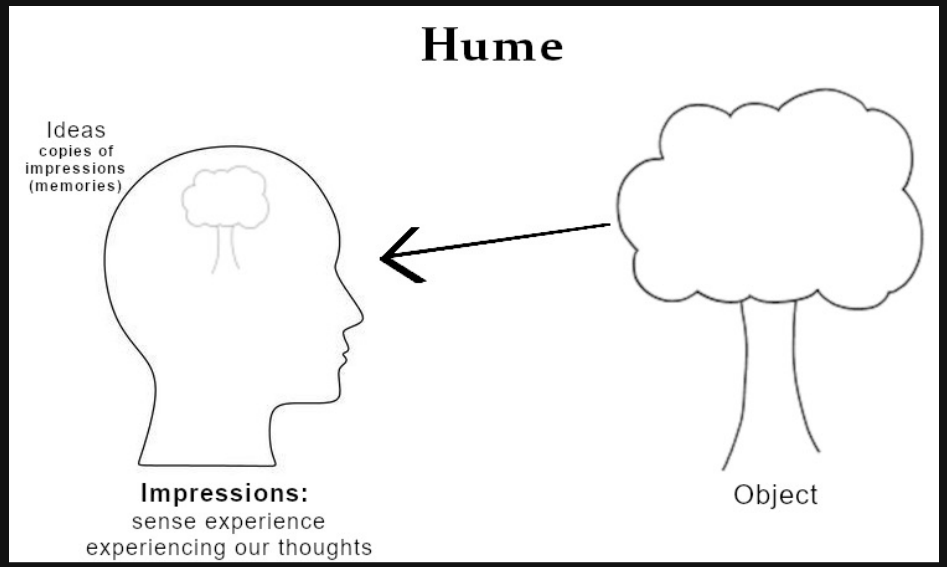
- Hume's conclusion: Causality is not a necessary connection in the objective world, but a subjective **belief** in our minds.

The challenge: what do we see?

- *What do we observe?*
  - **Spatial Contiguity**: The two balls are close in space.
  - **Temporal Priority**: The white ball moves first, then the red ball.
- *What do we not observe?*
  - We never observe the "necessary connection" itself.
  - This "necessary connection" is an **imagination** of the mind, not an **objective** fact.

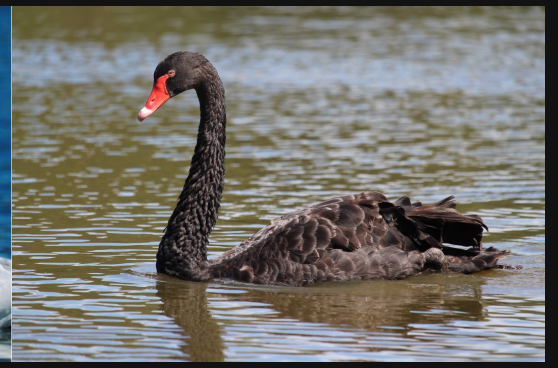
Hume's solution: Habit and Association

- Constant Conjunction: When we repeatedly observe two events happening in the same sequence, our minds begin to associate them.
- Custom: This repeated experience creates a psychological habit.
- Subjective Belief: The habit leads us to believe that when we see "A" again, "B" will surely follow. This is a belief, not a rational deduction.



Core Concept: The Illusion of Necessary Connection

- Necessary Connection: The heart of what we call “causality." Hume argues it is an illusion.
- Problem of Induction: We cannot use a limited number of past experiences to deduce a necessary future outcome.
- Non-Rational Foundation: The foundation of causality is not logic or reason, but our psychological habits and associations. This profoundly challenged rationalist philosophy.

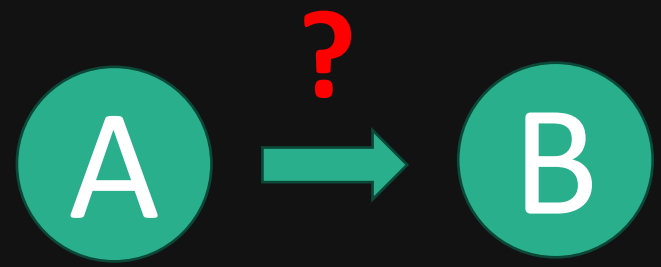


Immanuel Kant: The Philosophical Giant

- Immanuel Kant (1724-1804)
  - a German philosopher and one of the central thinkers of the Enlightenment.
  - Core Contribution: Founded critical philosophy, widely regarded as the starting point of modern philosophy.
  - The "Awakener": Kant admitted that Hume's skepticism "awakened me from my dogmatic slumber."
  - Philosophical Mission: To find a solid foundation for the necessity and universality of human knowledge, bridging the gap between rationalism and empiricism.

Hume's Challenge: The Dilemma of Knowledge

- Hume's Conclusion: We cannot find a necessary connection between cause and effect through experience. We only observe the spatial contiguity and temporal succession of events.
- Kant's Puzzle: If Hume is right, how can scientific knowledge (e.g., the laws of physics) possess universal necessity?
- Core Question: How are propositions like "Every event has a cause" possible? These are not derived from experience, yet we hold them to be necessarily true. Kant called these synthetic a priori judgments.
- The Lead-in: Kant realized that to solve this dilemma, we must change how we think about knowledge itself.



Critik
der
reinen Vernunft
von
Immanuel Kant
Professor in Königsberg.
Riga,
verlegts Johann Friedrich Hartknoch
1781.

Critique of Pure Reason

PlatoZane bilibili

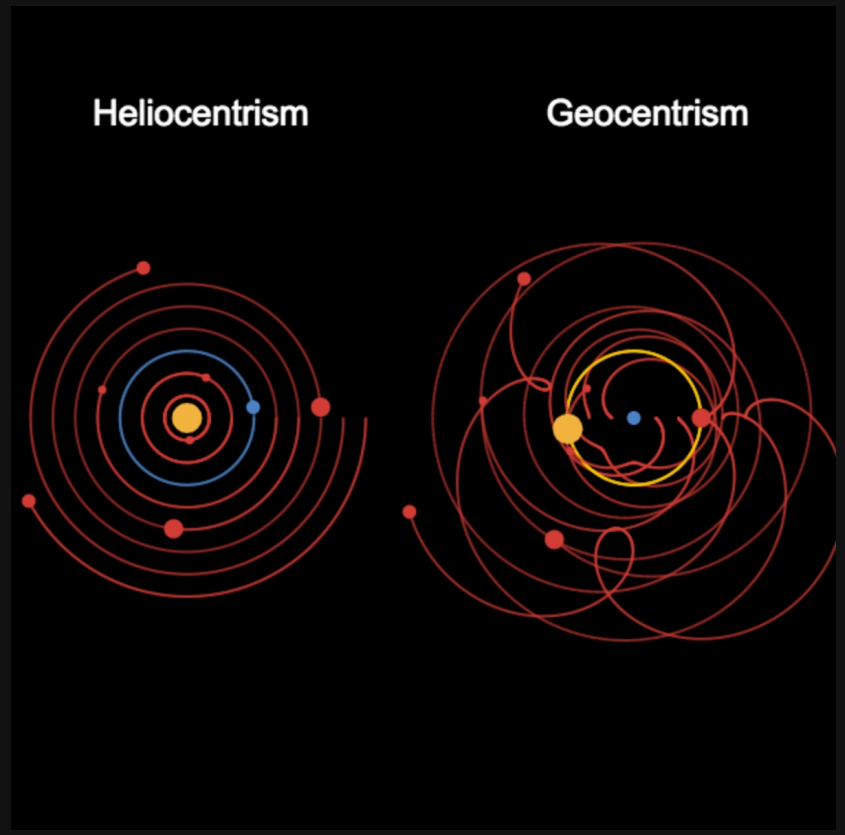
纯粹理性批判

Kritik der reinen Vernunft

伊曼努尔·康德

A Revolution in Thought: The Copernican Turn

- Traditional Epistemology: The mind passively conforms to the external world.
- Kant's Revolutionary View: The external world must conform to the mind's innate structure. Just as Copernicus overturned the geocentric model, Kant overturned the "object-centric" view of knowledge.
- The Analogy: Our minds are not blank slates; they come equipped with innate "organizing tools." Think of it as a pair of "causality glasses" we are born with.
- The Birth of Causality: We don't "discover" causality from the external world. Instead, our mind's innate structure automatically organizes sensory experience into cause-and-effect relationships. Without these "glasses," the world would be a chaotic stream of sensations.



Kant's Solution to Hume's Challenge

1. The Problem of Necessity

- Hume's Challenge: Causal connections are not necessary, only habitual associations derived from repeated experience.
- Kant's Response: Causality is not a product of experience, but a condition for experience. Just as we necessarily perceive through space-time, we necessarily organize experience through causality

2. The Problem of Universality

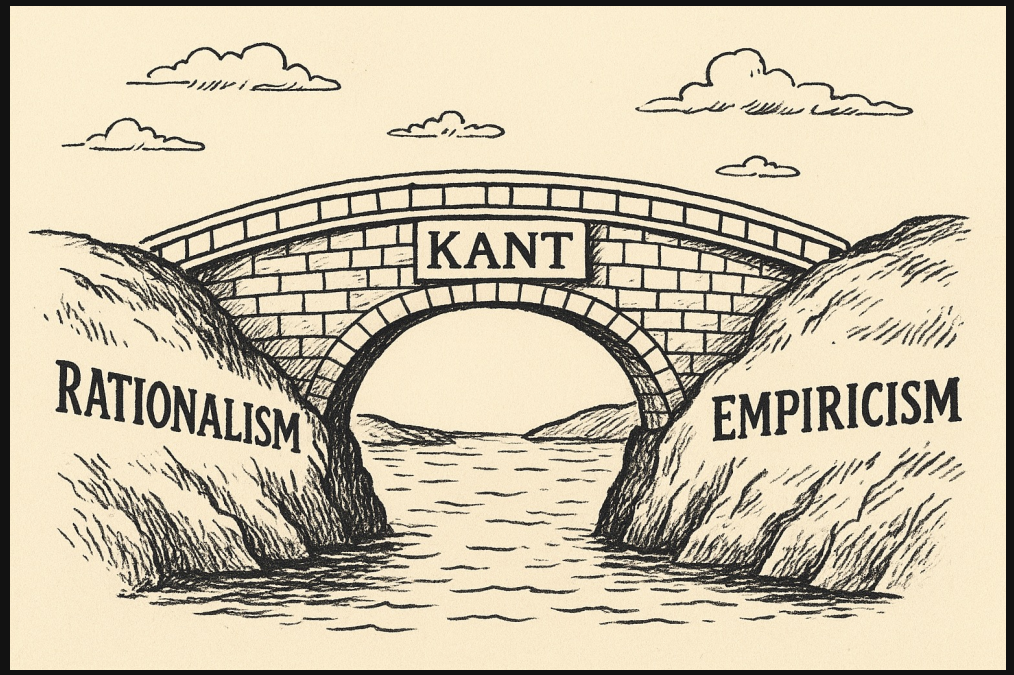
- Hume's Challenge: Different people may develop different habits, undermining universal validity.
- Kant's Response: All rational beings share the same cognitive structure.

3. The Problem of Induction

- Hume's Challenge: We cannot logically justify inferring future from past experiences.
- Kant's Response: Within the phenomenal realm, causal laws necessarily apply.

The Ultimate Answer and Historical Legacy

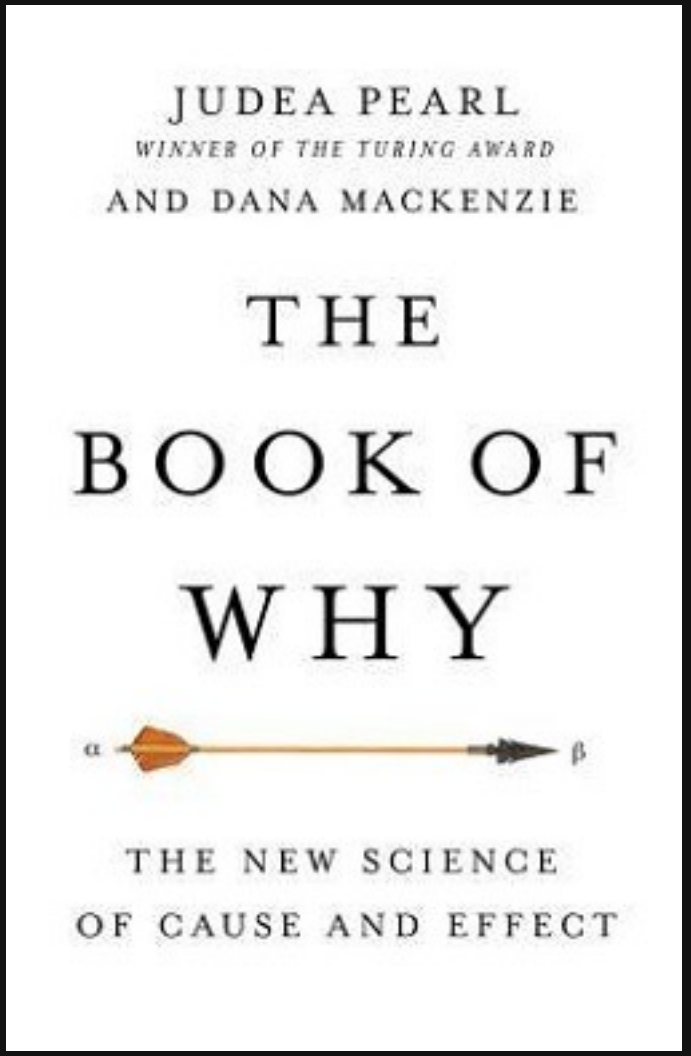
- The Nature of Causality: Causality is an inherent category of understanding in the human mind, a rule for organizing experience that is present from birth.
- The Distinction from Hume:
  - Hume: Causality is a posterior, non-necessary psychological habit.
  - Kant: Causality is an a priori, universally necessary structure of thought.
- Kant's Legacy:
  - His theory provided a strong philosophical defense for the universal necessity of scientific knowledge.
  - It also drew a crucial boundary between what we can know (the phenomenal world) and what we cannot (the noumenal world, or "things-in-themselves").



The Fundamental Divide: Hume vs. Kant

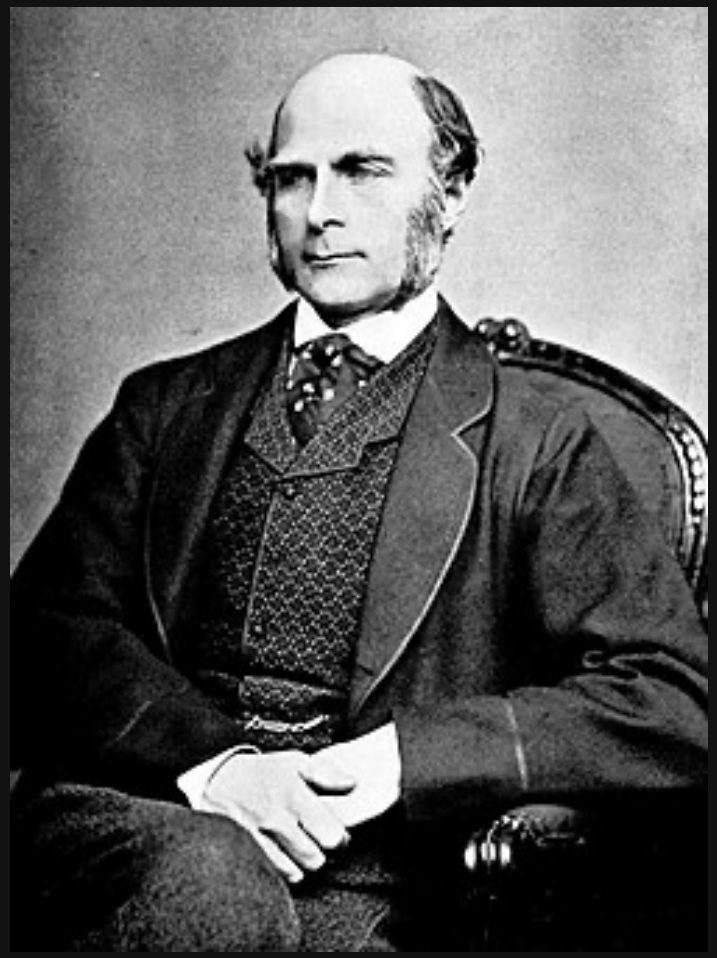
Aspect	Hume	Kant
Nature	A posteriori (derived from experience)	A priori (prior to experience)
Necessity	Non-necessary habit	Universally necessary structure
Source	Formed through repeated experiences	Innate endowment of reason
Status	Psychological phenomenon	Epistemological condition

Look into modern science

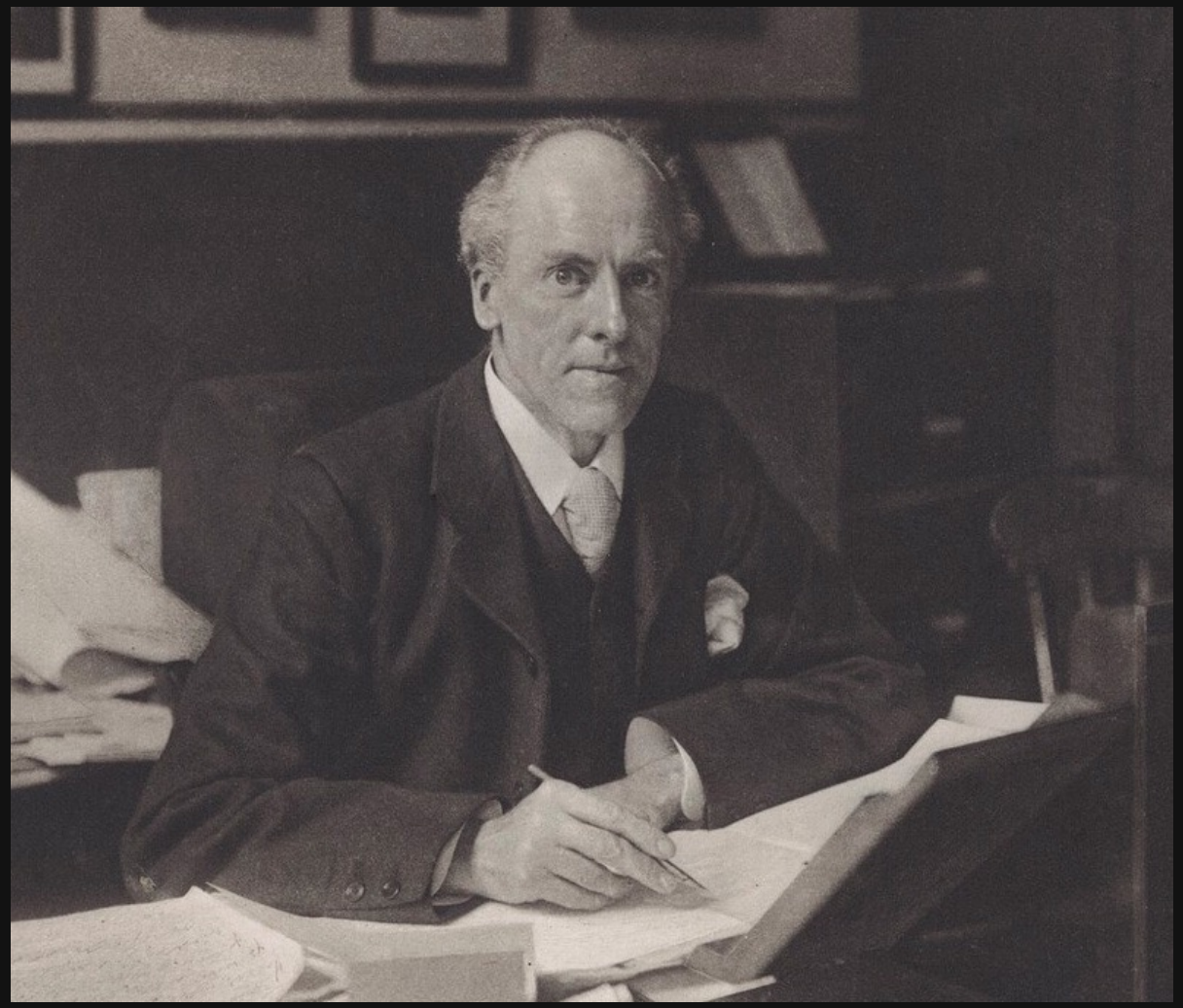


How causality becomes rigorous science?

Francis Galton: Discovery of correlation



Karl Pearson: Correlation



$$r_{xy} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

Simpson paradox

- Gender bias among graduate school admissions to University of California, Berkeley.

	All		Men		Women	
	Applicants	Admitted	Applicants	Admitted	Applicants	Admitted
Total	12,763	41%	8,442	44%	4,321	35%

Department	All		Men		Women	
	Applicants	Admitted	Applicants	Admitted	Applicants	Admitted
A	933	64%	**825**	62%	108	82%
B	585	63%	**560**	63%	25	68%
C	918	35%	325	37%	**593**	34%
D	792	34%	417	33%	375	35%
E	584	25%	191	28%	**393**	24%
F	714	6%	373	6%	341	7%
Total	**4526**	**39%**	**2691**	**45%**	**1835**	**30%**

Legend:

greater percentage of successful applicants than the other gender
greater number of applicants than the other gender

bold – the two 'most applied for' departments for each gender

Sewell Wright: Path diagram

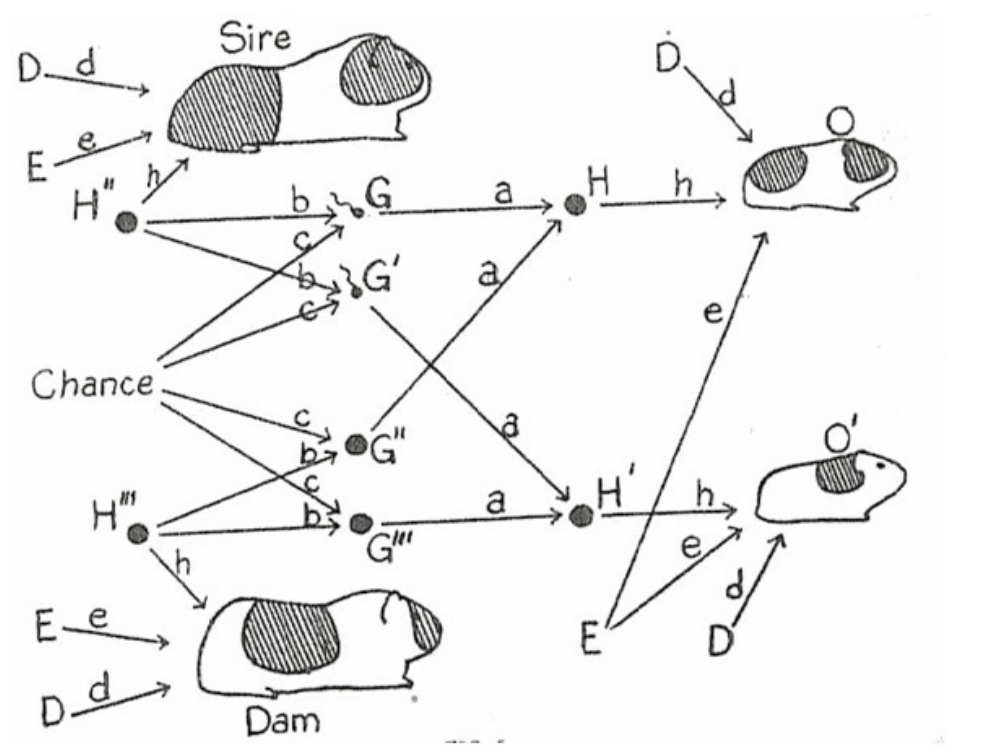
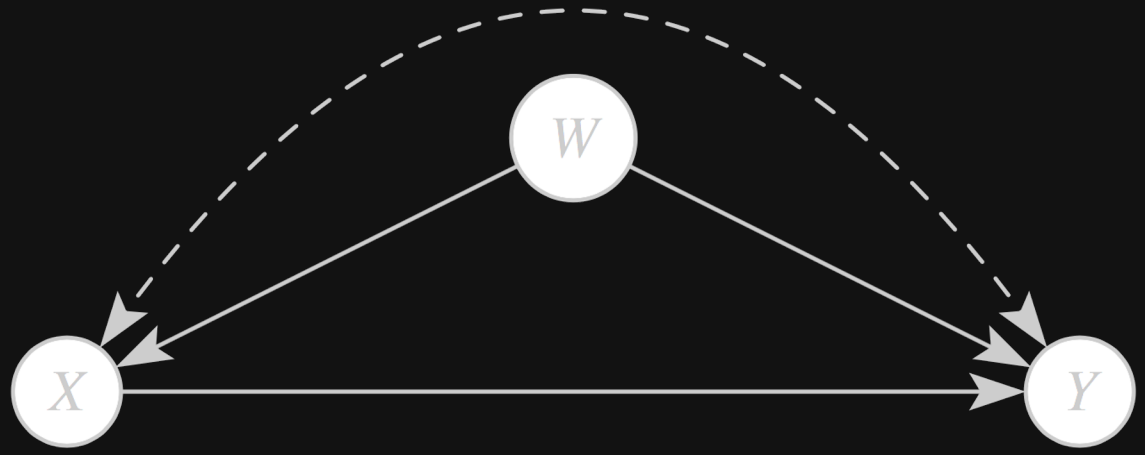


Figure I. Wright's First Path Diagram showing the influence of heredity and environment on the inheritance of color in the guinea pig (reproduced from Wright, 1920, p.328).

Donald Rubin: Potential outcome



An example of potential outcome

Question: Does College Education Increase Income?

Simple Comparison:

- College Group Average Income: $80,000
- No-College Group Average Income: $50,000
- Difference: $30,000

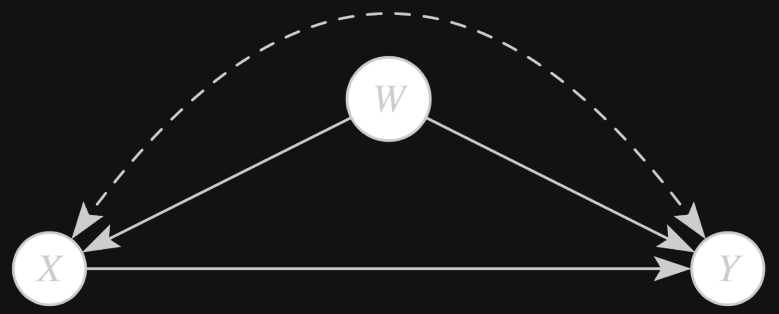
Question: Is this $30,000 really due to education?

An example of potential outcome

Question: Does College Education Increase Income?

Variable Setup:

- W: Family Background (Wealthy vs. Poor)
- X: College Education (1=College, 0=No College)
- Y: Annual Income at Age 30



The Confounding Problem

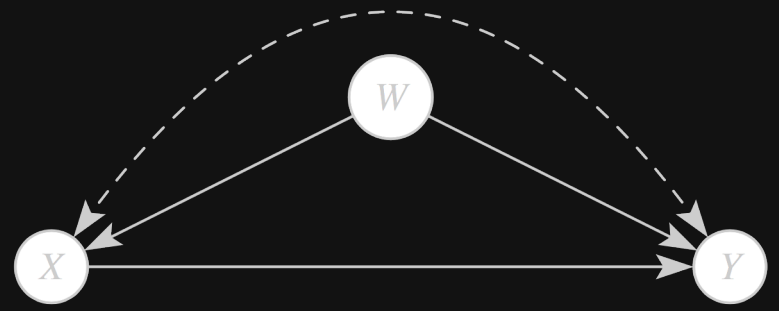
- W->X Wealthy families' children are more likely to attend college
- W->Y Wealthy families' children may earn more even without college (connections, resources)

An example of potential outcome

Step 1: Define Potential Outcomes

For each individual, define two potential outcomes:

- Y_1: Income if this person goes to college
- Y_0: Income if this person doesn't go to college



Example:

Xiao Ming (Wealthy): Y_1 = \$90k, Y_0 = \$60k → Individual Effect = \$30k

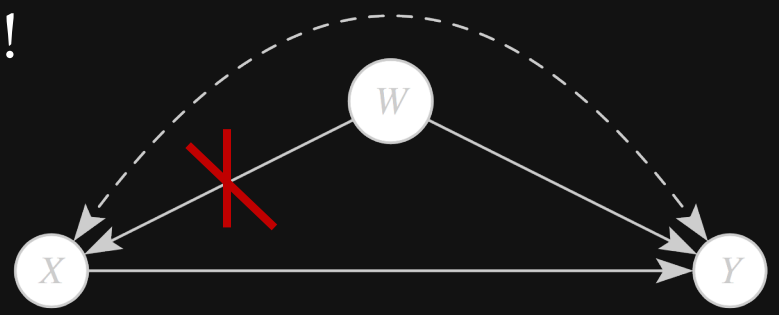
Xiao Hong (Poor): Y_1 = \$70k, Y_0 = \$40k → Individual Effect = \$30k

Step 2: The Fundamental Problem

- We only observe one outcome per person
- Xiao Ming's Y_0 is forever missing - this is the missing data problem!

An example of potential outcome

Magic of Randomization: The arrow W → X is severed!

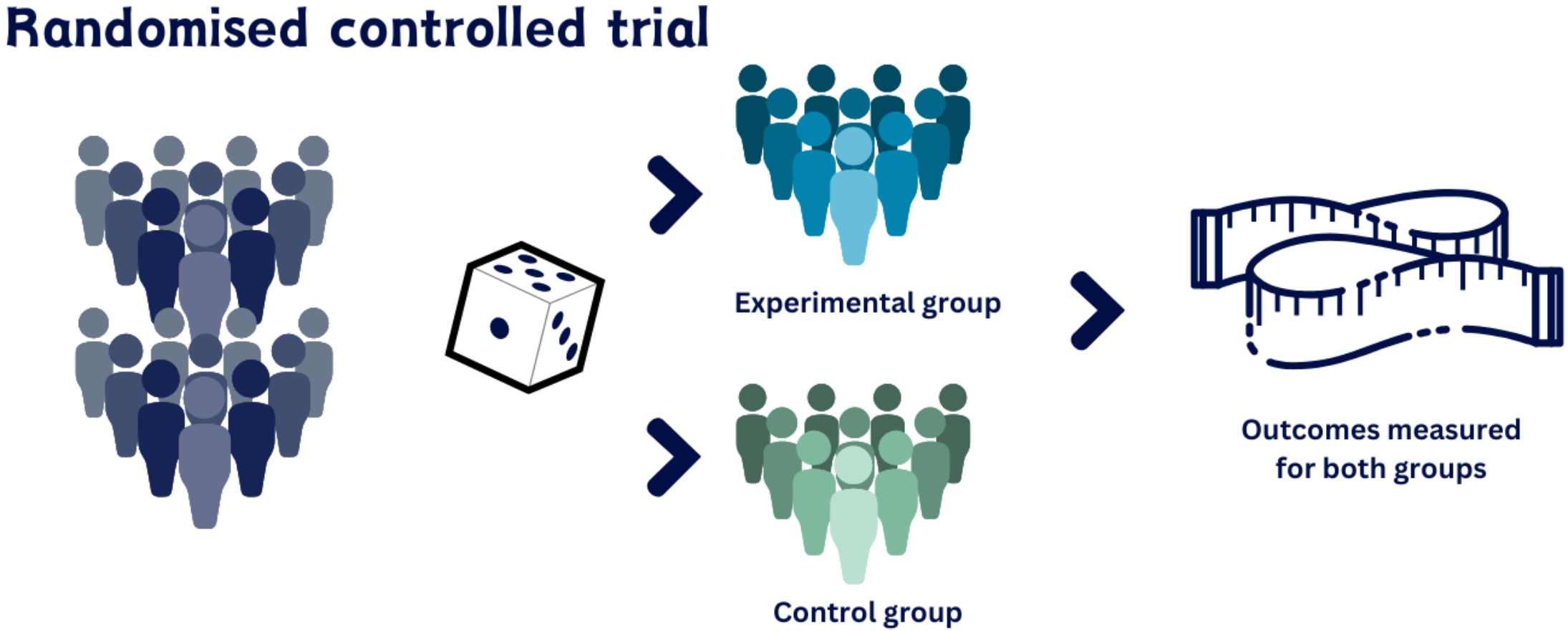


Treatment Group (Randomly assigned to college):
- 250 wealthy + 250 poor students
- Average income at 30: $75,000

Control Group (Randomly assigned no college):
- 250 wealthy + 250 poor students
- Average income at 30: $45,000

Average Treatment Effect = $75,000 - $45,000 = $30,000
This $30,000 is the TRUE causal effect of education!

Random controlled trial



The NEW ENGLAND JOURNAL of MEDICINE

ESTABLISHED IN 1812 MARCH 18, 2021 VOL. 384 NO. 11

Once-Weekly Semaglutide in Adults with Overweight or Obesity

RESULTS

The mean change in body weight from baseline to week 68 was –14.9% in the semaglutide group as compared with –2.4% with placebo, for an estimated treatment difference of –12.4 percentage points (95% confidence interval [CI], –13.4 to –11.5; P<0.001). More participants in the semaglutide group than in the placebo group achieved weight reductions of 5% or more (1047 participants [86.4%] vs. 182 [31.5%]), 10% or more (838 [69.1%] vs. 69 [12.0%]), and 15% or more (612 [50.5%] vs. 28 [4.9%]) at week 68 (P<0.001 for all three comparisons of odds). The change in body weight from baseline to week 68 was –15.3 kg in the semaglutide group as compared with –2.6 kg in the placebo group (estimated treatment difference, –12.7 kg; 95% CI, –13.7 to –11.7). Participants who received semaglutide had a greater improvement with respect to cardiometabolic risk factors and a greater increase in participant-reported physical functioning from baseline than those who received placebo. Nausea and diarrhea were the most common adverse events with semaglutide; they were typically transient and mild-to-moderate in severity and subsided with time. More participants in the semaglutide group than in the placebo group discontinued treatment owing to gastrointestinal events (59 [4.5%] vs. 5 [0.8%]).

The NEW ENGLAND JOURNAL of MEDICINE

ESTABLISHED IN 1812 APRIL 21, 2022 VOL. 386 NO. 16

Calorie Restriction with or without Time-Restricted Eating in Weight Loss

Deying Liu, M.D., Yan Huang, M.S., Chensihan Huang, M.D., Shunyu Yang, M.D., Xueyun Wei, M.D., Peizhen Zhang, M.D., Dan Guo, M.D., Jiayang Lin, M.D., Bingyan Xu, M.D., Changwei Li, Ph.D., Hua He, Ph.D., Jiang He, M.D., Ph.D., Shiqun Liu, M.D., Linna Shi, M.D., Yaoming Xue, M.D., and Huijie Zhang, M.D., Ph.D.

RESULTS

Of the total 139 participants who underwent randomization, 118 (84.9%) completed the 12-month follow-up visit. The mean weight loss from baseline at 12 months was –8.0 kg (95% confidence interval [CI], –9.6 to –6.4) in the time-restriction group and –6.3 kg (95% CI, –7.8 to –4.7) in the daily-calorie-restriction group. Changes in weight were not significantly different in the two groups at the 12-month assessment (net difference, –1.8 kg; 95% CI, –4.0 to 0.4; P=0.11). Results of analyses of waist circumferences, BMI, body fat, body lean mass, blood pressure, and metabolic risk factors were consistent with the results of the primary outcome. In addition, there were no substantial differences between the groups in the numbers of adverse events.

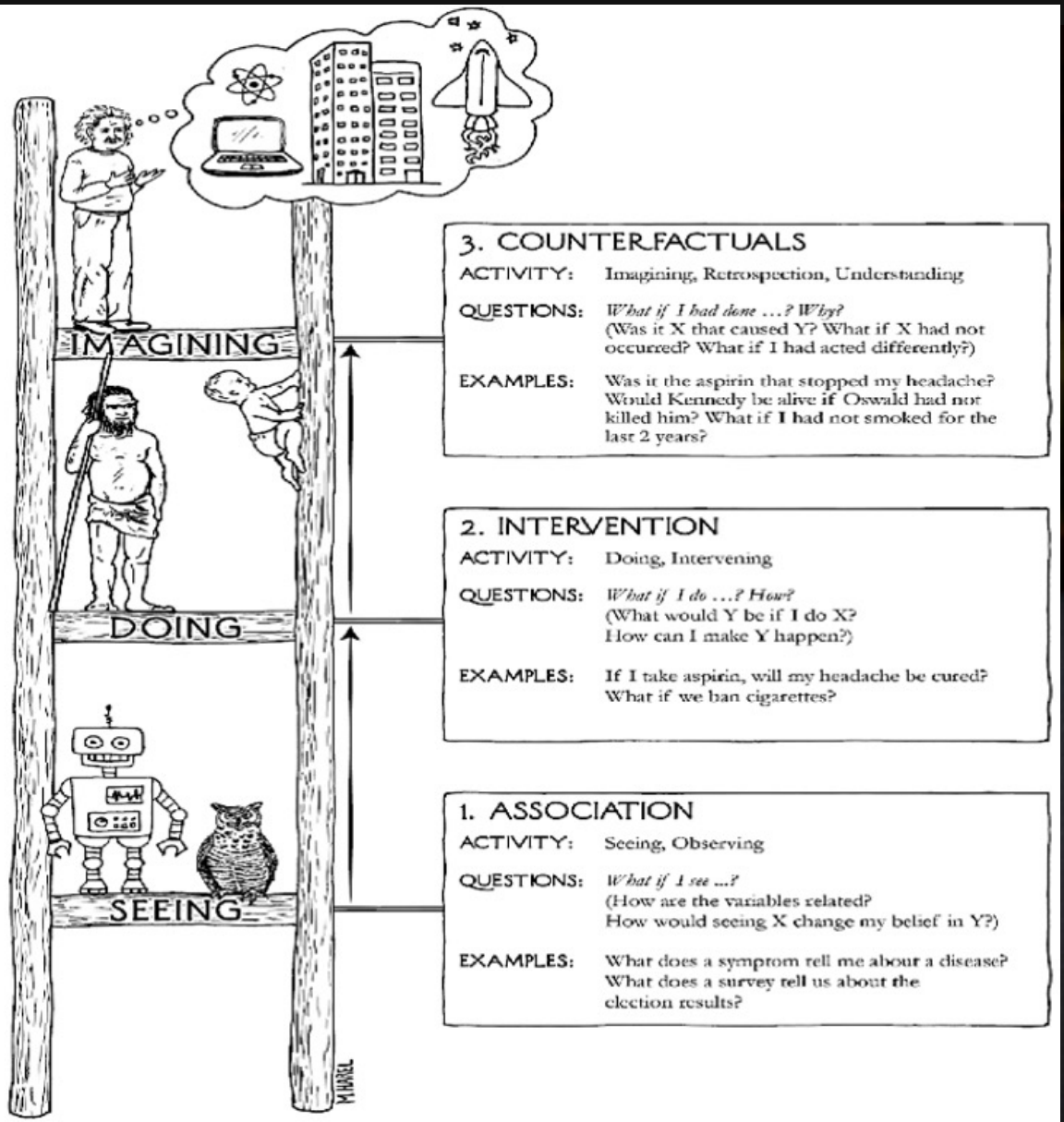
Judea Pearl: Do calculus and counterfactuals

The Ladder of Causality

"Actual" Causality

"Causality-in-mean"

Statistics



3. COUNTERFACTUALS

ACTIVITY: Imagining, Retrospection, Understanding

QUESTIONS: *What if I had done ...? Why?*
(Was it X that caused Y? What if X had not occurred? What if I had acted differently?)

EXAMPLES: Was it the aspirin that stopped my headache? Would Kennedy be alive if Oswald had not killed him? What if I had not smoked for the last 2 years?

2. INTERVENTION

ACTIVITY: Doing, Intervening

QUESTIONS: *What if I do ...? How?*
(What would Y be if I do X?
How can I make Y happen?)

EXAMPLES: If I take aspirin, will my headache be cured? What if we ban cigarettes?

1. ASSOCIATION

ACTIVITY: Seeing, Observing

QUESTIONS: *What if I see ...?*
(How are the variables related?
How would seeing X change my belief in Y?)

EXAMPLES: What does a symptom tell me about a disease? What does a survey tell us about the election results?

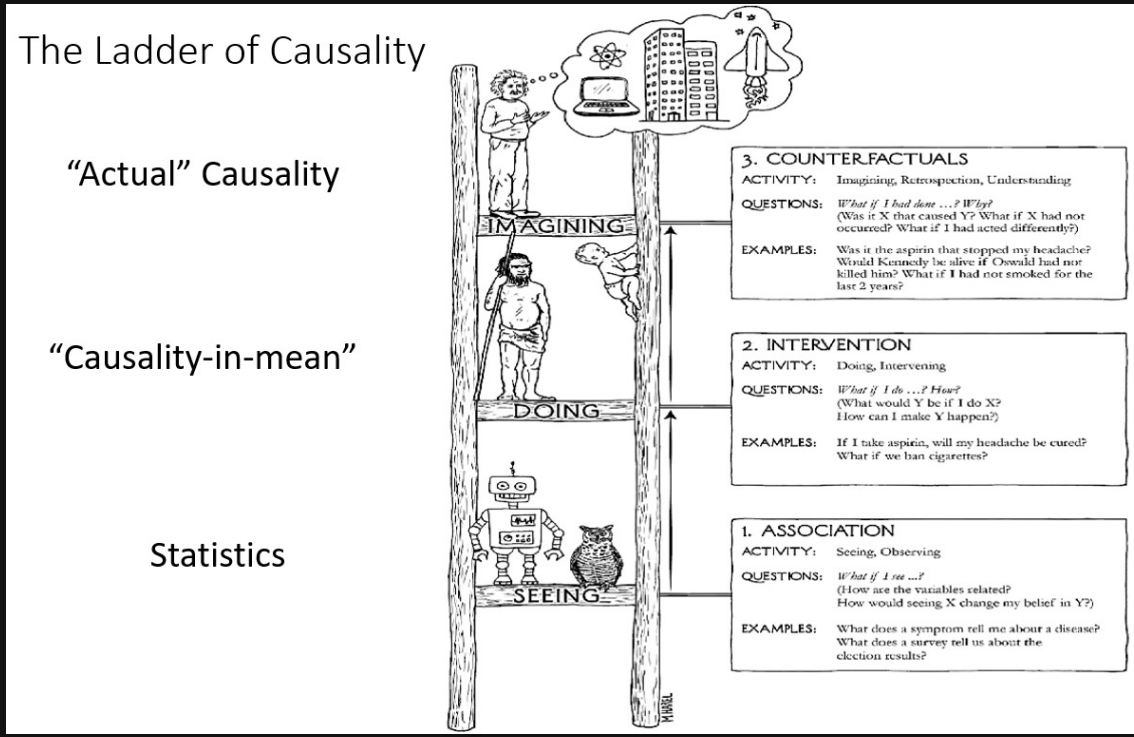
Pearl's Ladder of Causation

Layer 1: Association

- Definition: Observational association - passive observation of statistical relationships in data
- Mathematical Notation: P(Y | X)
- Question Answered:

"What is the probability of Y when we observe X?"

- Examples:
  - Lung cancer rates among smokers
  - Average income of college graduates
  - Recovery rates of patients taking medication
- Key Limitation: Cannot distinguish correlation from causation



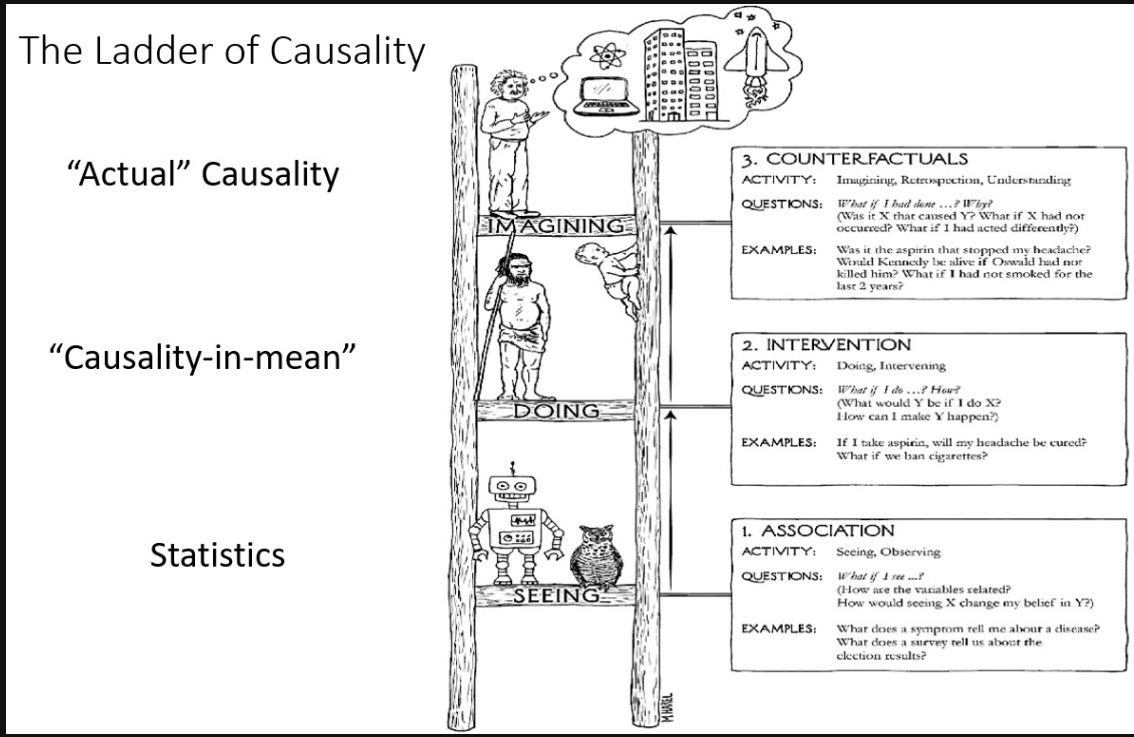
Pearl's Ladder of Causation

Layer 2: Intervention

- Definition: Active manipulation - external intervention on the system
- Mathematical Notation: P(Y | do(X))
- Question Answered:

"What would be the probability of Y if we force X to take a specific value?"

- Examples:
  - If we force everyone to quit smoking, how would lung cancer rates change?
  - If we implement free education policy, how much would national income increase?
  - If we give all patients the new drug, what would be the cure rate?
- Key Feature: Breaks confounding paths through intervention



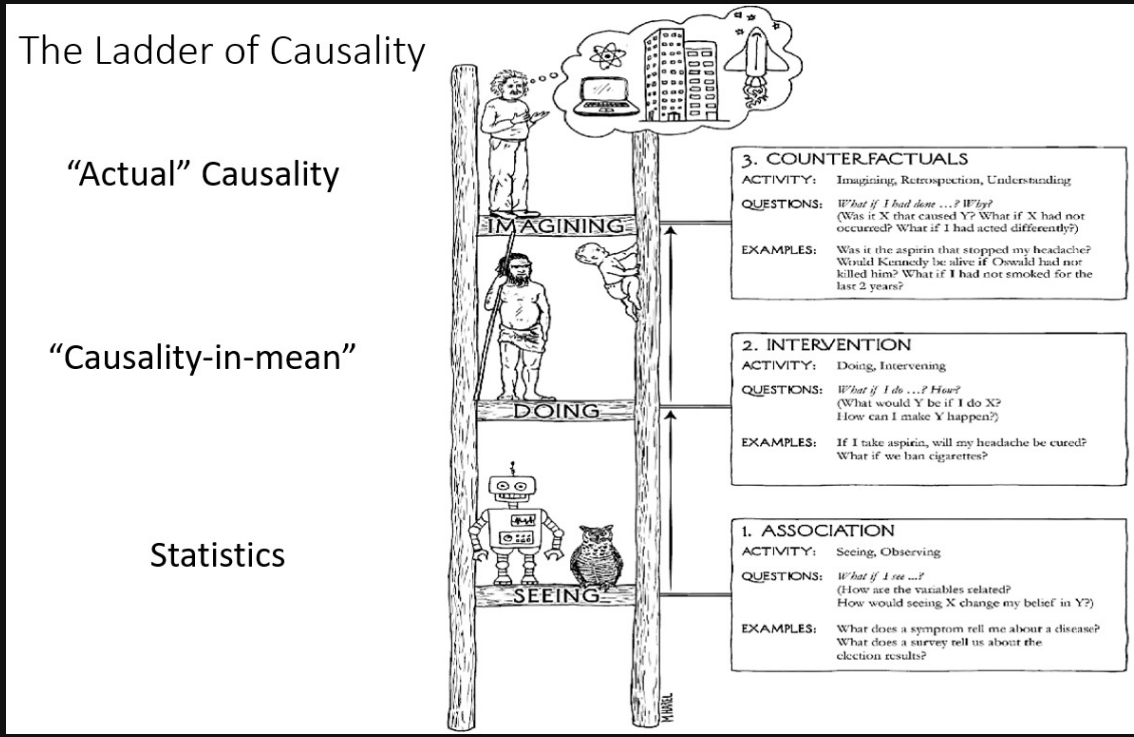
Pearl's Ladder of Causation

Layer 3: Counterfactual

- Definition: Individual-level hypothetical scenarios - "what if..." questions about specific individuals
- Mathematical Notation: P(Y_x | X', Y')
- Question Answered:

"What would have happened to this specific individual if circumstances had been different?"

- Examples:
  - If Zhang San had never smoked, would he still have gotten lung cancer?
  - If Li Ming had gone to college, what would his current income be?
  - If patient A had taken a different drug, would they have recovered?
- Key Feature: Individual-specific causal reasoning



Pearl's Ladder of Causation

Layer	Focus	Scope	Verifiability
Association	Observation	Statistical relationships	✅ Fully verifiable
Intervention	Population manipulation	Policy/treatment effects	✅ Experimentally verifiable
Counterfactual	Individual hypothesis	Specific scenario reasoning	❌ Fundamentally unverifiable

Outline

- Part 1: A historical tour

  How do pioneers understand causality from different perspectives?

- Part 2: Cognitive phenomena

  How people understand and perceive causation?

- Part 3: Computational models

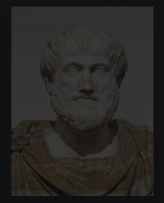
  How to describe the causal cognition in computation?

- Part 4: Modern approaches

  What are researchers doing nowadays?

Outline

Philosophy

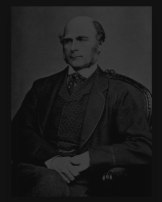


Aristotle (384–322 BC)
Four causes

Hume (1711-1776)
Regularity

Kant (1724-1804)
Necessity

Statistics & Economics & Epidemiology



Francis Galton (1877)
Regression analysis

Karl Pearson (1904)
Correlation

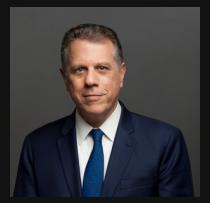
Sewall Wright (1921)
Path analysis

Donald Rubin (1980)
Rubin causal model

Judea Pearl (2000)
Pearl causal model

Psychology & Cognition

Albert Michotte (1963)
Causal perception



Michael Waldmann (1992)
Causal structure

New eras of causality research

Causal inference

Causal discovery

Causal learning

Causal reasoning

Causal judgment

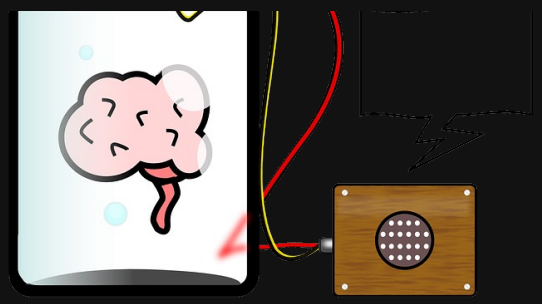
Causal reinforcement learning

Causal representation learning

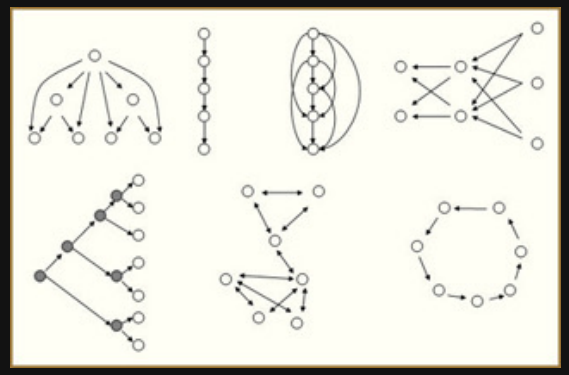
…

Why cognitive phenomena first?

Human as research subject



Cognitive Experiment
Description from the outer side



Computational modeling
Description from the inner side

What is the difference?

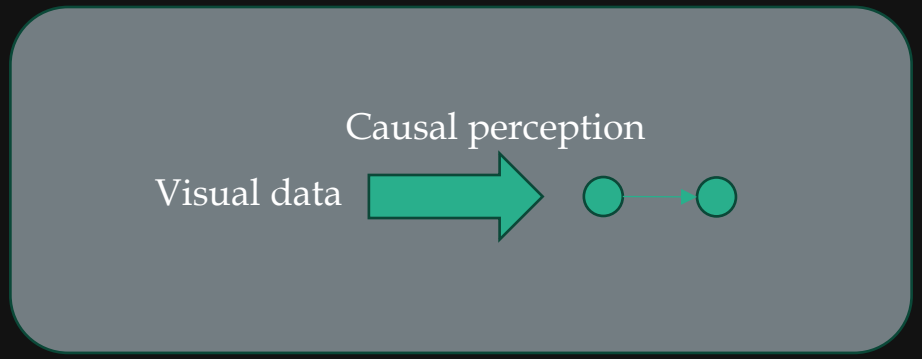
- Causality in cognitive science
  - "Causality" is a subjective cognitive phenomenon.
  - How do humans perceive it?

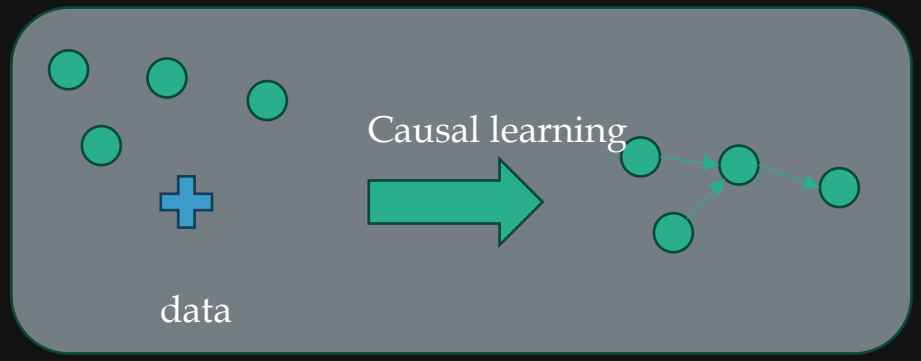
- Causality in statistics
  - "Causality" is objective within the real physical world.
  - How to teach AI/Machine/Agent discover it from data?

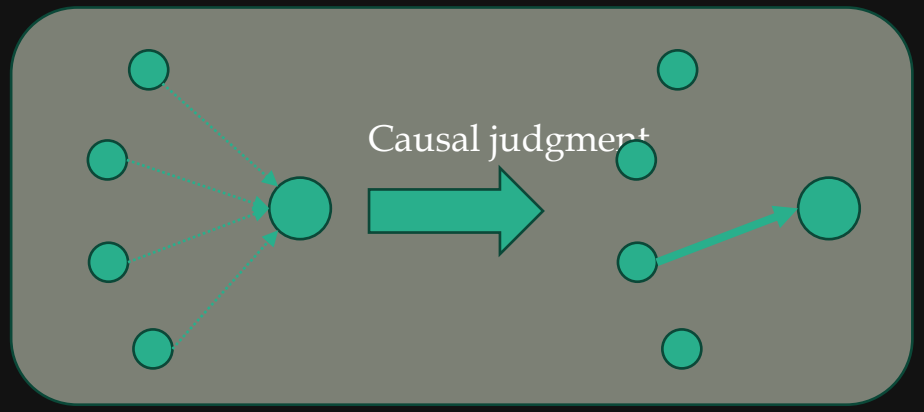
Some topics in cognitive science we select…

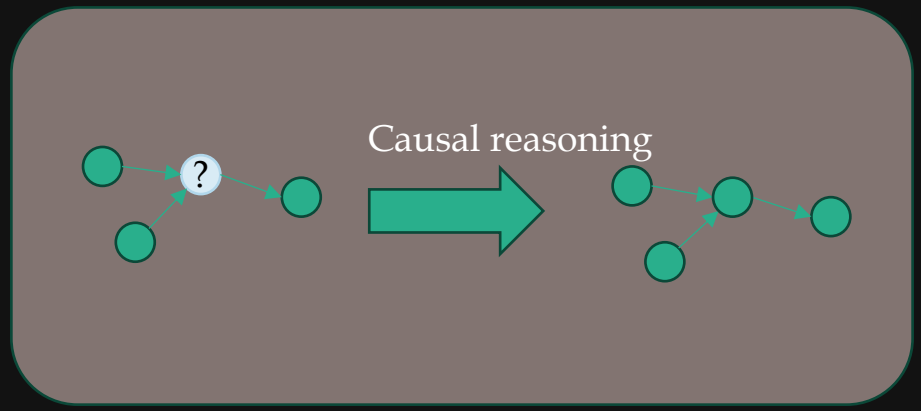
- Causal perception
- Causality reasoning
- Causal learning
- Causal judgment

Outline

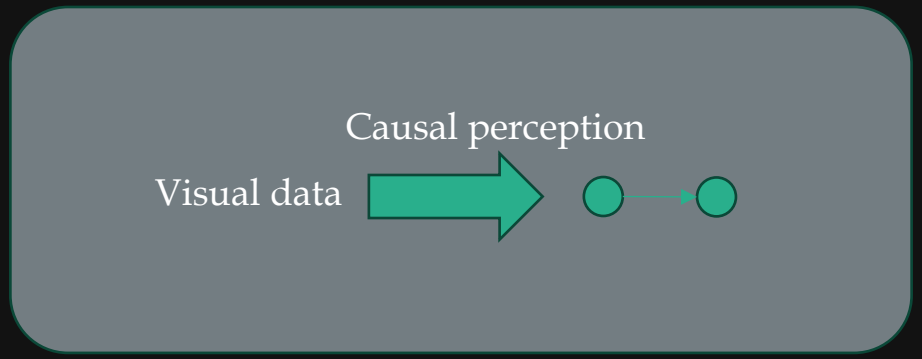

Causal perception
Visual data

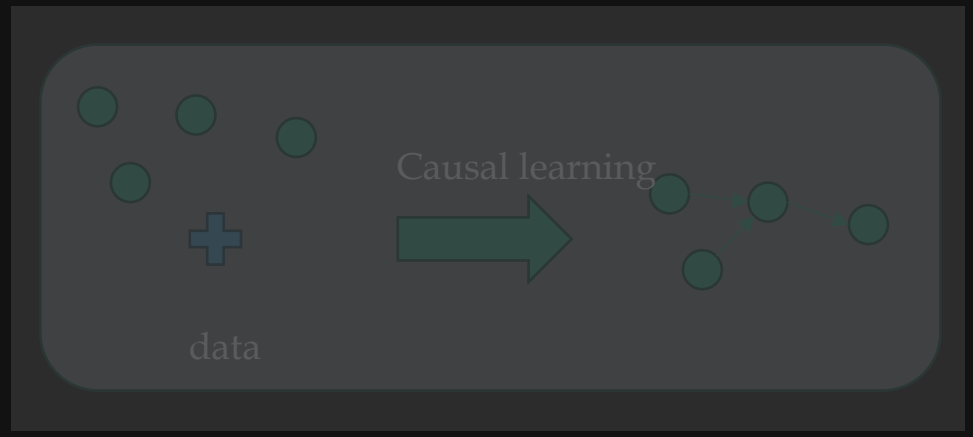

Causal learning
data

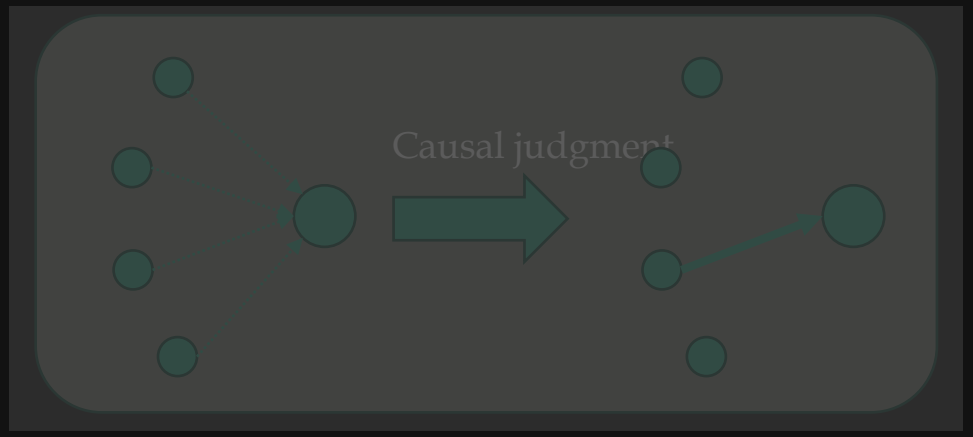

Causal judgment

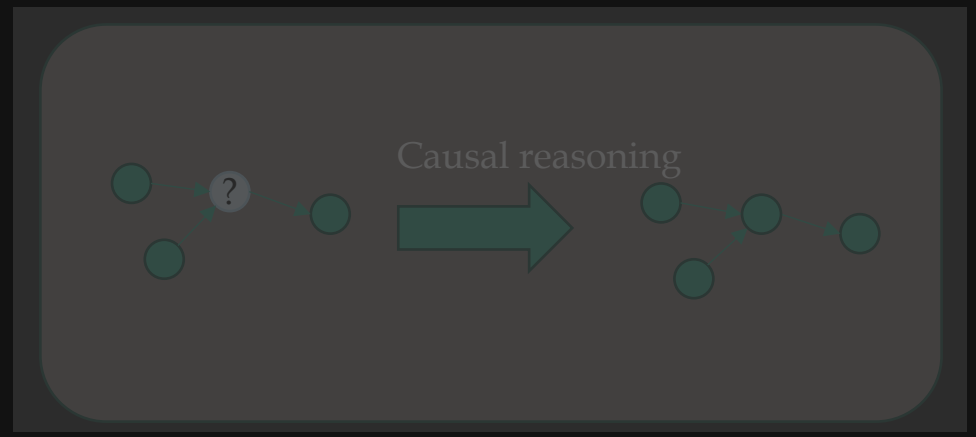

?
Causal reasoning

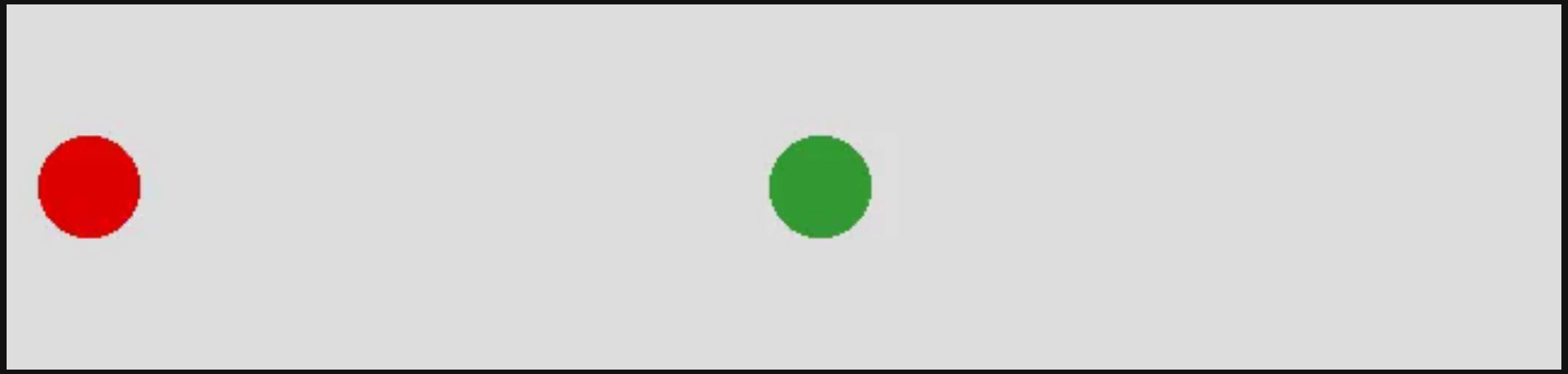
Outline

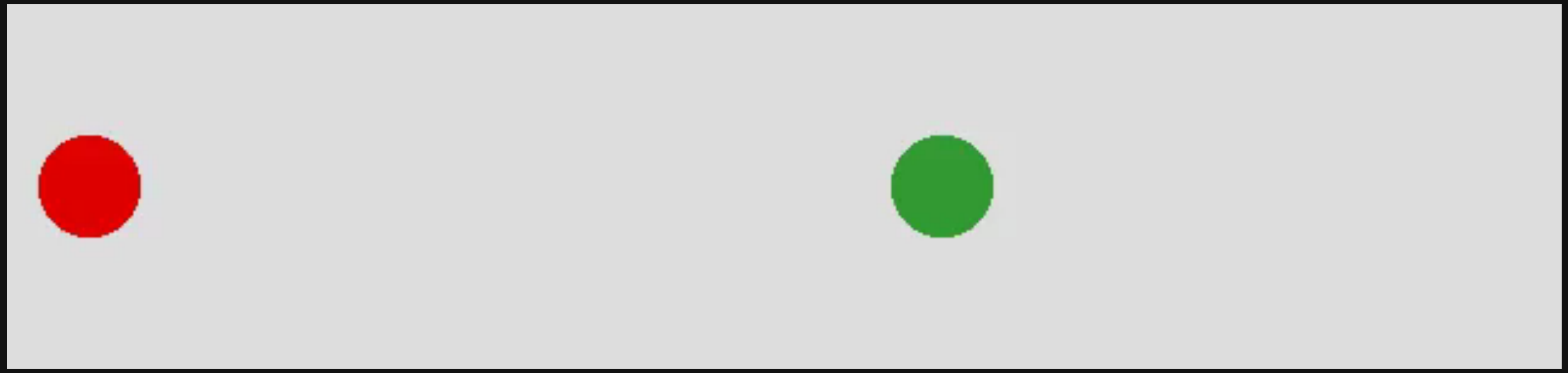

Causal perception
Visual data

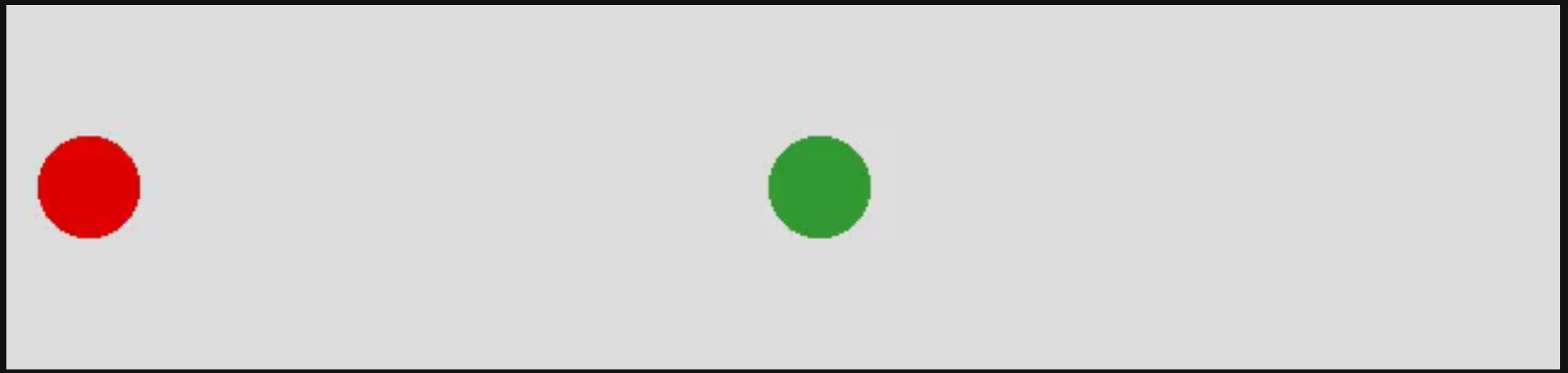

Causal learning
data

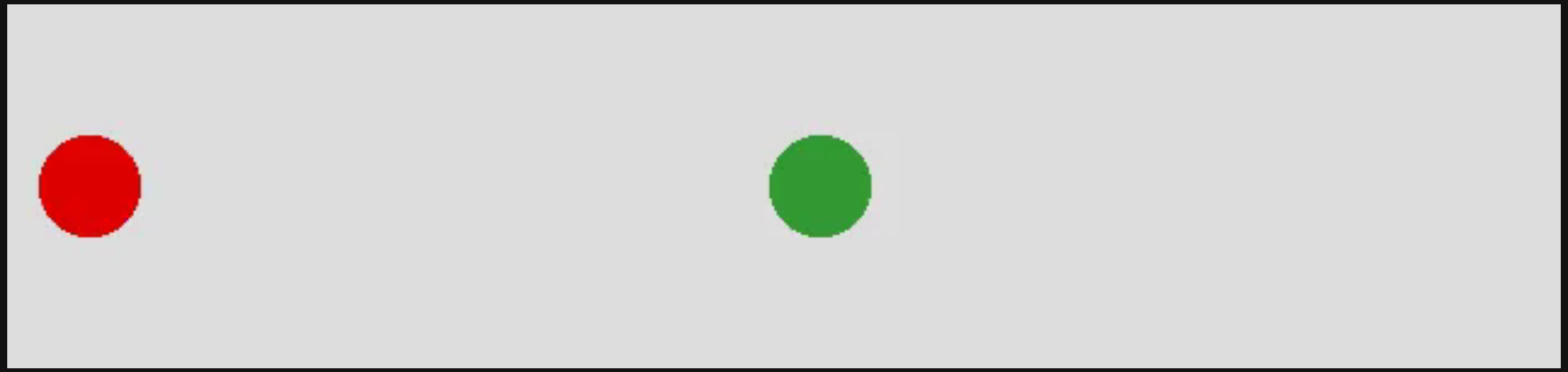

Causal judgment

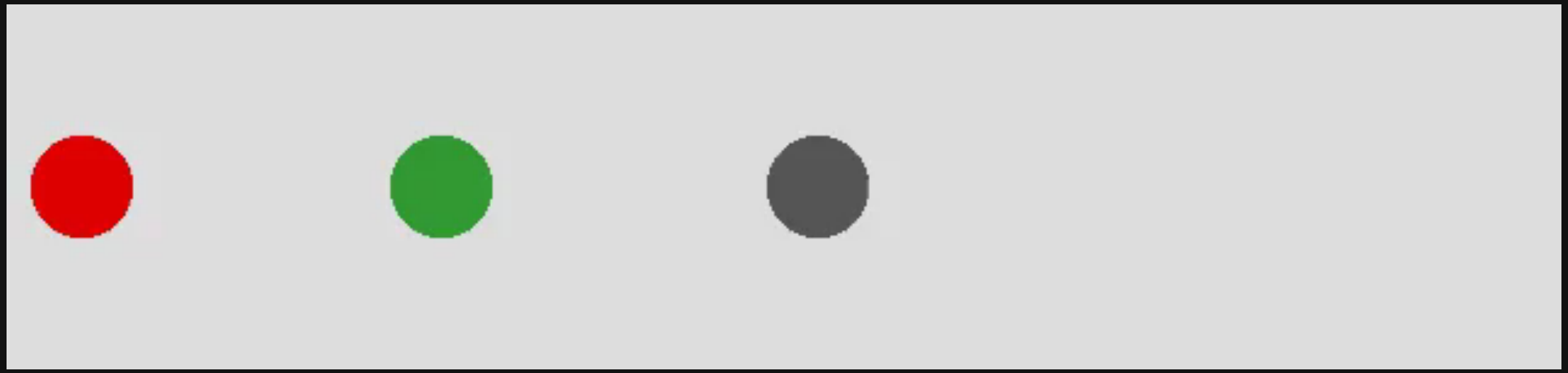

Causal reasoning
?

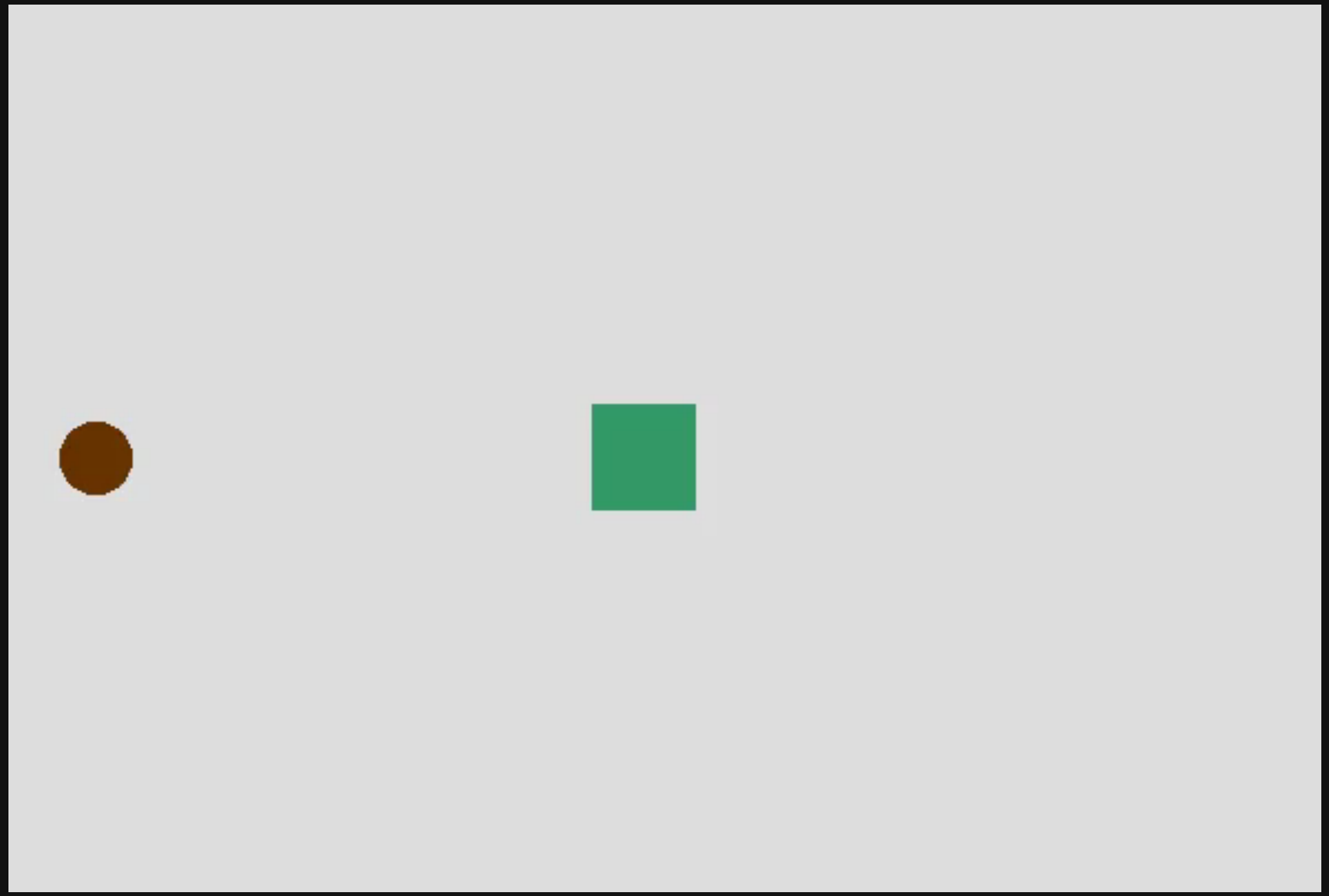












Perception or cognition?

- Do we directly perceive causality, or do we only perceive the cooccurrence, and infer the causality in the high-level cognition?

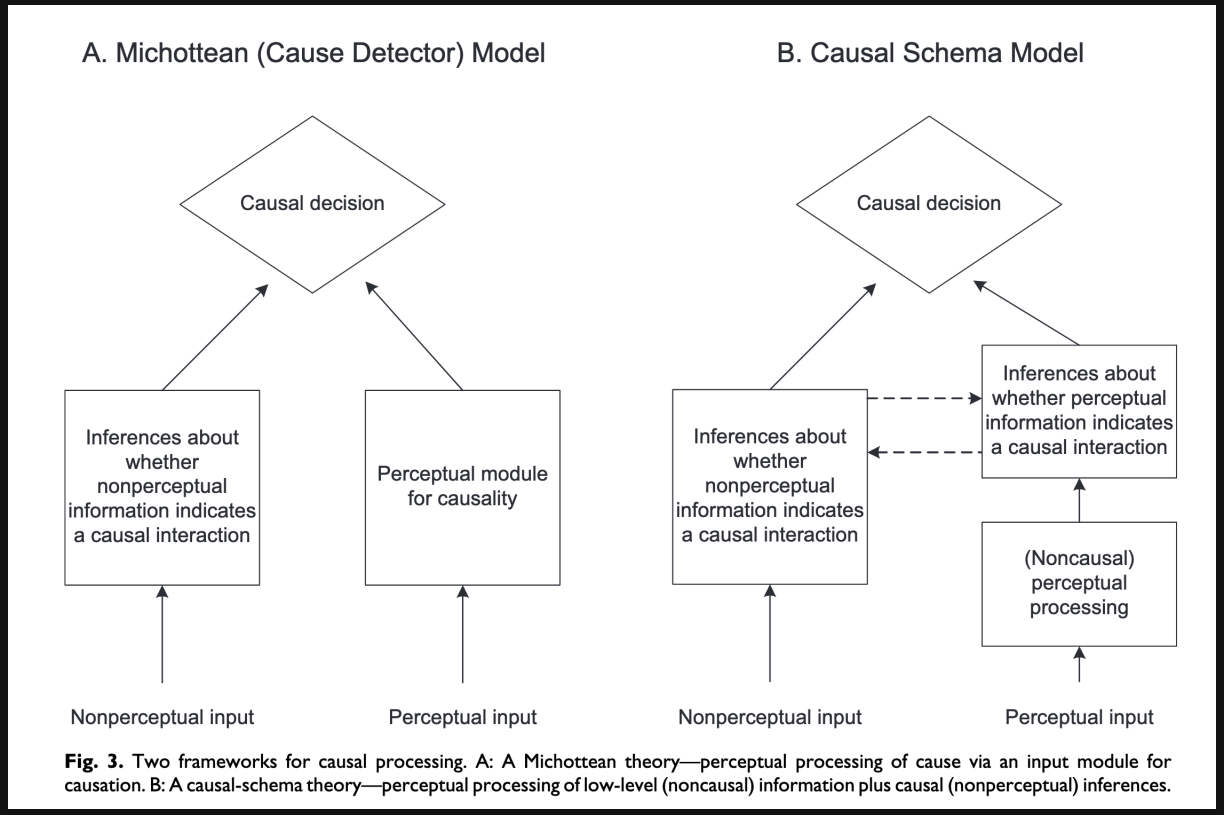
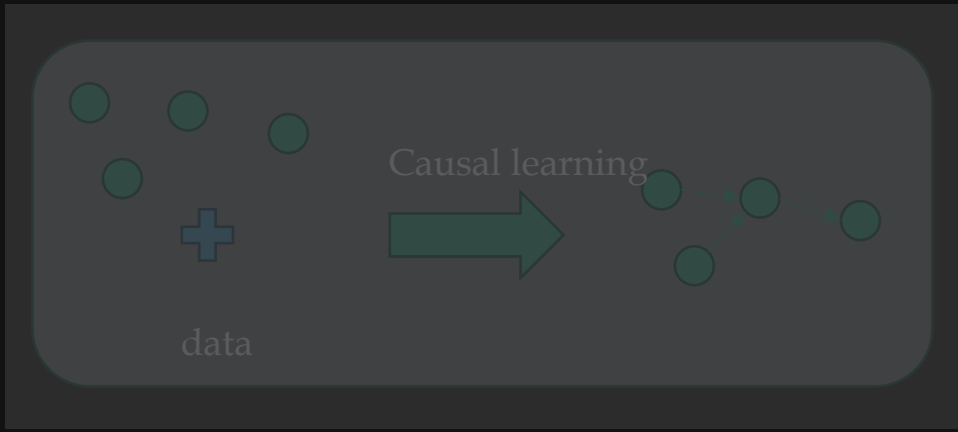
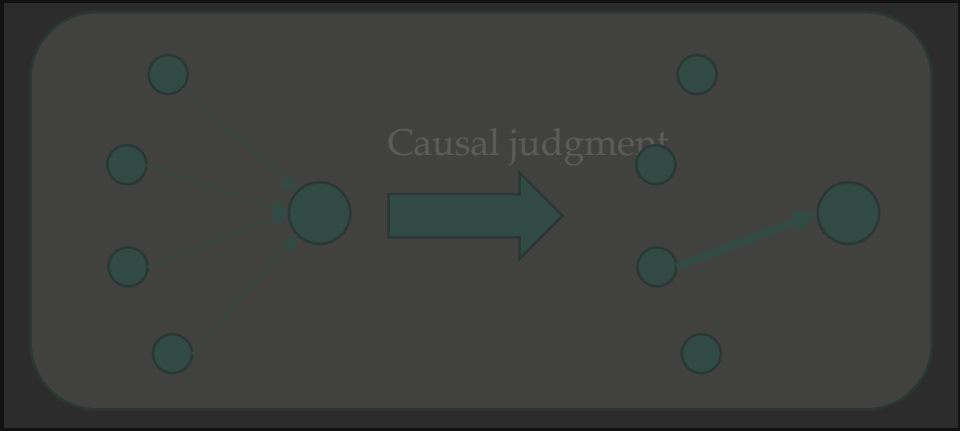


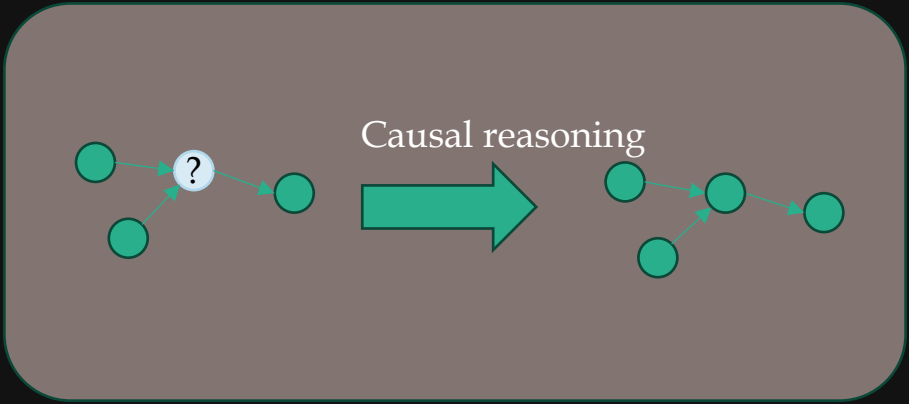
Fig. 3. Two frameworks for causal processing. A: A Michottean theory—perceptual processing of cause via an input module for causation. B: A causal-schema theory—perceptual processing of low-level (noncausal) information plus causal (nonperceptual) inferences.

Outline

Causal perception
Visual data


Causal learning
data


Causal judgment


?
Causal reasoning

REPORTS

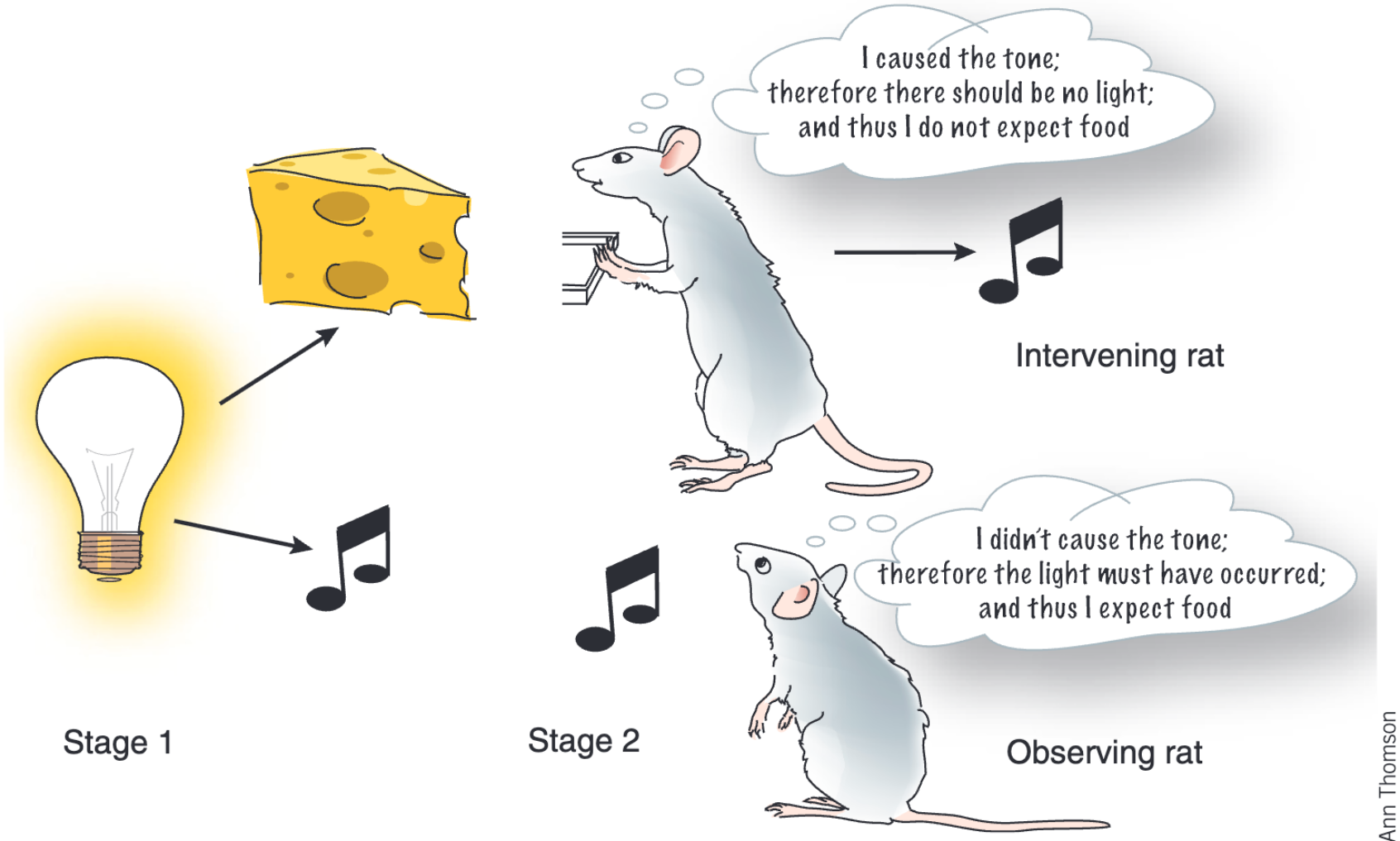
Causal Reasoning in Rats

AARON P. BLAISDELL, KOSUKE SAWA, KENNETH J. LEISING, AND MICHAEL R. WALDMANN Authors Info & Affiliations

SCIENCE • 17 Feb 2006 • Vol 311, Issue 5763 • pp. 1020-1022 • DOI: 10.1126/science.1121872

Abstract

Empirical research with nonhuman primates appears to support the view that causal reasoning is a key cognitive faculty that divides humans from animals. The claim is that animals approximate causal learning using associative processes. The present results cast doubt on that conclusion. Rats made causal inferences in a basic task that taps into core features of causal reasoning without requiring complex physical knowledge. They derived predictions of the outcomes of interventions after passive observational learning of different kinds of causal models. These competencies cannot be explained by current associative theories but are consistent with causal Bayes net theories.



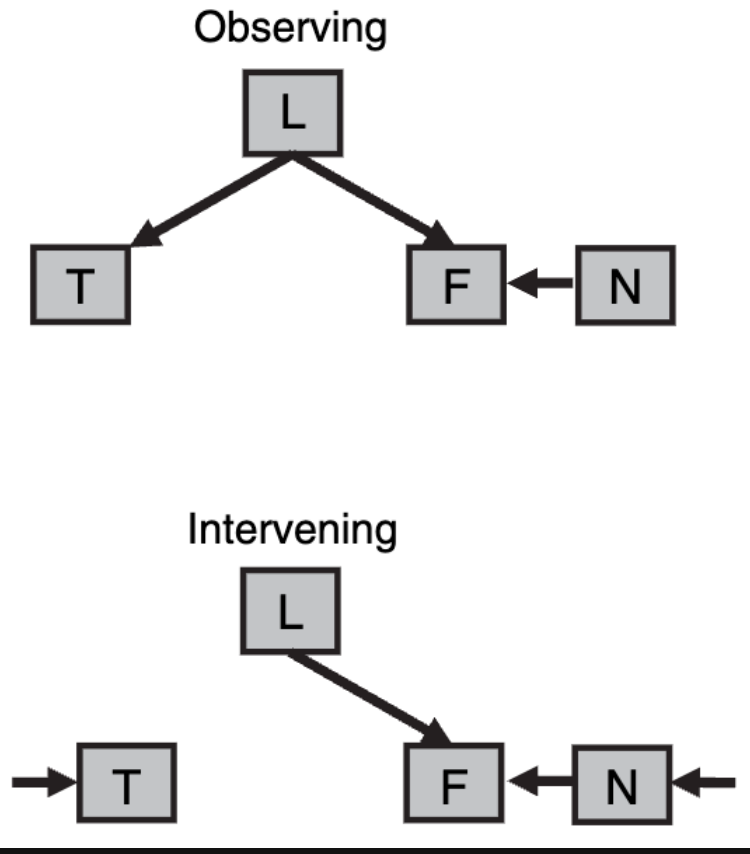
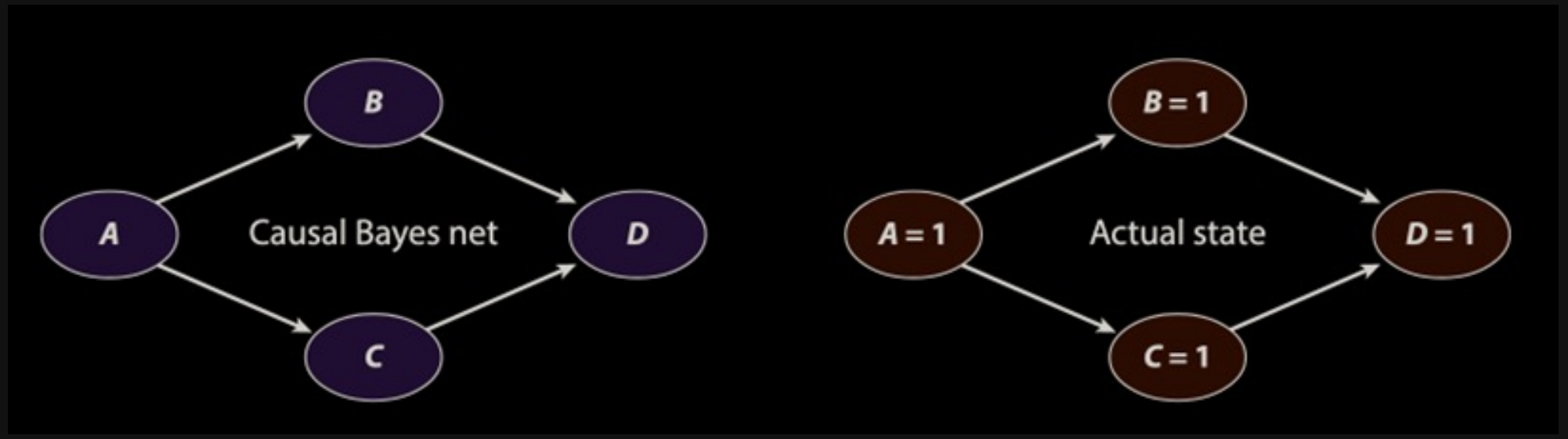


Figure 1 Design of the experiments of Blaisdell and colleagues. In the first stage (left), the rats learned that a light predicted a tone and the appearance of food. In the second stage (right), rats that heard the tone proceeded to search for food, while rats that pressed a lever to cause the tone did not search for food.

Example 1

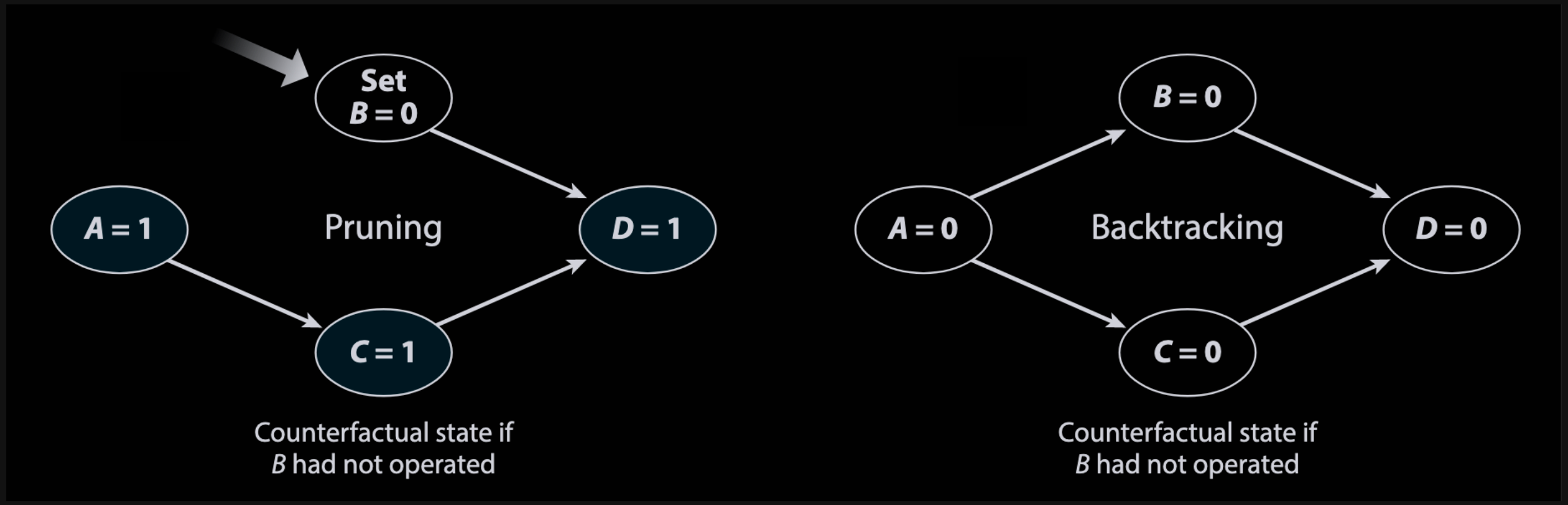
- Movement of Ball A causes Ball B to move. In other words, if A, then B. Both are moving.
  - Suppose Ball A were prevented from moving. Would component B still be moving?
    - No! Movement of B depends on movement of A
  - Suppose Ball B were prevented from moving. Would component A still be moving?
    - Yes! Movement of A doesn't depend on movement of B
  - Suppose A were observed to not be moving, will B move?
    - No!
  - Suppose B were observed to not be moving, will A move?
    - Unknown!
- Causality is tightly coupled with order of time and intervention

Example 2



- What would the value of D be, if we set B=0?

Example 2



Pruning Theory

Minimal Network Theory

Causal reasoning is difficult!

Example 3

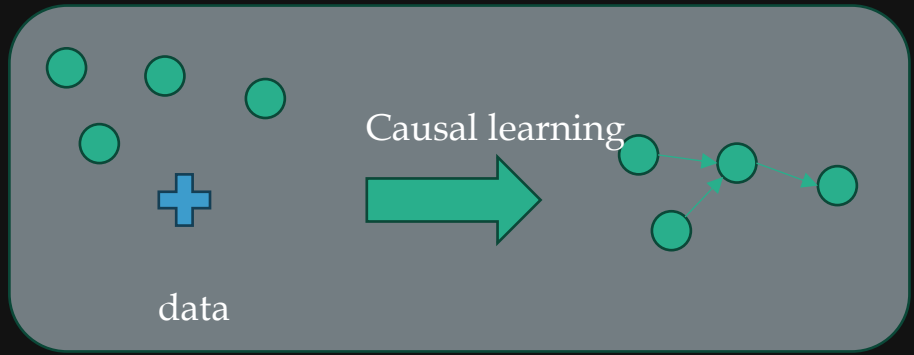
- Recent research has shown that of 100 men who help with chores, 82 are in good health, whereas only 32 of 100 men who do not help with chores are in good health. Imagine that a friend of yours is married and is concerned about his health.

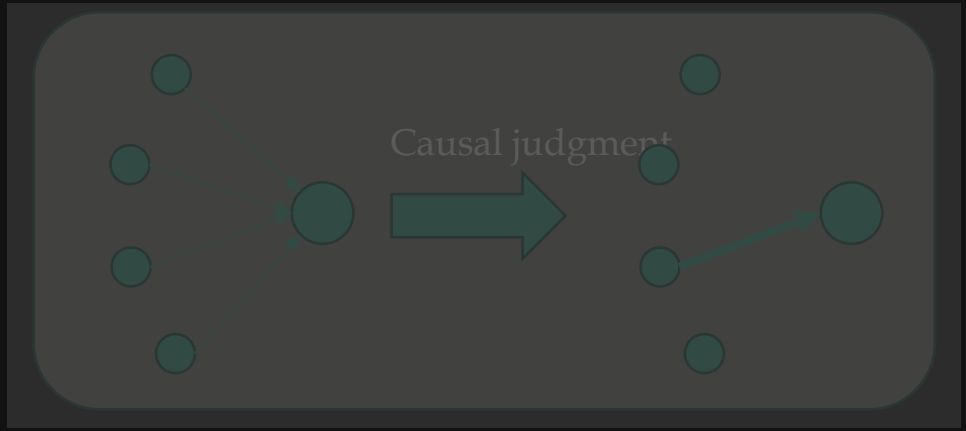
Should he start to do the chores or not?

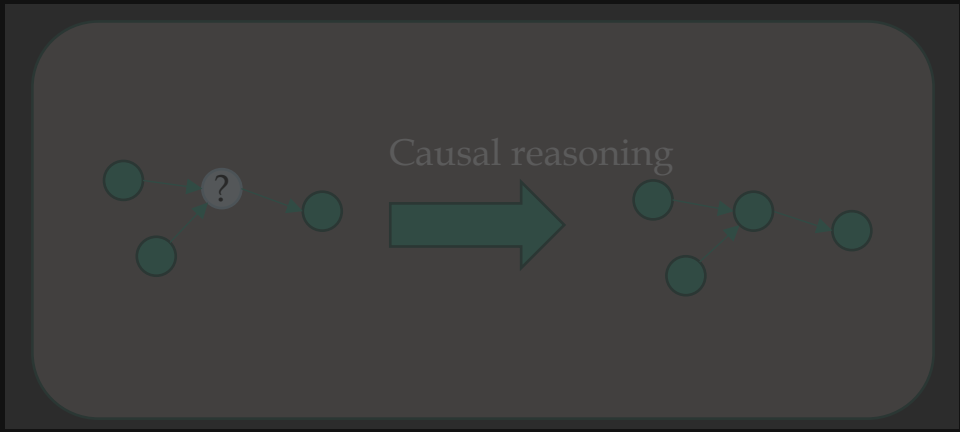
- Case 1: Doing the chores is an additional exercise every day. – He should
- Case 2: Men who are concerned about equality issues are also concerned about health issues and therefore both help to do the chores and eat healthier food. -> He shouldn’t

Outline

Causal perception
Visual data


Causal learning
data


Causal judgment


Causal reasoning
?

Review Article | Published: 26 April 2024

The development of human causal learning and reasoning

Mariel K. Goddu & Alison Gopnik

Nature Reviews Psychology **3**, 319–339 (2024) | Cite this article

4649 Accesses | **23** Citations | **238** Altmetric | Metrics

Abstract

Causal understanding is a defining characteristic of human cognition. Like many animals, human children learn to control their bodily movements and act effectively in the environment. Like a smaller subset of animals, children intervene: they learn to change the environment in targeted ways. Unlike other animals, children grow into adults with the causal reasoning skills to develop abstract theories, invent sophisticated technologies and imagine alternate pasts, distant futures and fictional worlds. In this Review, we explore the development of human-unique causal learning and reasoning from evolutionary and ontogenetic perspectives. We frame our discussion using an 'interventionist' approach. First, we situate causal understanding in relation to cognitive abilities shared with non-human animals. We argue that human causal understanding is distinguished by its depersonalized (objective) and decontextualized (general) representations. Using this framework, we next review empirical findings on early human causal learning and reasoning and consider the naturalistic contexts that support its development. Then we explore connections to related abilities. We conclude with suggestions for ongoing collaboration between developmental, cross-cultural, computational, neural and evolutionary approaches to causal understanding.

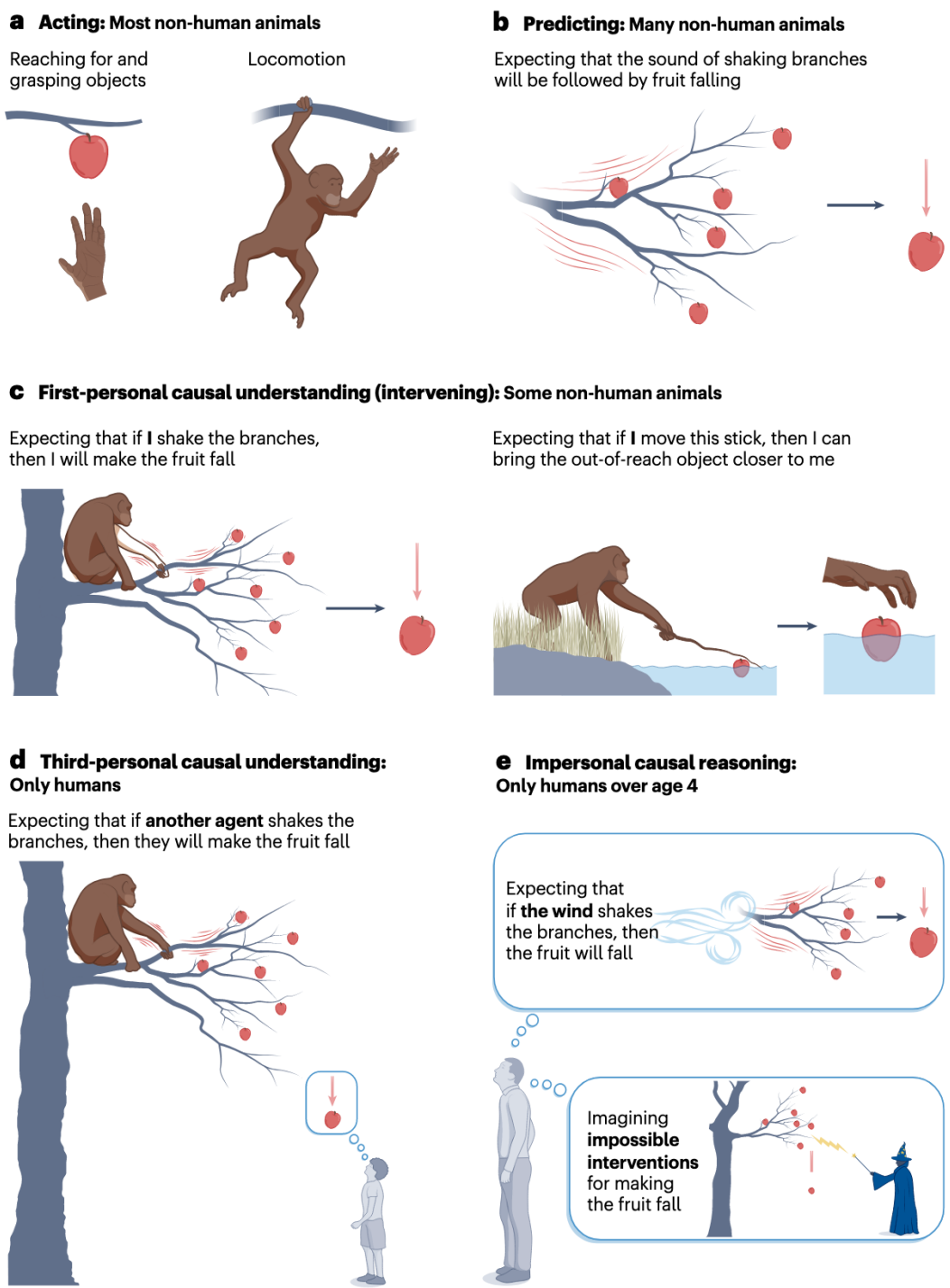
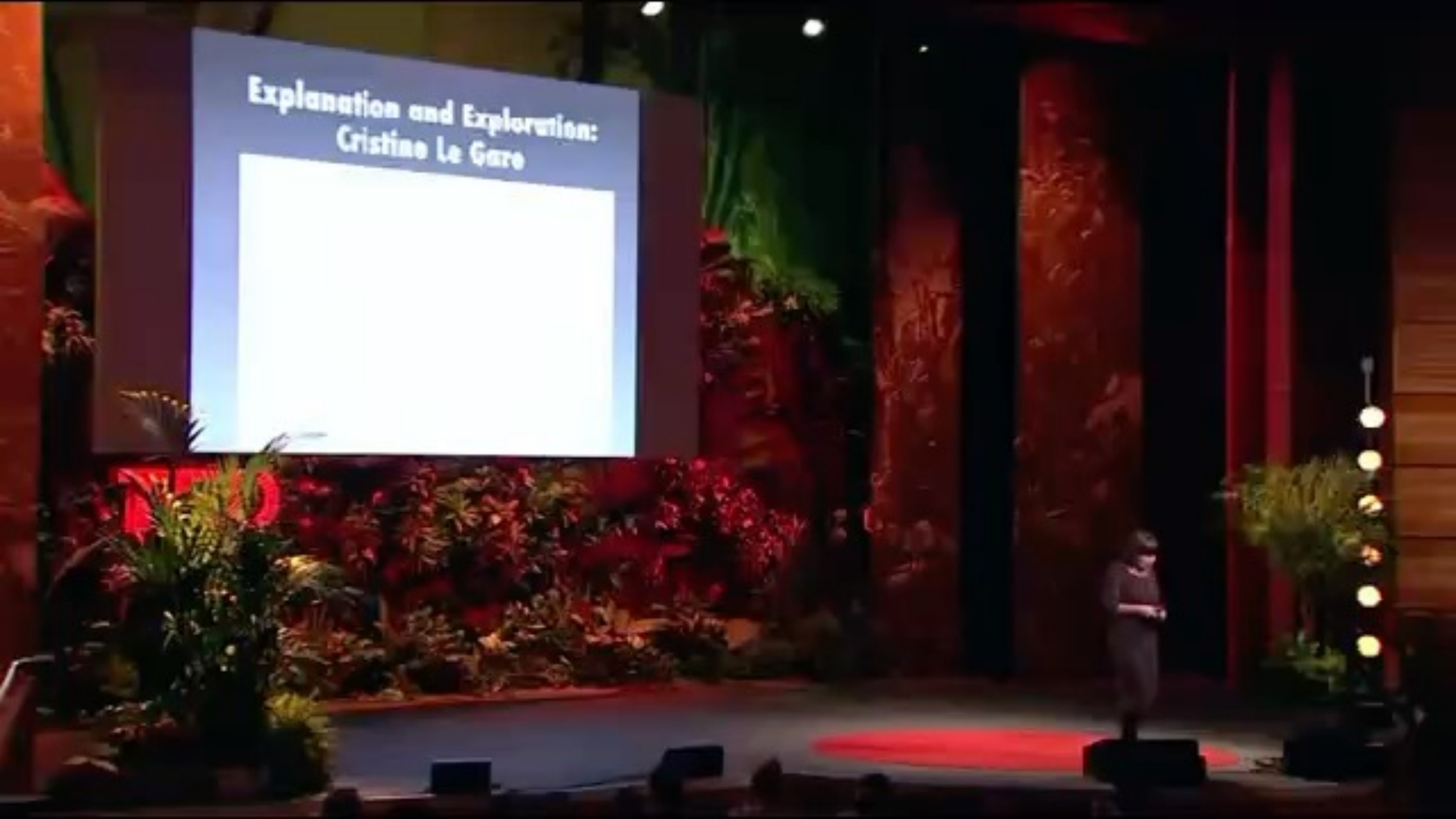


Fig. 1 | Causal reasoning and antecedent abilities in human and non-human animals. a, Acting effectively involves applying knowledge of the differences that one's bodily movements make to one's perceptions and goals. These skills involve interacting with the environment, but not manipulating it to generate an external effect. **b**, Predicting contingencies between external events, such as expecting a positive outcome after hearing a sound, does not involve causal reasoning. **c**, Predicting external events resulting from one's own interventions, such as learning to manipulate one object to change the position of another object, is first-personal causal understanding. **d**, Predicting external events resulting from others' interventions is third-personal causal understanding. **e**, Imagining specific causal events that are independent of agents' actions, and might even be impossible, is impersonal causal understanding. Adult human causal understanding is depersonalized and decontextualized (involves general representations of causal variables). The theoretical distinctions and examples in this figure are inspired by previous accounts[4,9,20,23,41].


Explanation and Exploration:
Cristina Le Gare

Psychological Review
2004, Vol. 111, No. 1, 000–000

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0033-295X/04/$12.00 DOI: 10.1037/0033-295X.111.1.000

A Theory of Causal Learning in Children: Causal Maps and Bayes Nets

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Institute for Human Machine Cognition

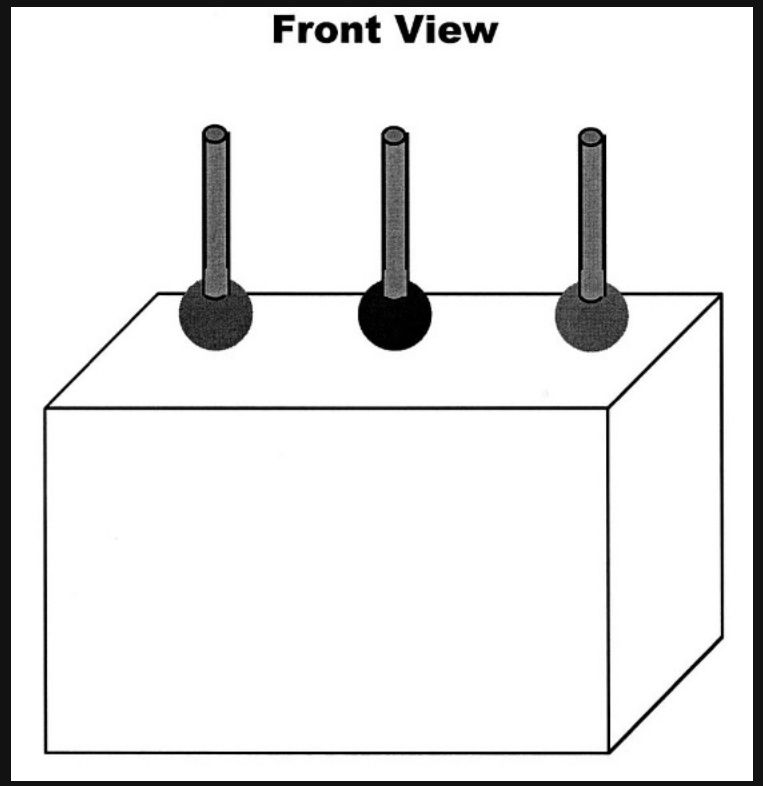
David M. Sobel
Brown University

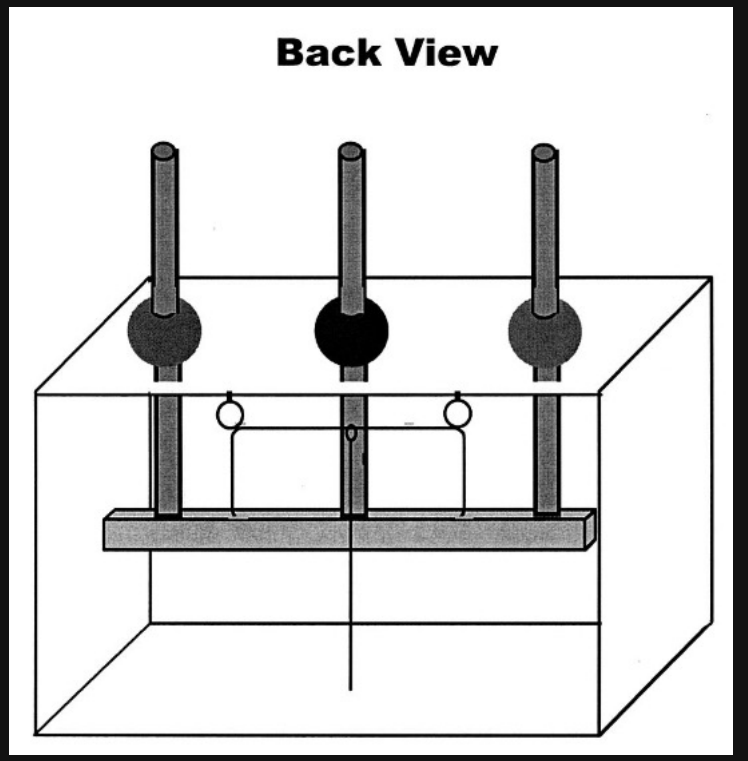
Laura E. Schulz and Tamar Kushnir
University of California, Berkeley

David Danks
Carnegie Mellon University and Institute for Human Machine Cognition

The authors outline a cognitive and computational account of causal learning in children. They propose that children use specialized cognitive systems that allow them to recover an accurate "causal map" of the world: an abstract, coherent, learned representation of the causal relations among events. This kind of knowledge can be perspicuously understood in terms of the formalism of directed graphical causal models, or Bayes nets. Children's causal learning and inference may involve computations similar to those for learning causal Bayes nets and for predicting with them. Experimental results suggest that 2- to 4-year-old children construct new causal maps and that their learning is consistent with the Bayes net formalism.

The puppet machine



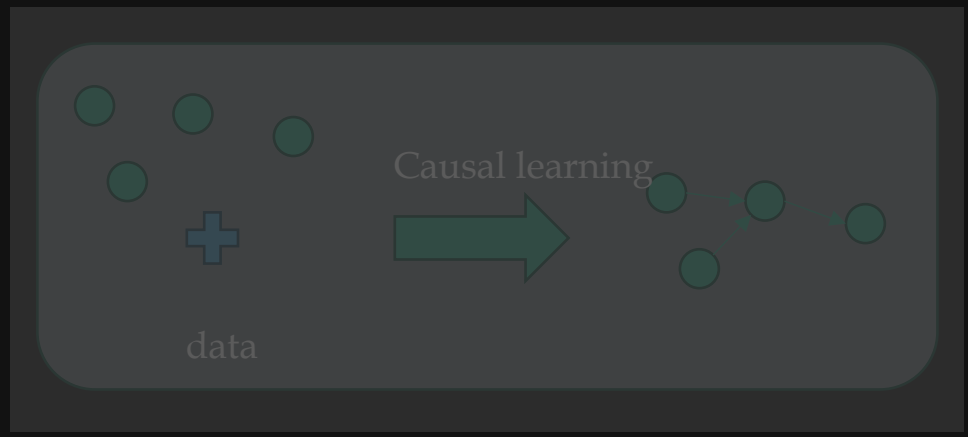


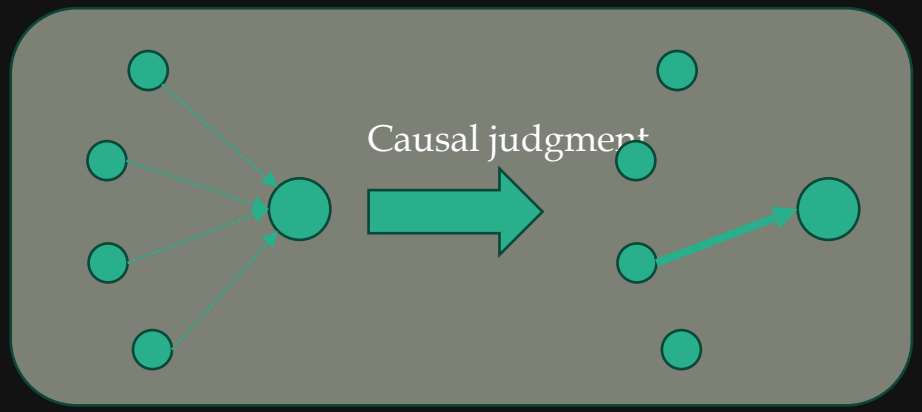
OpenLock

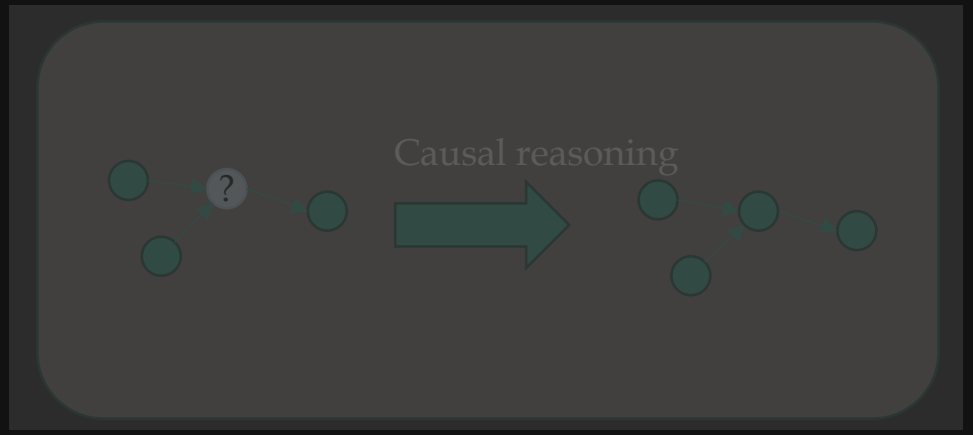
Human execution: Common Cause 3

Outline

Causal perception
Visual data


Causal learning
data


Causal judgment


Causal reasoning
?

Causal judgment

Example

- Both Suzy and Billy throw a rock at the bottle. They hit the bottle at the same time, and it breaks.
- Both Suzy and Billy throw a rock at the bottle. Suzy's rock hits the bottle first, and the bottle breaks. Billy's throw misses but would have broken the bottle if Suzy's throw had not.

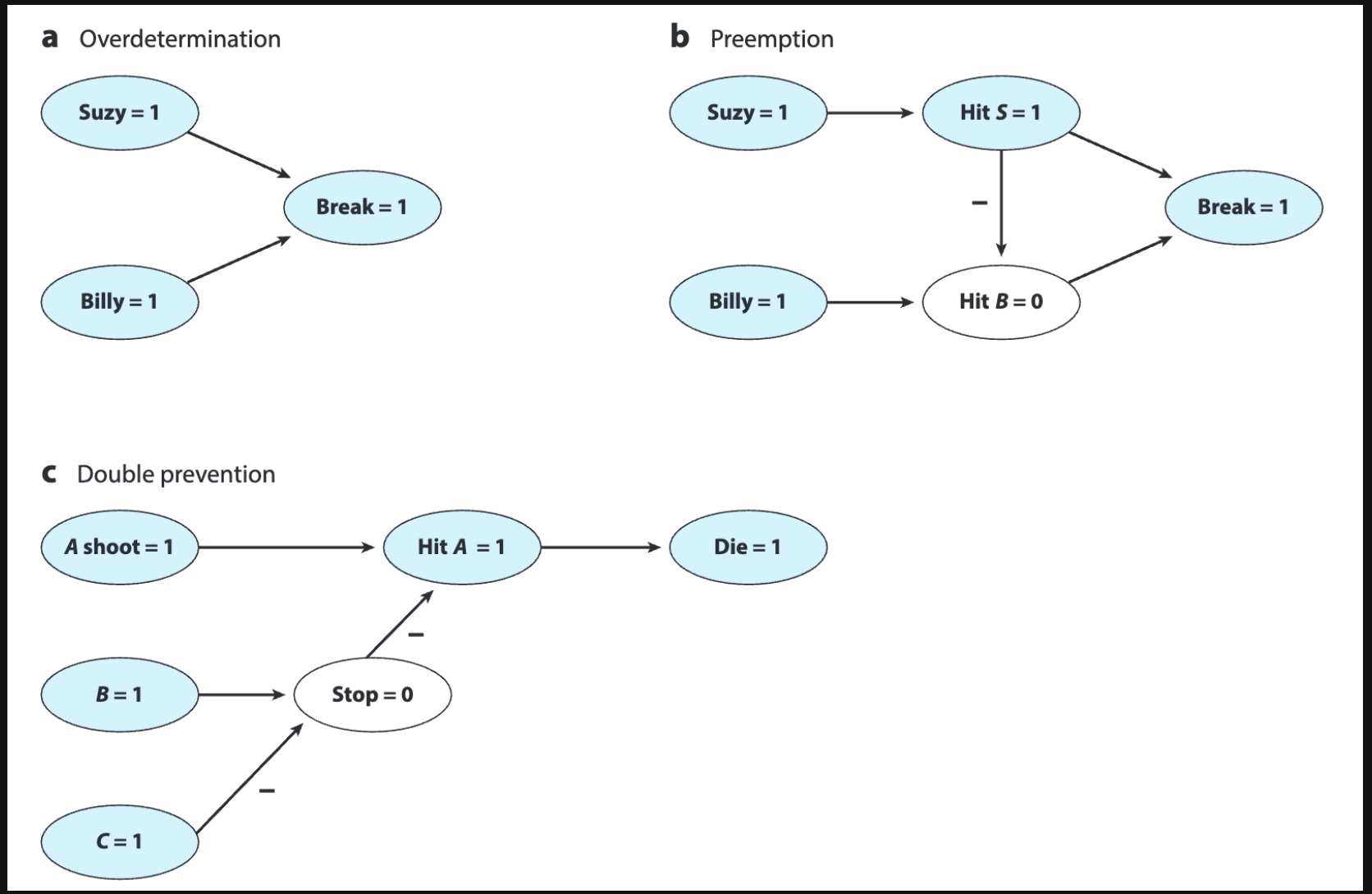
What is the cause of the breaking bottle?

How to define this causation?

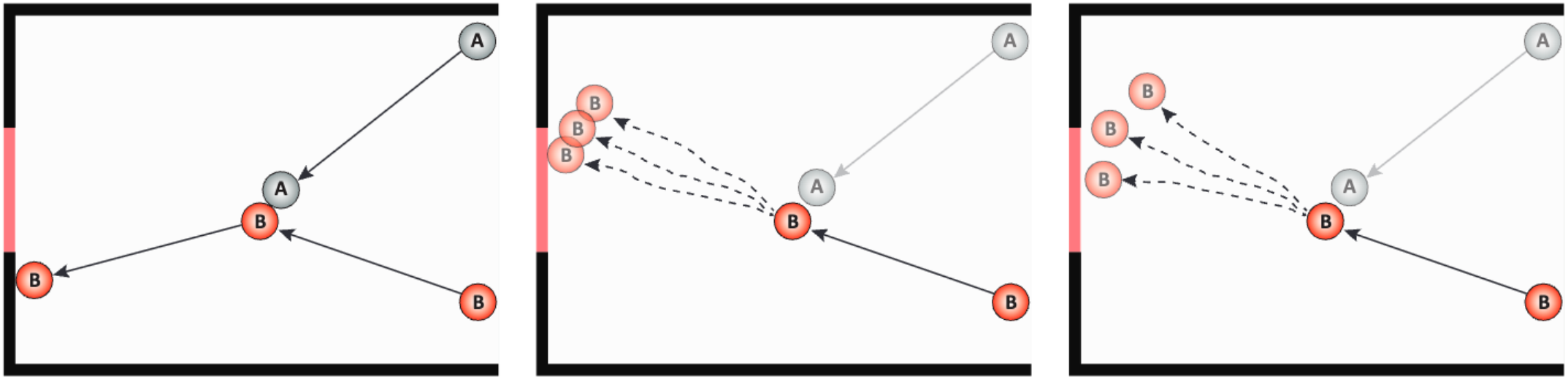
Causal theory

- Process theory: causality is viewed as a “chain of causation”, in which some conserved quantities such as force or information are transferred.
- Failure case of process theory
  - Consider a prankster who pulls away Granny’s chair just before she sits down. Granny falls painfully. What caused the fall? Intuitively it was the prankster’s action, even though no active process linked this action to the fall.
- Counterfactual theory: causality is defined as “difference making”, in which the causality is determined by a comparison between what actually happened and what (counterfactually) could have happened.

Causal theory



Sloman, S. A. & Lagnado, D. Causality in thought. *Annu. Rev. Psychol.* **66**, 223–247 (2015)

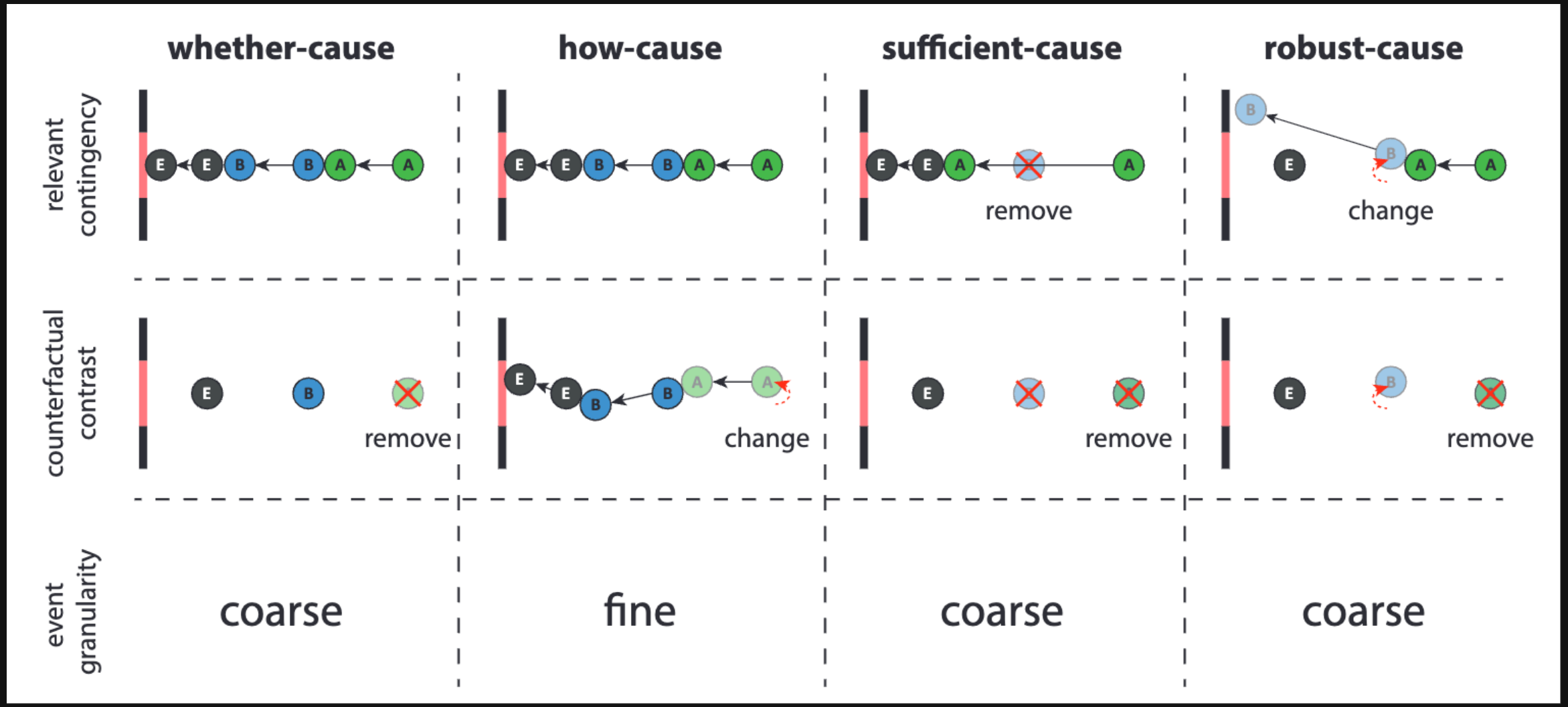


(a) actual situation

(b) counterfactual simulations with small degree of noise

(c) counterfactual simulations with large degree of noise

Figure 1. a) Illustration of what actually happened. b) and c) Illustrations of approximate simulations of where ball B would have ended up if ball A had been removed from the scene with small and large degrees of noise in the counterfactual simulations.


whether-cause
how-cause
sufficient-cause
robust-cause
relevant contingency
counterfactual contrast
event granularity
E
B
A
remove
change
coarse
fine
coarse
coarse

Outline

- Part 1: A historical tour

  How do pioneers understand causality from different perspectives?

- Part 2: Cognitive phenomena

  How people understand and perceive causation?

- Part 3: Computational models

  How to describe the causal cognition in computation?

- Part 4: Modern approaches

  What are researchers doing nowadays?

Computational models

- Statistical models
  - Rubin causal model
  - Pearl causal model
  - Granger causality
- Cognitive science
  - Associative account
  - Bayesian account
  - Darwinism account

Donald B Rubin

FOLLOW

YCMS Tsinghua; Senior Research Fellow at Fox / Temple U, Emeritus at Harvard U
Verified email at temple.edu

Statistics Economics Psychology Biostatistics Epidemiology

TITLE	CITED BY	YEAR
Maximum Likelihood from Incomplete Data Via the *EM* Algorithm AP Dempster, NM Laird, DB Rubin Journal of the royal statistical society: series B (methodological) 39 (1), 1-22	77304	1977
The central role of the propensity score in observational studies for causal effects PR Rosenbaum, DB Rubin Biometrika 70 (1), 41-55	43172	1983
Bayesian data analysis A Gelman, J Carlin, H Stern, DB Rubin CRC press	42945 *	2004

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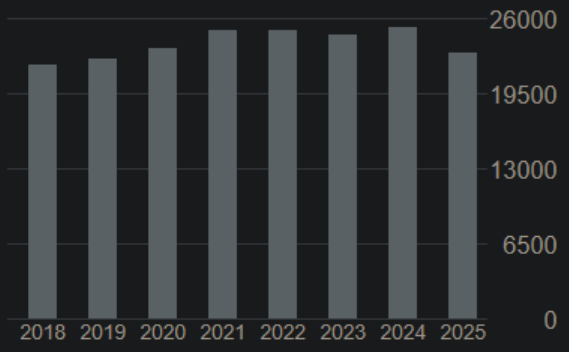
h-index	146	91
i10-index	405	275



What are potential outcomes?

The fundamental problem of causal inference

Getting around the fundamental problem of causal inference

A complete example with estimation

RMIT
UNIVERSITY

3ie
International Initiative
for Impact Evaluation

BetterEvaluation

unicef

Judea Pearl

FOLLOWING

Professor of Computer Science, University of California, Los Angeles
Verified email at cs.ucla.edu - Homepage

causality artificial intelligence structural equations counterfactuals

TITLE	CITED BY	YEAR
Probabilistic reasoning in intelligent systems: Networks of plausible inference J Pearl Synthese-Dordrecht 104 (1), 161	34999 *	1995
Causality J Pearl Cambridge university press	30854	2009
Heuristics: intelligent search strategies for computer problem solving J Pearl Addison-Wesley Longman Publishing Co., Inc.	5516	1984

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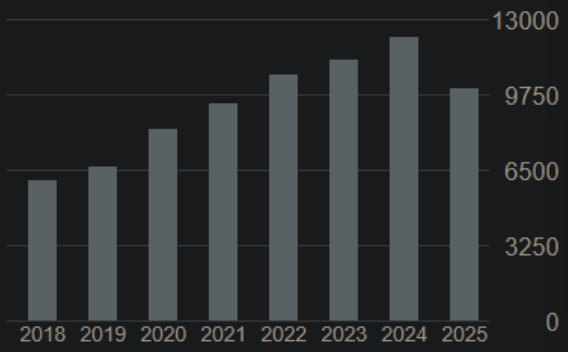
h-index	127	81
i10-index	388	235

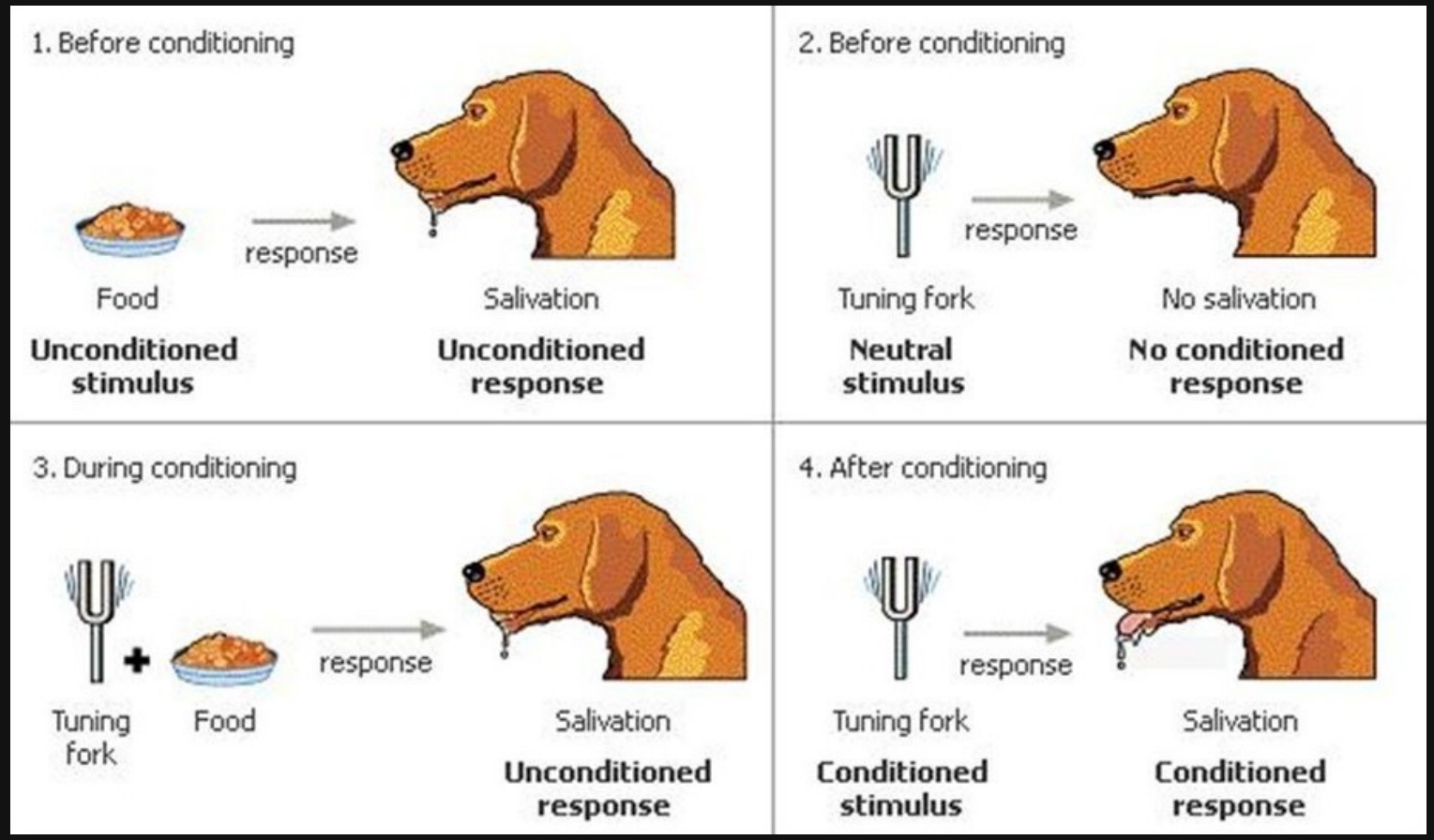


Pearl's *do*-calculus

Will allow us to identify any causal quantity that is identifiable

Associative Account of Causality

- Main argument: human causal learning can be reduced into the association links between cues and effects.



Bayesian Account

- Causal power theory: learning the causal strength
- Causal support theory: learning the causal structure
- Theory-based causal induction: learning the causal theory
- Theory-based causal transfer: active learning the dual causal theories
- Active causal learning: active learning with constrained computational resources

Causal Power Theory

- The classic "statistical" measure of the strength of a contingency:

$$\Delta P = P(e^+ \mid c^+) - P(e^+ \mid c^-).$$

- However, in some situations in which multiple causal factors co-occur, this ΔP rule fails to explain human and animal causal judgement.

Psychological Review
1997, Vol. 104, No. 2, 367–405

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0033-295X/97/$3.00

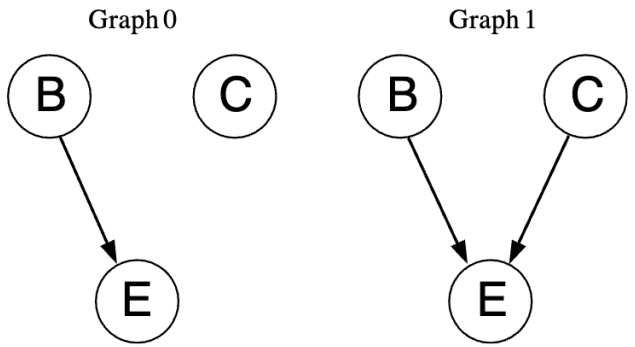
$$q_G = \frac{\Delta P}{1 - P(e^+ \mid c^-)}$$

From Covariation to Causation: A Causal Power Theory

Patricia W. Cheng
University of California, Los Angeles

Causal support theory

- Causal support: the evidence that data D provide in favor of Graph 1 over Graph 0



$$\log \frac{P(\text{Graph } 1|D)}{P(\text{Graph } 0|D)} = \log \frac{P(D|\text{Graph } 1)}{P(D|\text{Graph } 0)} + \log \frac{P(\text{Graph } 1)}{P(\text{Graph } 0)},$$

$$\text{support} = \log \frac{P(D \mid Graph1)}{P(D \mid Graph0)}.$$

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Cognitive Psychology 51 (2005) 334–384

Cognitive Psychology

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Structure and strength in causal induction ☆

Thomas L. Griffiths [a,*], Joshua B. Tenenbaum [b]

[a] *Department of Psychology, Stanford University, USA*
[b] *Department of Brain and Cognitive Sciences, Massachusetts Institute of Technology, 77 Massachusetts Avenue, Cambridge, MA 02139-4307, USA*

Theory-based causal induction

Theory-based causal induction: Bayesian model to integrate prior knowledge and statistical relations.

Causal theory

↓

Causal structure (graphical model)

↓

Data

Ontology:

Types	Number	Predicates	Values	
Chemical	$N_C \sim P_C$	**Injected**(**Chemical**, **Mouse**)	**Boolean:**	$\{\mathbf{T}, \mathbf{F}\}$
Gene	$N_G \sim P_G$	**Expressed**(**Gene**, **Mouse**)	**Boolean:**	$\{\mathbf{T}, \mathbf{F}\}$
Mouse	$N_M \sim P_M$			

Plausible relations:

Injected$(\mathbf{C}, \mathbf{M}) \rightarrow$ **Expressed**$(\mathbf{G}, \mathbf{M})$

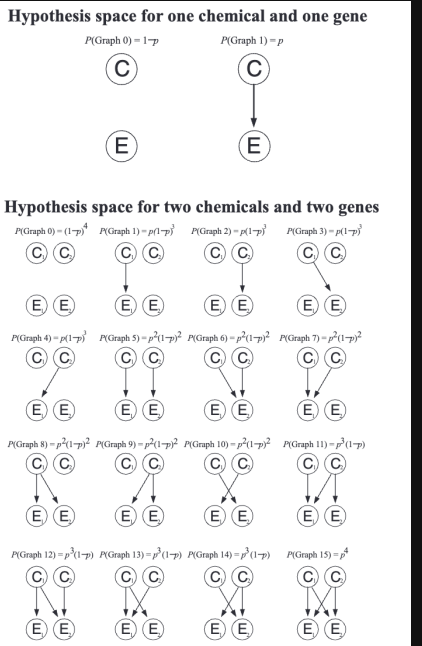
True for all $\mathbf{M}$ with probability p for each $\mathbf{C}$, $\mathbf{G}$ pair

Functional form:

Injected$(\mathbf{C}, \mathbf{M}) \sim$ Bernoulli$(\cdot)$

Expressed$(\mathbf{G}, \mathbf{M}) \sim$ Bernoulli(ν) for ν from a noisy-OR:

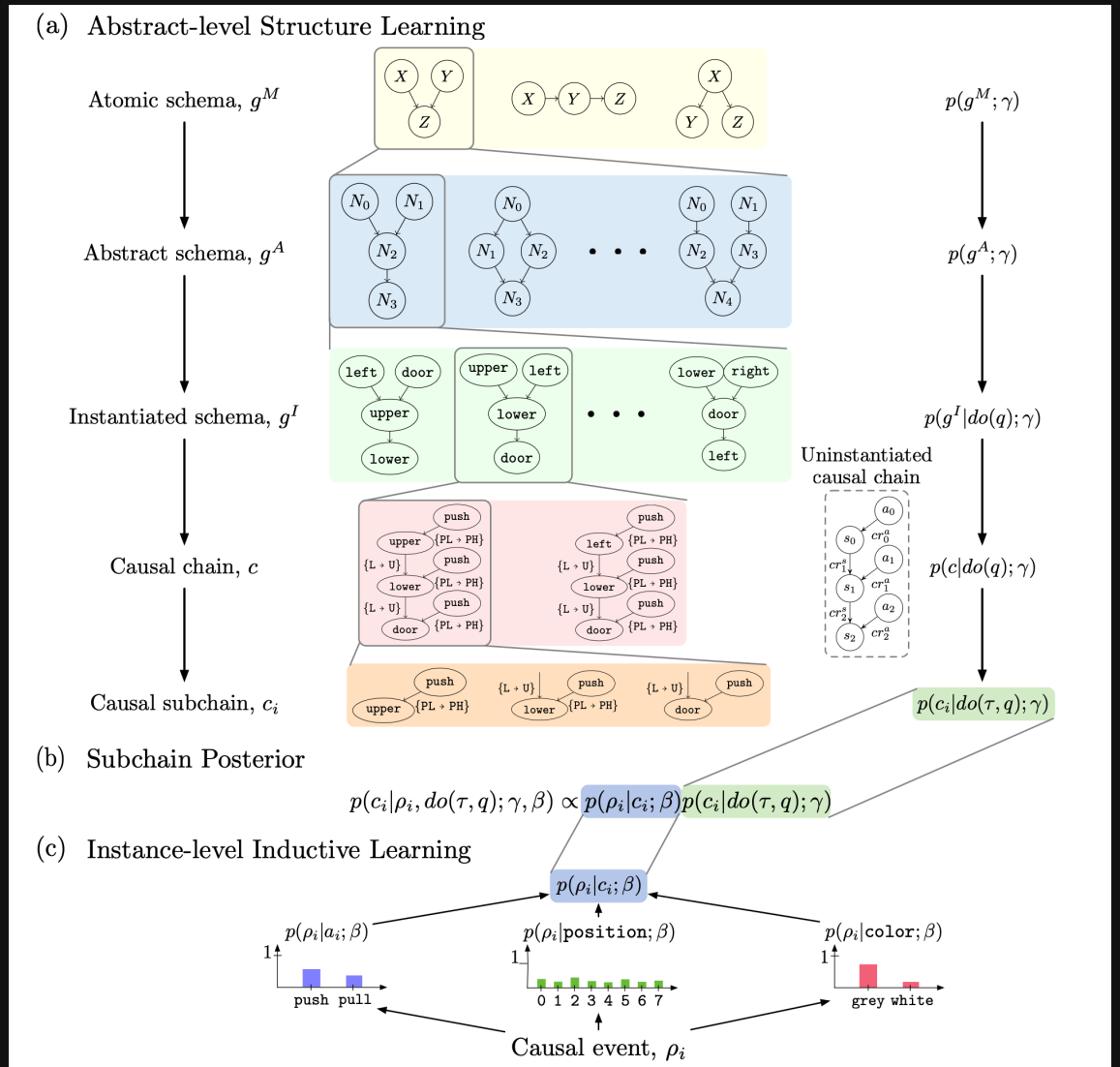
Cause	Strength
(Background)	$w_0 \sim \text{Uniform}(0, 1)$
Injected$(\mathbf{C}, \mathbf{M})$	$w_i \sim \text{Uniform}(0, 1)$



Griffiths, T. L. & Tenenbaum, J. B. Theory-based causal induction. Psychol. Rev. 116, 661–716 (2009)

Theory-based causal transfer

- Hierarchical Bayesian Model



$$p(g^A;\gamma) = \sum_{g^M \in \Omega_{g^M}} p(g^A|g^M;\gamma)p(g^M;\gamma)$$

$$p(g^I|do(q);\gamma) = \sum_{g^A \in \Omega_{g^A}} p(g^I|g^A, do(q))p(g^A;\gamma)$$

$$p(c|do(q);\gamma) = \sum_{g^I \in \Omega_{g^I}} p(c|g^I, do(q))p(g^I|do(q);\gamma)$$

$$p(c_i|do(\tau,q);\gamma) = \sum_{c \in \Omega_c} p(c_i|c, do(\tau,q))p(c|do(q);\gamma)$$

$$p(\rho|c;\beta) = \prod_{c_i \in c} p(\rho_i|c_i;\beta)$$

$$p(\rho_i|c_i;\beta) \propto p(\rho_i|a_i, f_i;\beta) \prod_{\substack{\phi_{ij} \in s_i \\ s_i \in c_i}} p(\rho_i|\phi_{ij};\beta)$$

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  How to describe the causal cognition in computation?

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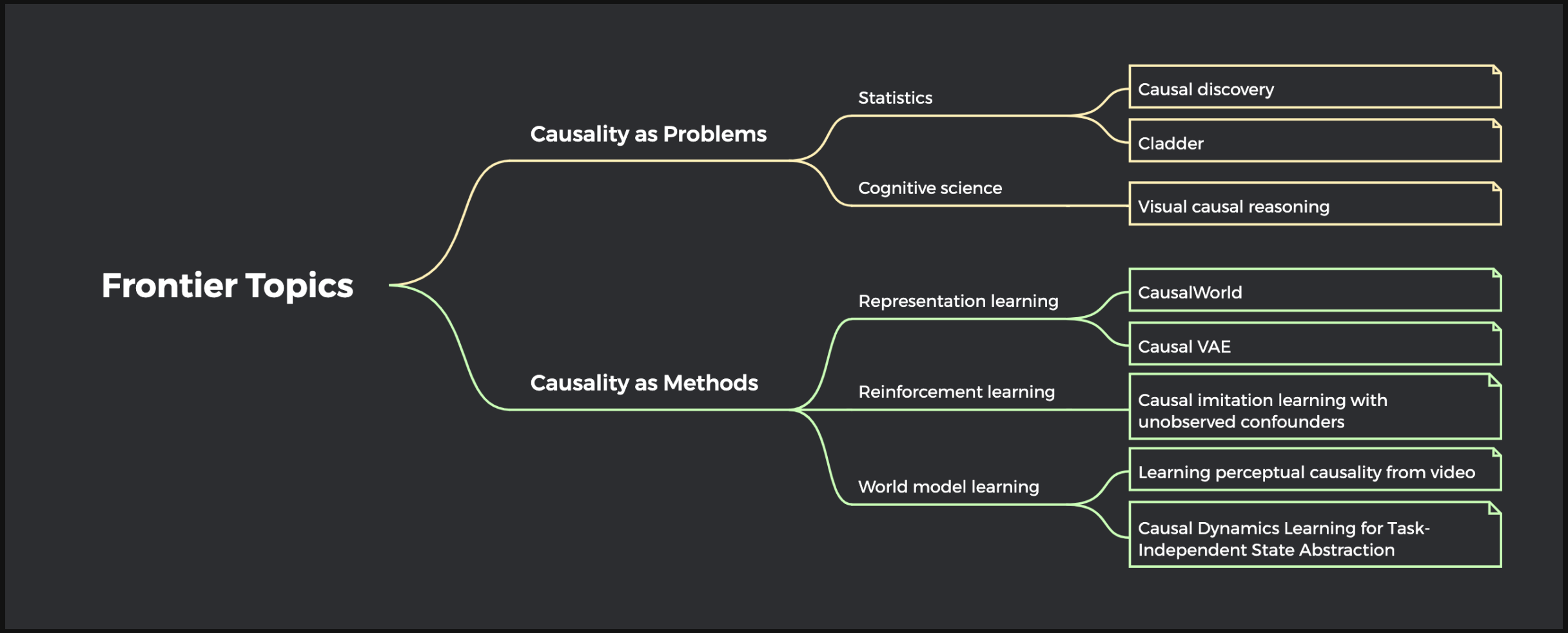
  What are researchers doing nowadays?

Modern approaches

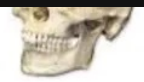


Causal discovery

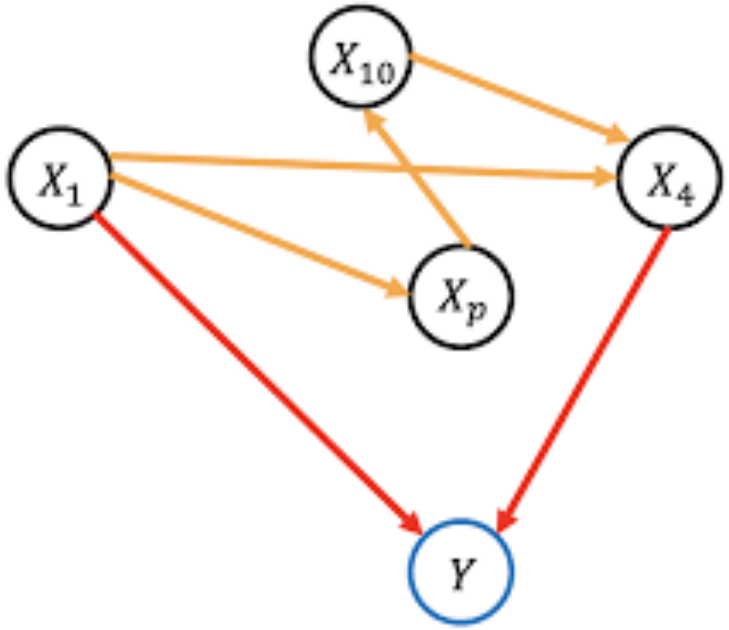
Causal discovery

Definition: finding the causal model from observational data.

- 8 variables of 250 skeletons collected from different locations



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
1	Id	Population	Sex	Cranial size	Diet or subsistence					Paramastic	Dental wear		Geographic location per population			Climate per population					
2			(Male, fem	(Centroid S	Gathering	Hunting	Fishing	Pastoralism	Agriculture	Yes=1, no=	Average att	Attrition pe	Distance to	Longitude	Latitude	Tmean	Tmin	Tmax	Vpmean	Vpmin	Vpmax
3	AINU31_1	Ainu	Unknown	713.2942	2	3	4	0	1	0	1.5	2	16464	43.548548	142.639159	2.86	-11.19	17.01	7.43	2.27	16.83
4	AINU7_1	Ainu	Unknown	676.148	2	3	4	0	1	0	1.5	1	16464	43.548548	142.639159	2.86	-11.19	17.01	7.43	2.27	16.83
5	AINU7_2	Ainu	Unknown	675.4924	2	3	4	0	1	0	1.5	1	16464	43.548548	142.639159	2.86	-11.19	17.01	7.43	2.27	16.83
6	AINU_1016	Ainu	Male	684.3304	2	3	4	0	1	0	1.5	2.5	16464	43.548548	142.639159	2.86	-11.19	17.01	7.43	2.27	16.83
7	AINU_1016	Ainu	Female	686.285	2	3	4	0	1	0	1.5	4	16464	43.548548	142.639159	2.86	-11.19	17.01	7.43	2.27	16.83
8	AUSM245	Australia	Male	673.8749	6	4	0	0	0	1	2.5	1	20164	-24.287027	135.615234	22.46	13.33	30.27	11.10	7.55	15.96
9	AUSM246	Australia	Male	647.4586	6	4	0	0	0	1	2.5	4	20164	-24.287027	135.615234	22.46	13.33	30.27	11.10	7.55	15.96
10	AUSM8217	Australia	Male	658.6616	6	4	0	0	0	1	2.5	2	20164	-24.287027	135.615234	22.46	13.33	30.27	11.10	7.55	15.96
11	AUSM8177	Australia	Male	667.5444	6	4	0	0	0	1	2.5	4	20164	-24.287027	135.615234	22.46	13.33	30.27	11.10	7.55	15.96
12	AUSM8173	Australia	Male	629.7138	6	4	0	0	0	1	2.5	3.5	20164	-24.287027	135.615234	22.46	13.33	30.27	11.10	7.55	15.96
13	AUSM8173	Australia	Male	648.7064	6	4	0	0	0	1	2.5	3.5	20164	-24.287027	135.615234	22.46	13.33	30.27	11.10	7.55	15.96
14	AUSM8171	Australia	Male	643.0378	6	4	0	0	0	1	2.5	2	20164	-24.287027	135.615234	22.46	13.33	30.27	11.10	7.55	15.96
15	AUSM8165	Australia	Male	616.55	6	4	0	0	0	1	2.5	3.5	20164	-24.287027	135.615234	22.46	13.33	30.27	11.10	7.55	15.96
16	AUSM8154	Australia	Male	635.0605	6	4	0	0	0	1	2.5	2	20164	-24.287027	135.615234	22.46	13.33	30.27	11.10	7.55	15.96
17	AUSM8153	Australia	Male	650.6959	6	4	0	0	0	1	2.5	3	20164	-24.287027	135.615234	22.46	13.33	30.27	11.10	7.55	15.96
18	AUSF1412	Australia	Female	618.4781	6	4	0	0	0	1	2.5	1	20164	-24.287027	135.615234	22.46	13.33	30.27	11.10	7.55	15.96
19	AUSF8179	Australia	Female	634.3122	6	4	0	0	0	1	2.5	3.5	20164	-24.287027	135.615234	22.46	13.33	30.27	11.10	7.55	15.96
20	AUSF8175	Australia	Female	605.1759	6	4	0	0	0	1	2.5	1.5	20164	-24.287027	135.615234	22.46	13.33	30.27	11.10	7.55	15.96
21	AUSF8172	Australia	Female	613.8324	6	4	0	0	0	1	2.5	3	20164	-24.287027	135.615234	22.46	13.33	30.27	11.10	7.55	15.96
22	AUSF8169	Australia	Female	619.1206	6	4	0	0	0	1	2.5	2.5	20164	-24.287027	135.615234	22.46	13.33	30.27	11.10	7.55	15.96
23	AUSF8157	Australia	Female	628.2819	6	4	0	0	0	1	2.5	2	20164	-24.287027	135.615234	22.46	13.33	30.27	11.10	7.55	15.96
24	AUSF8155	Australia	Female	628.4609	6	4	0	0	0	1	2.5	3.5	20164	-24.287027	135.615234	22.46	13.33	30.27	11.10	7.55	15.96
25	AUSF1578	Australia	Female	640.6311	6	4	0	0	0	1	2.5	2	20164	-24.287027	135.615234	22.46	13.33	30.27	11.10	7.55	15.96
26	AUSF243	Australia	Female	606.164	6	4	0	0	0	1	2.5	2.5	20164	-24.287027	135.615234	22.46	13.33	30.27	11.10	7.55	15.96
27	AUSF8158	Australia	Female	631.6258	6	4	0	0	0	1	2.5	2	20164	-24.287027	135.615234	22.46	13.33	30.27	11.10	7.55	15.96
28	DENM1432	Denmark	Male	663.6198	0	0	1	3	6	0	2.1	2	10440	55.717055	11.711426	8.01	-0.02	16.66	9.67	5.59	15.27
29	DENM1011	Denmark	Male	651.4847	0	0	1	3	6	0	2.1	3	10440	55.717055	11.711426	8.01	-0.02	16.66	9.67	5.59	15.27
30	DENM1205	Denmark	Male	636.9831	0	0	1	3	6	0	2.1	1.5	10440	55.717055	11.711426	8.01	-0.02	16.66	9.67	5.59	15.27
31	DENM116_	Denmark	Male	642.9192	0	0	1	3	6	0	2.1	3	10440	55.717055	11.711426	8.01	-0.02	16.66	9.67	5.59	15.27
32	DENM116_	Denmark	Male	646.6609	0	0	1	3	6	0	2.1	2.5	10440	55.717055	11.711426	8.01	-0.02	16.66	9.67	5.59	15.27
33	DENM116_	Denmark	Male	674.9799	0	0	1	3	6	0	2.1	2	10440	55.717055	11.711426	8.01	-0.02	16.66	9.67	5.59	15.27
34	DENM7_77	Denmark	Male	666.53	0	0	1	3	6	0	2.1	2.5	10440	55.717055	11.711426	8.01	-0.02	16.66	9.67	5.59	15.27



Causal discovery: a bit of history

- Reichenbach's common cause principle ("The Direction of Time", 1956)
- Markov condition (Kiiveri et al, 1984)
- "Causation, Prediction, and Search" (Spirtes, Glymour, & Scheines, 1993)
  - Faithfulness condition, PC algorithm, SGS, FCI, Tetrad program ….
- "Causality, Models, Reasoning and Inference" (Pearl, 2000)
- Greedy equivalence search (GES) (Chickering, 2003)
- Functional causal model-based methods (LiNGAM, PNL… since 2005)
- Latent variable recovery: Factor analysis (Spearman, 1904), Tetrad condition (Spearman & CMU), Laten tree structure (Pearl et al., 1989), measurement model (CMU, 2006), GIN (GDUT & CMU), rank deficiency (CMU)…

Causal discovery

- Three dimensions of the problem:
  - i.i.d data?
  - Parametric constraints?
  - Latent confounders?

Causal Discovery

DAGs with NO TEARS: Continuous Optimization for Structure Learning

Xun Zheng, Bryon Aragam, Pradeep Ravikumar, and Eric P. Xing

Carnegie Mellon University

November 6, 2018

Causal Discovery

$$\min_{W \in \mathbb{R}^{d \times d}} F(W) \quad \text{subject to} \quad \mathsf{G}(W) \in \mathsf{DAGs} \iff \min_{W \in \mathbb{R}^{d \times d}} F(W) \quad \text{subject to} \quad h(W) = 0,$$

(a) $h(W) = 0$ if and only if W is acyclic (i.e. $\mathsf{G}(W) \in \mathbb{D}$);

(b) The values of h quantify the "DAG-ness" of the graph;

(c) h is smooth;

(d) h and its derivatives are easy to compute.

Causal Discovery

Theorem 1. *A matrix $W \in \mathbb{R}^{d\times d}$ is a DAG if and only if*

$$h(W) = \operatorname{tr}\left(e^{W\circ W}\right) - d = 0, \tag{7}$$

where $\circ$ is the Hadamard product and e^A is the matrix exponential of A. Moreover, $h(W)$ has a simple gradient

$$\nabla h(W) = \left(e^{W\circ W}\right)^T \circ 2W, \tag{8}$$

and satisfies all of the desiderata (a)-(d).

Causal Discovery

Algorithm 1 NOTEARS algorithm

1. Input: Initial guess (W_0, α_0), progress rate $c \in (0, 1)$, tolerance $\epsilon > 0$, threshold $\omega > 0$.
2. For $t = 0, 1, 2, \ldots$:
   (a) Solve primal $W_{t+1} \leftarrow \arg\min_W L^\rho(W, \alpha_t)$ with ρ such that $h(W_{t+1}) < ch(W_t)$.
   (b) Dual ascent $\alpha_{t+1} \leftarrow \alpha_t + \rho h(W_{t+1})$.
   (c) If $h(W_{t+1}) < \epsilon$, set $\widetilde{W}_{\mathsf{ECP}} = W_{t+1}$ and break.
3. Return the thresholded matrix $\widehat{W} := \widetilde{W}_{\mathsf{ECP}} \circ 1(|\widetilde{W}_{\mathsf{ECP}}| > \omega)$.

Causality + LLM

Causality + LLM

- Motivation: LLM for causal reasoning
  - How to evaluate?

    Need a problem dataset.
  - How to solve?

    Need a method.

CLADDER

CLADDER: Assessing Causal Reasoning in Language Models

Zhijing Jin[1,2,*] **Yuen Chen**[1,*] **Felix Leeb**[1,*] **Luigi Gresele**[1,*]
Ojasv Kamal[3], **Zhiheng Lyu**[4], **Kevin Blin**[2], **Fernando Gonzalez**[2], **Max Kleiman-Weiner**[5],
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fleeb@tue.mpg.de luigi.gresele@tue.mpg.de

CLADDER dataset

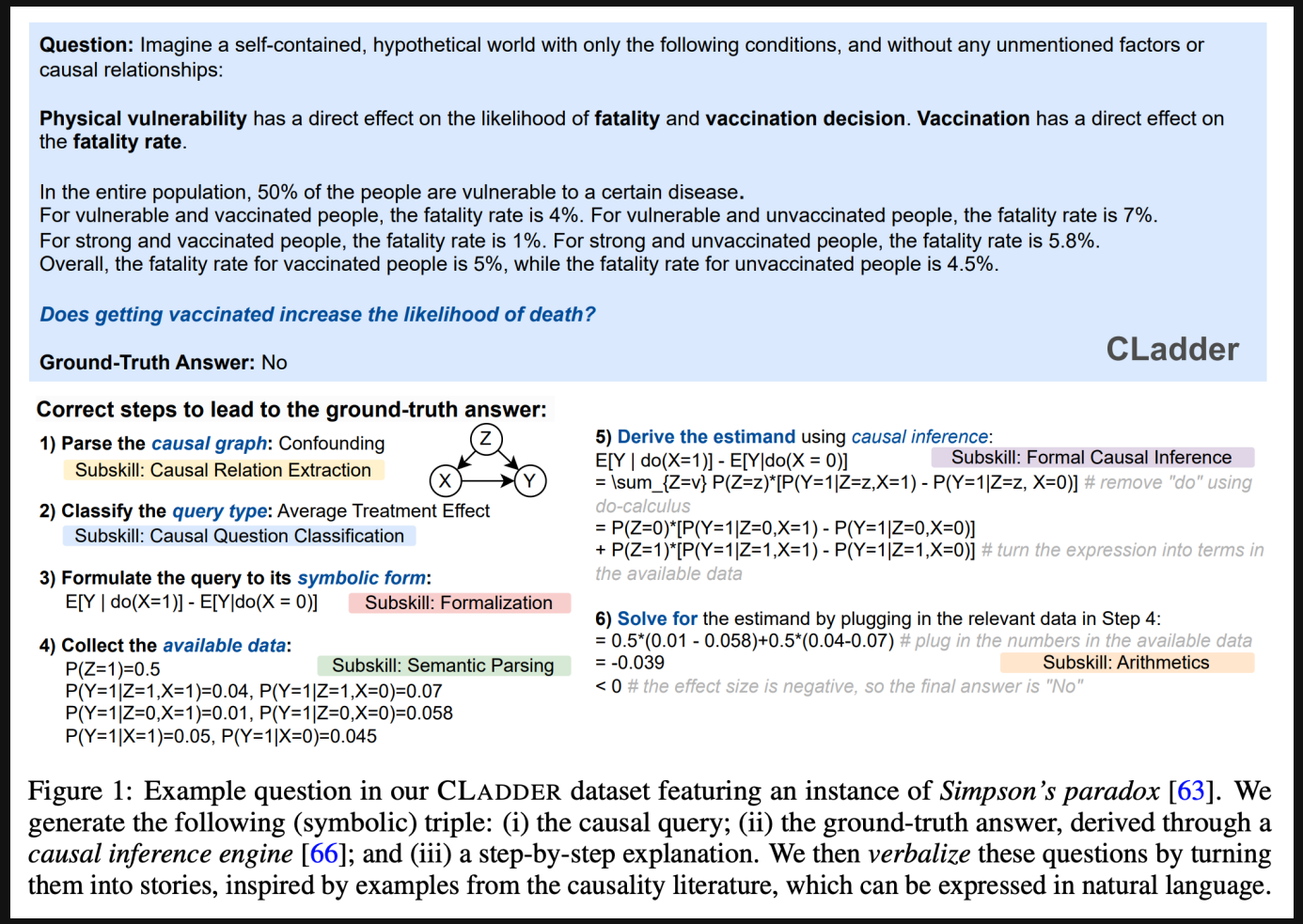


Figure 1: Example question in our CLADDER dataset featuring an instance of *Simpson's paradox* [63]. We generate the following (symbolic) triple: (i) the causal query; (ii) the ground-truth answer, derived through a *causal inference engine* [66]; and (iii) a step-by-step explanation. We then *verbalize* these questions by turning them into stories, inspired by examples from the causality literature, which can be expressed in natural language.

- **Task Formulation**: Natural language causal reasoning grounded in symbolic queries and ground-truth answers
- **Ladder of Causation**: 3 rungs
  - Association (Rung 1)
  - Intervention (Rung 2)
  - Counterfactuals (Rung 3)
- **Data Generation Pipeline**:
  - Formal phase
  - Verbalization phase
  - Story Variants
  - Balanced Statistics
  - Explanation Generation

Causal CoT

Six steps guided reasoning

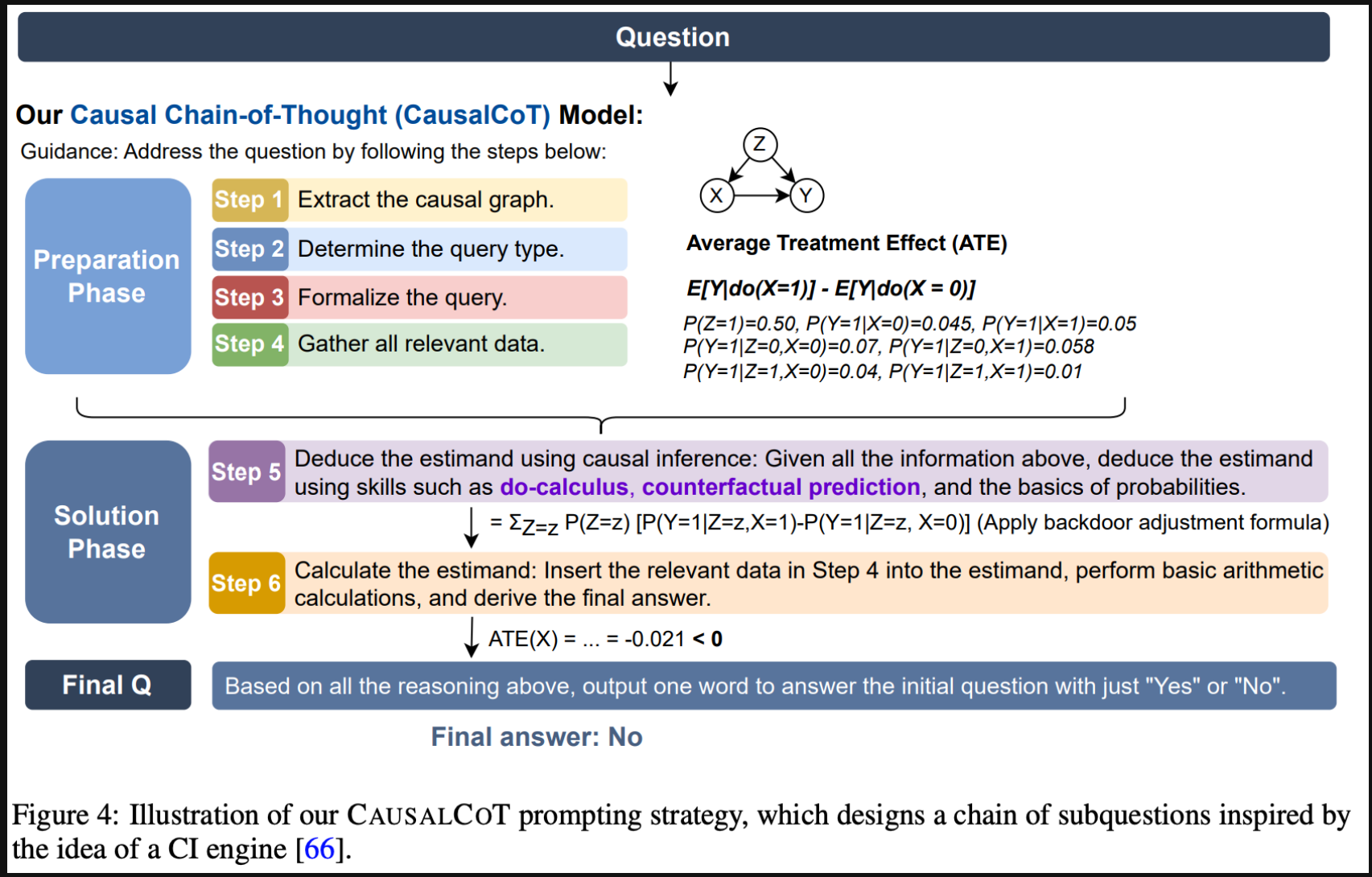


Figure 4: Illustration of our CAUSALCOT prompting strategy, which designs a chain of subquestions inspired by the idea of a CI engine [66].

Results

	Overall Acc.	Acc. by Rung			Acc. by Commonsense Alignment		
		1	2	3	Comm.	Nonsens.	Anti-C.
Random	49.27	50.28	48.40	49.12	49.01	49.69	49.12
LLaMa	44.03	48.23	29.46	52.66	45.14	44.22	42.67
Alpaca	44.66	52.03	29.53	51.13	44.86	44.40	44.77
GPT-3 Non-Instr. (davinci)	49.92	50.00	49.75	50.00	49.06	49.97	50.72
GPT-3 Instr. (text-davinci-001)	51.40	51.30	52.63	50.47	54.31	50.13	50.05
GPT-3 Instr. (text-davinci-002)	53.15	50.85	56.96	51.90	55.33	52.47	51.81
GPT-3 Instr. (text-davinci-003)	56.26	51.11	62.97	54.96	56.83	54.79	57.49
GPT-3.5	52.18	51.80	54.78	50.32	54.09	50.68	52.09
GPT-4	62.03	63.01	62.82	60.55	62.27	63.09	60.47
+ CausalCoT	**70.40**	**83.35**	**67.47**	**62.05**	**69.25**	**71.58**	**70.12**

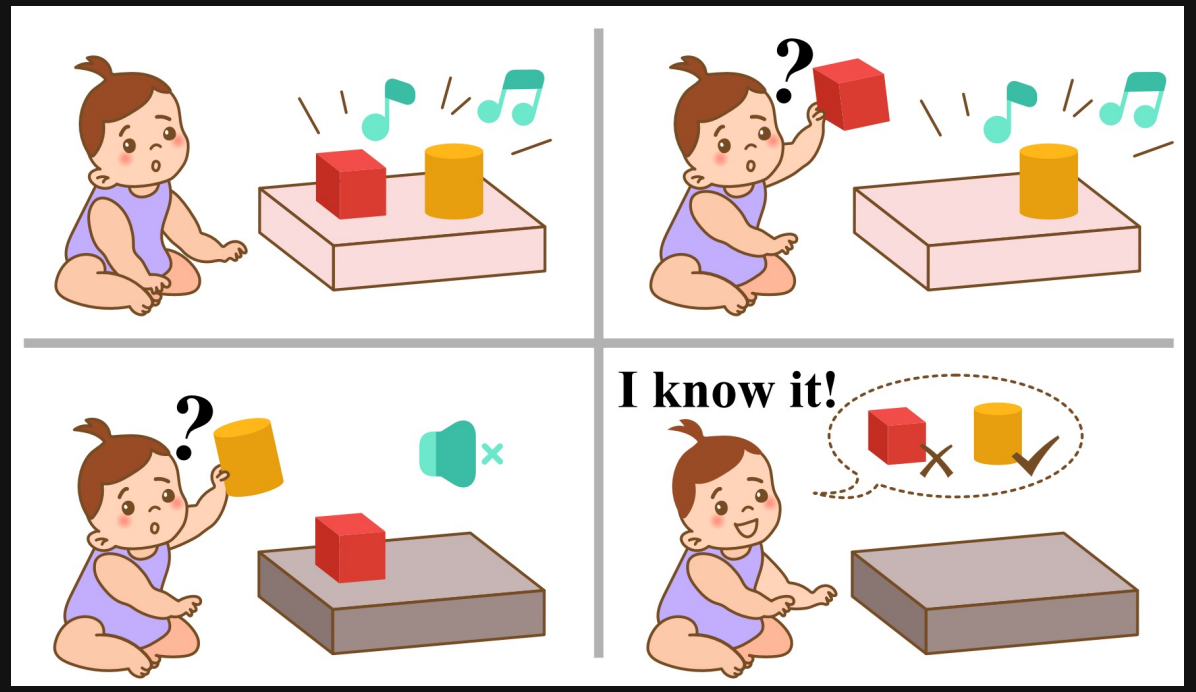
Table 2: Performance of all models on our CLadder dataset v1.5. We report the overall accuracy (Acc.), and also fine-grained accuracy by rung, and by degree of commonsense alignment, from commonsensical (Comm.), nonsensical (Nonsens.), to anti-commonsensical (Anti-C.).

- All models struggle: GPT-4 only 62 %, others near random.
- CAUSALCOT lifts GPT-4 to 70 % (+8 pp).
- Gains biggest on anti-commonsense / nonsense stories.
- Hardest steps: query typing, formalizing, deriving estimand.

Visual causal reasoning

Visual causal reasoning

- One of the hallmarks of human intelligence:
  - Hume's quest for causality
  - scientific discovery
  - drawing knowns from unknowns
- Children's quickly uncertainty resolution:
  - children's exploratory play
  - an example: the Blicket machine

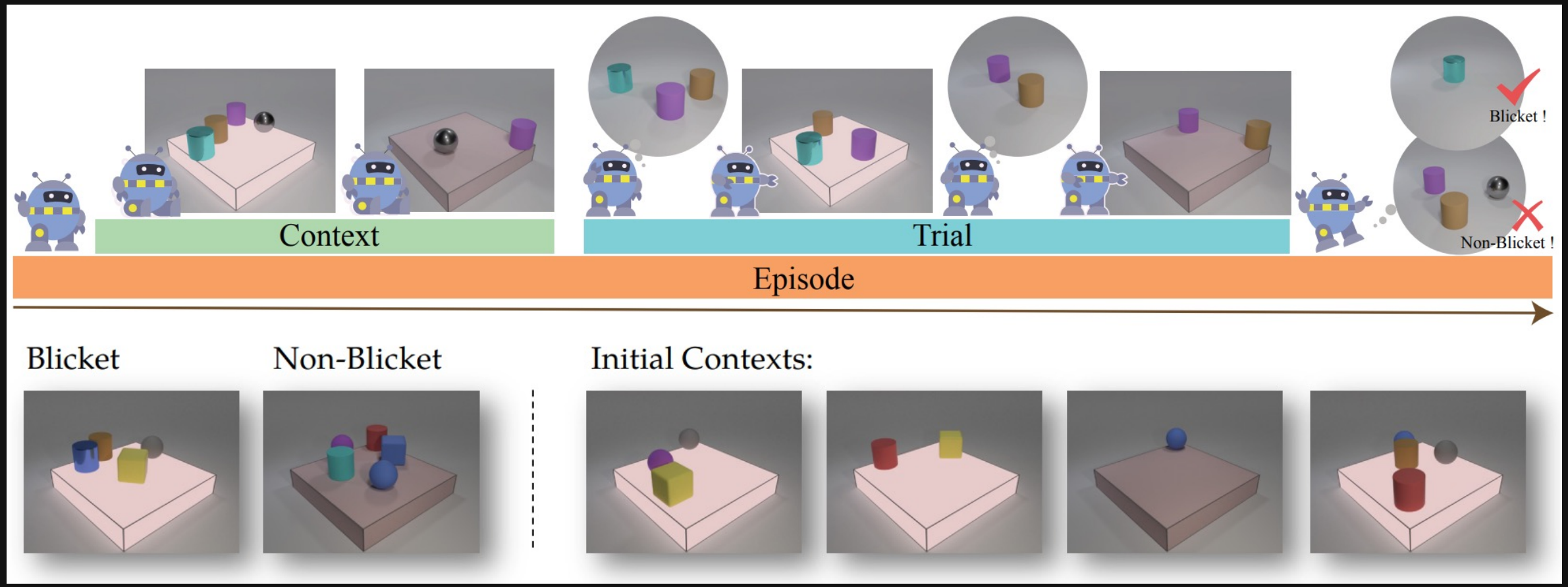


Visual causal reasoning

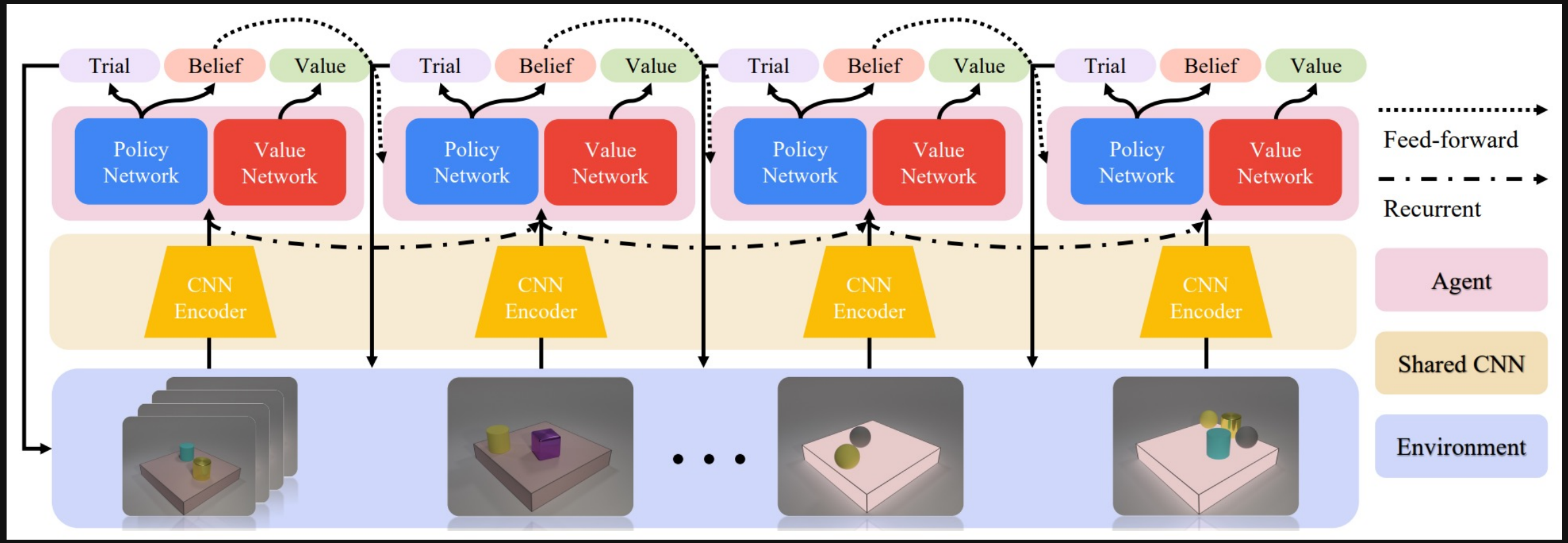
Benchmarks	Task	Size	Format	Temporal	Interactive	Uncertainty	Few-shot
CLEVR (Johnson et al., 2017)	vqa	100k	image	✗	✗	✗	✗
CLEVRER (Yi et al., 2019)	vqa	20k	video	✓	✗	✗	✗
CATER (Girdhar and Ramanan, 2019)	cls	5.5k	video	✓	✗	✗	✗
CURI (Vedantam et al., 2021)	cls	990k	image	✓	✗	✓	✓
ACRE (Zhang et al., 2021a)	cls	30k	image	✓	✗	✓	✓
Alchemy (Wang et al., 2021)	game	-	image/symbol	✓	✓	✓	✗
IVRE (Ours)	game	-	image/symbol	✓	✓	✓	✓

- Despite the breakthroughs from natural language understanding to visual reasoning, those quests for machine reasoning are far from realized.
- Machines still lag behind how we, humans, even infants, understand the world, reason cause-and-effect relations, and make scientific discoveries.
- IVRE differs from previous works by introducing uncertainty into visual reasoning in an interactive environment.

IVRE


Blicket !
Non-Blicket !
Context
Trial
Episode
Blicket
Non-Blicket
Initial Contexts:

Reinforcement agents



Results

Model	Context		Episode		Model	Context		Episode	
	Acc	R	Acc	R		Acc	R	Acc	R
Random	0.86%	-5.42	1.87%	-14.14	DDPG-FF	22.47%	-0.17	32.47%	-3.7
Bayes	15.60%	-2.98	43.03%	-3.78	DDPG-Re	13.55%	-2.03	46.03%	-0.29
Naive	3.50%	-3.91	43.62%	-1.69	TD-3-Re	12.57%	-2.40	36.83%	-2.71
NOTEARS	9.10%	-4.66	12.70%	-13.02	TD-3-FF	21.91%	-0.42	30.05%	-4.48
Search-Naive	1.51%	-3.68	83.80%	9.39	PPO	6.87%	-3.87	28.56%	-5.85
Search-Random	1.80%	-3.62	34.15%	-1.87	DDPG-V	0.35%	-5.02	0.72%	-13.40
GPT-3.5	3%	-5.91	11%	-13.39	TD-3-V	0.27%	-5.04	0.31%	-13.51
GPT-4	10%	-3.36	26%	-7.88	Human	33.33%	5.01	98.15%	12.70

Causal representation learning

Causal representation learning

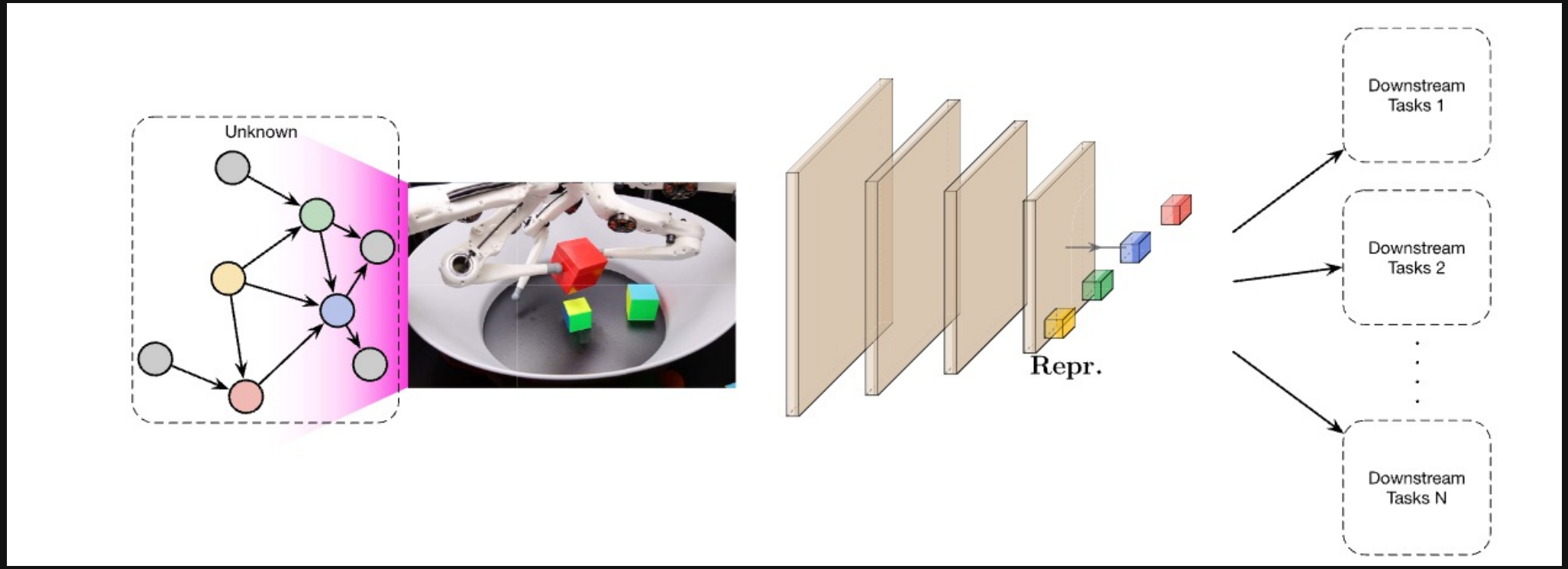
- Problem:

  How to integrate the structural information into the machine learning methods?

- Why do we need structural information?
  - Robustness
  - Learning reusable mechanisms
  - Active intervention
- Intuition
- Environment
- Method

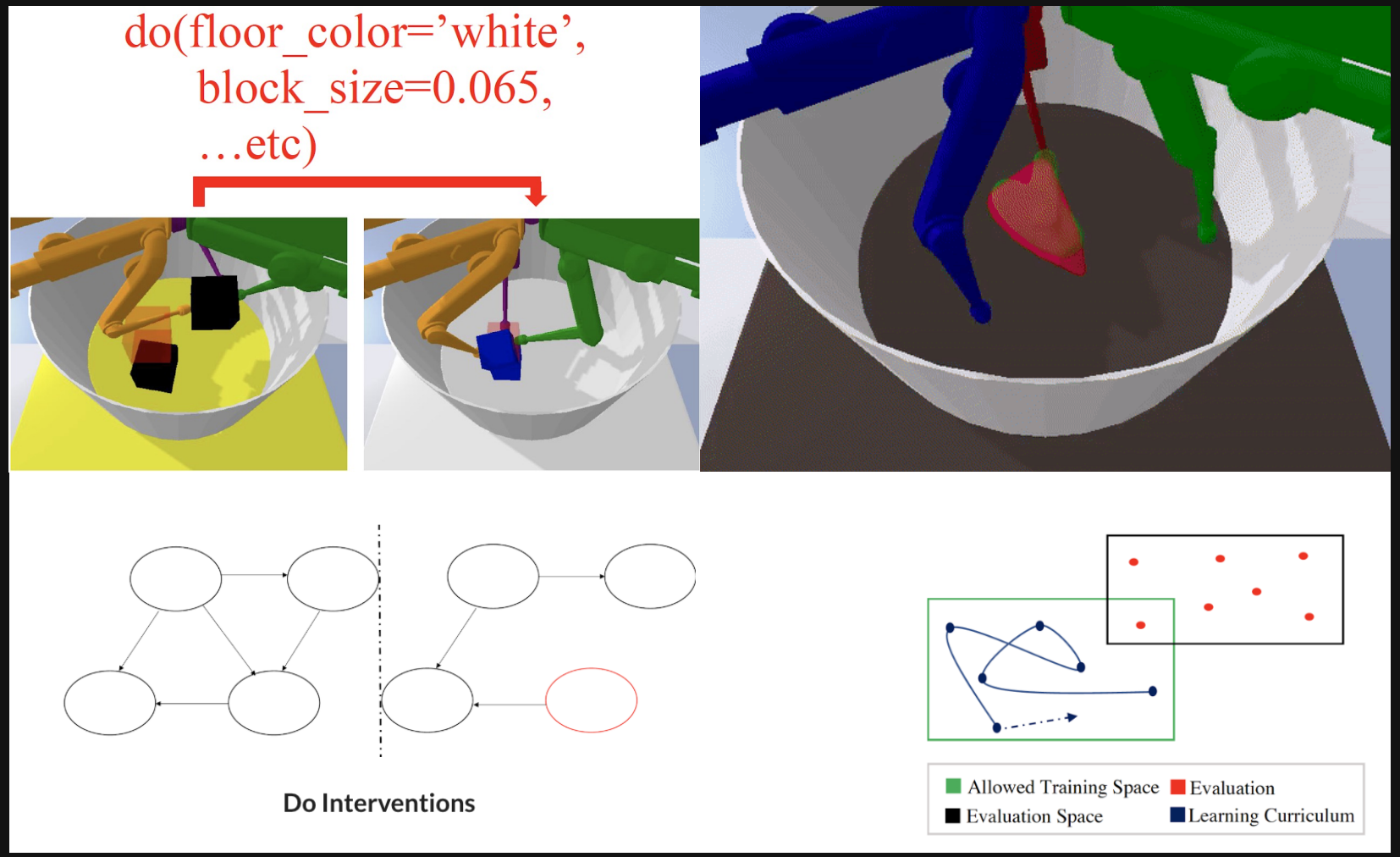
Causal representation learning

- Discover high-level causal variables from low-level observations.
  - Independent causal mechanism (ICM)
  - Smallest mechanism shift (SMS)



Schölkopf, B. et al. Toward Causal Representation Learning. Proc. IEEE 109, 612–634 (2021)

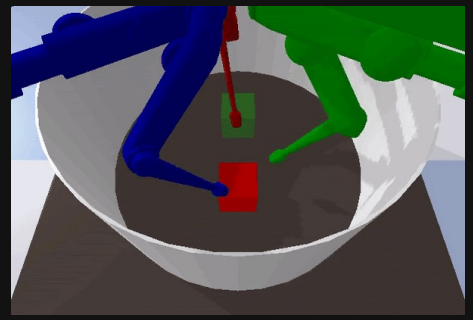
CausalWorld: Benchmark for causal structure and transfer learning



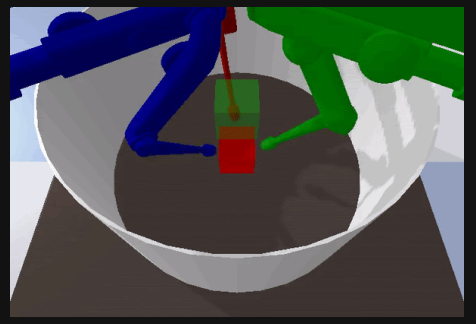
Ahmed, Ossama, et al. "Causalworld: A robotic manipulation benchmark for causal structure and transfer learning." arXiv preprint arXiv:2010.04296 (2020).

CausalWorld

Reaching

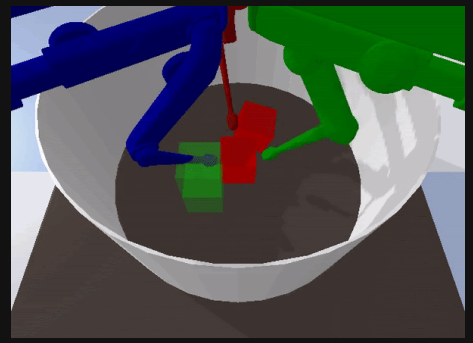


Pushing

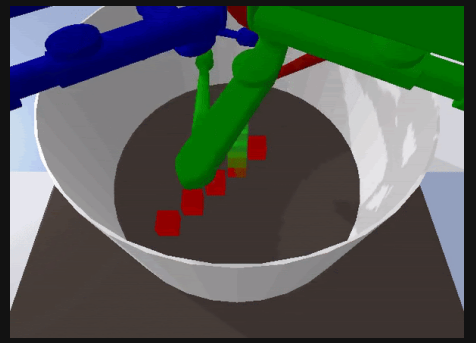


Picking

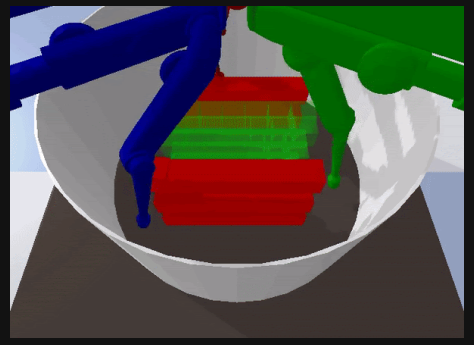
Pick and Place



Stacking



Tower



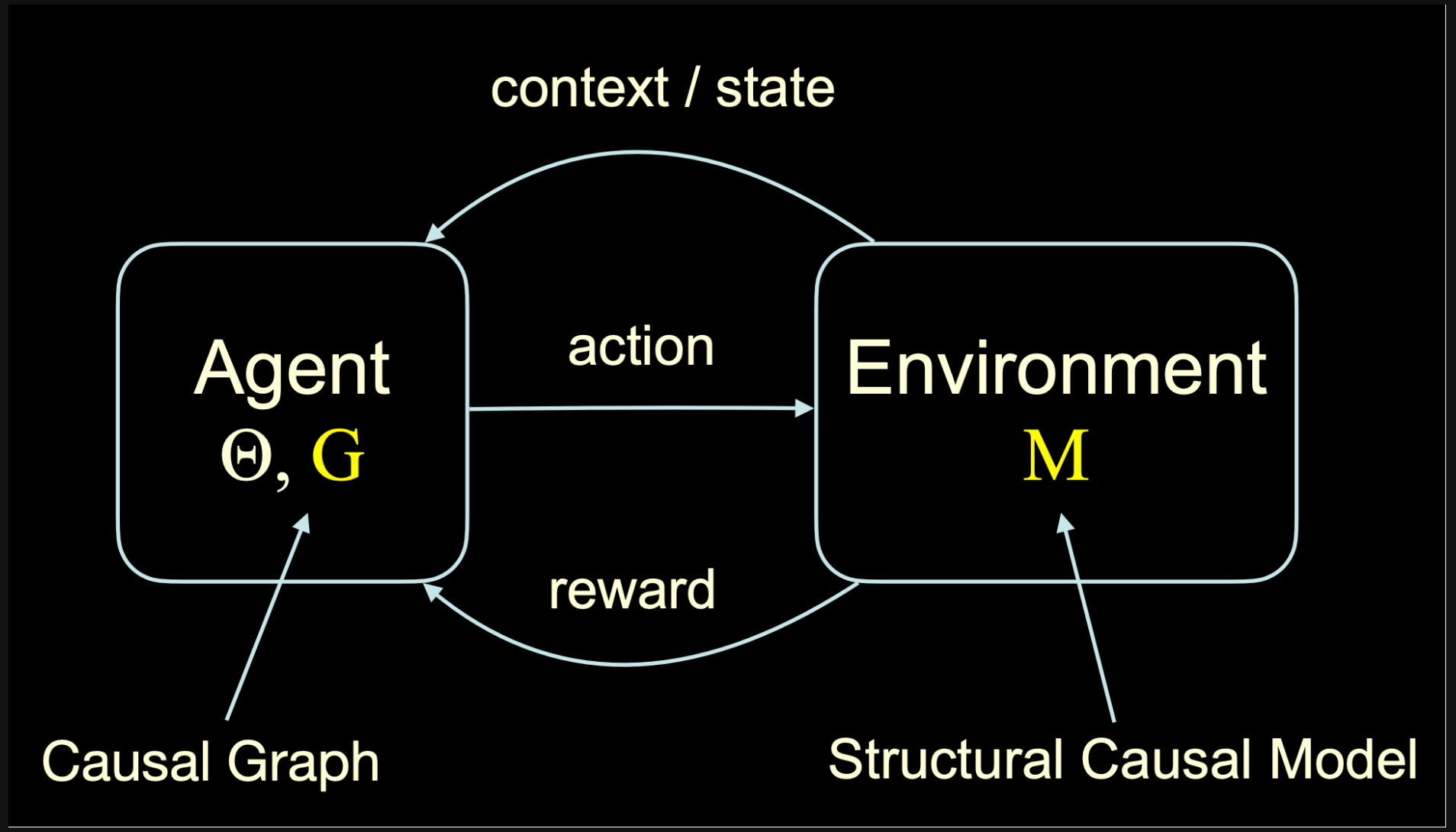
Stacked blocks

Creative stacked blocks

General

Causal reinforcement learning

Causal reinforcement learning



Causal Imitation Learning with Unobserved Confounders

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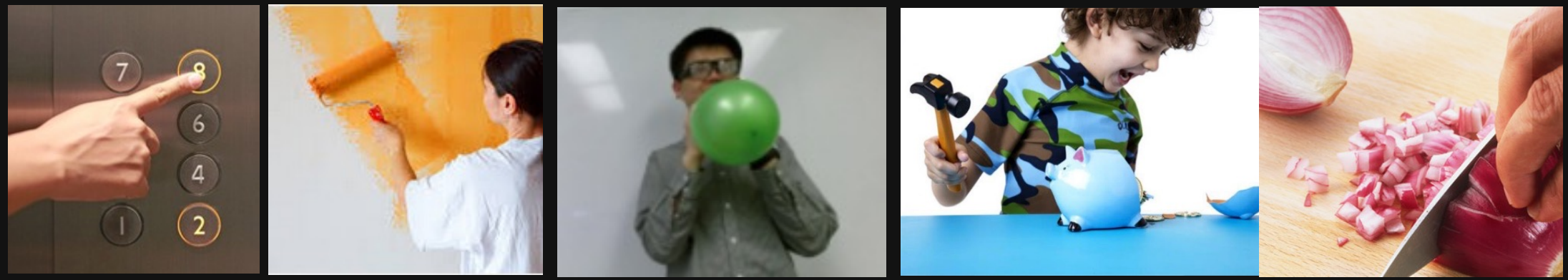
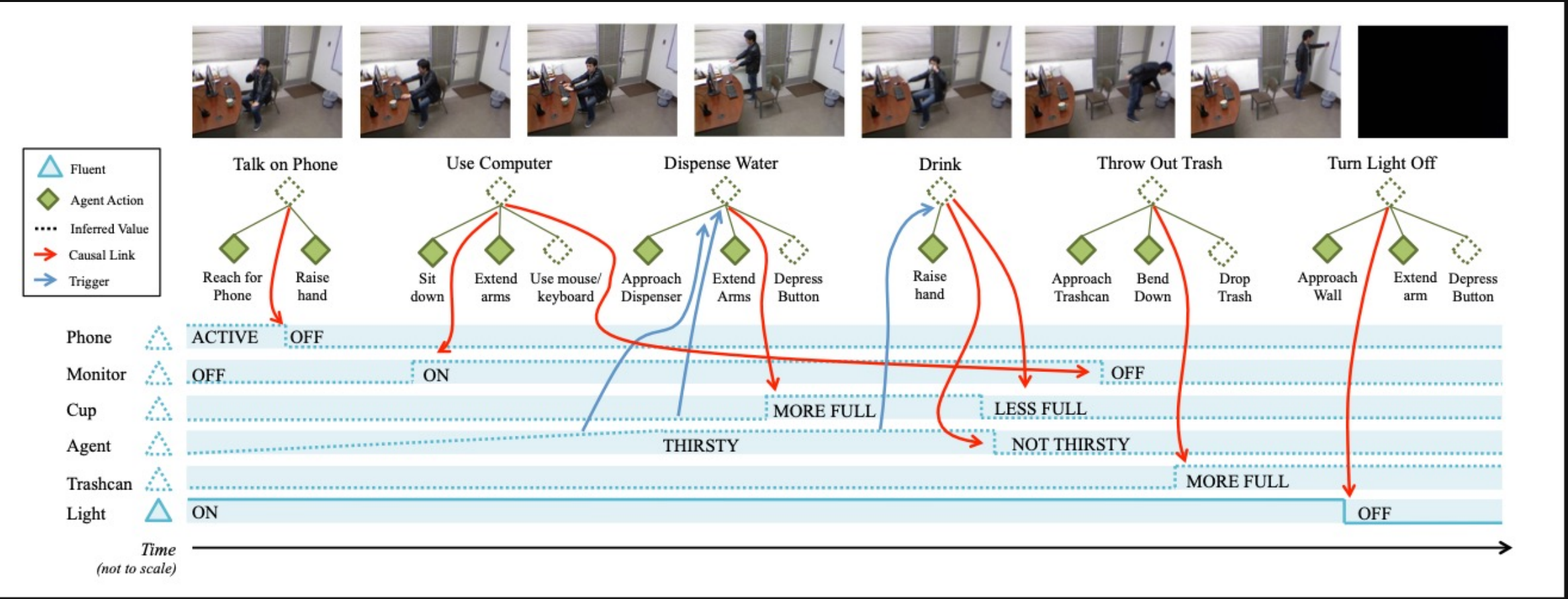
World model learning

Learning causal relations from video

- Problem: infer causal relations from video data.



Learning causal relations from video



Learning abstract causal transitions

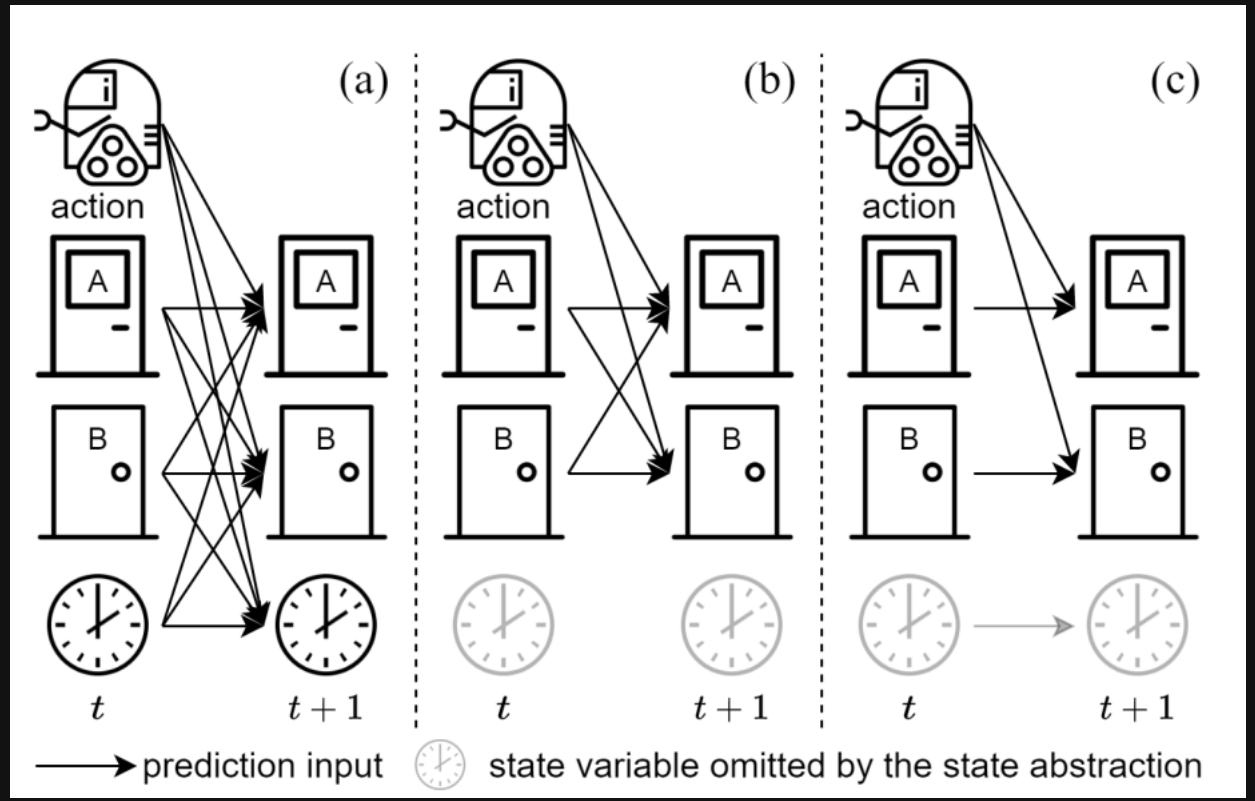
Causal Dynamics Learning for Task-Independent State Abstraction

Zizhao Wang [1] **Xuesu Xiao** [2] **Zifan Xu** [2] **Yuke Zhu** [2] **Peter Stone** [2 3]

Learning abstract causal transitions

- Problem:

  How to learn a generalizable and robust dynamic model for RL?



Intuition: learns a causal dynamics model that **removes unnecessary dependencies** between state variables and the action, thus generalizing well to unseen states.

Learning abstract causal transition

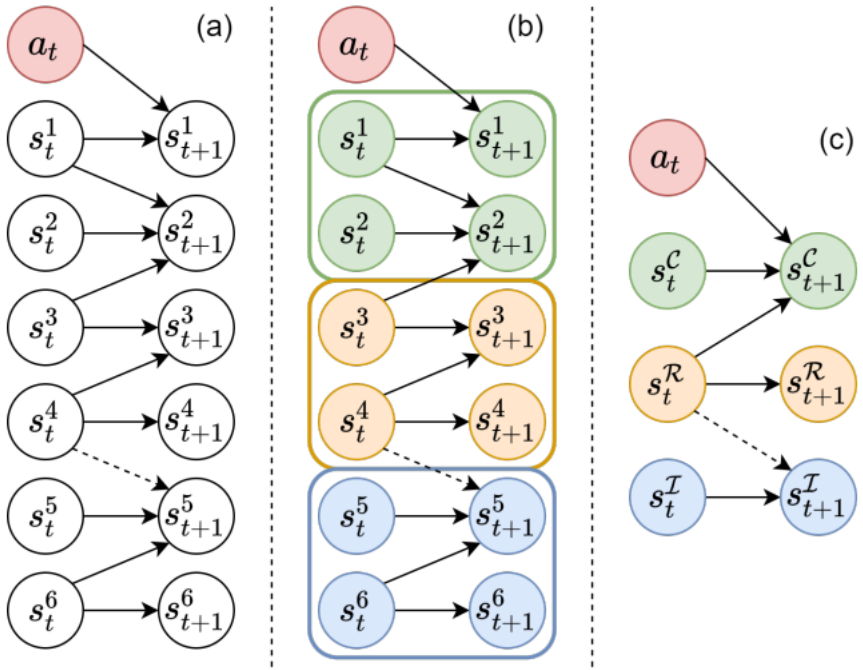


Figure 2. **(a)** an example causal dynamics model. **(b)** state variables can be split into three types: controllable (green), action-relevant (orange), and action-irrelevant (blue). The dashed arrow represents whether it exists does not affect s^5 to be action-irrelevant. **(c)** the causal graph can be split into three subgraphs, one for each type of state variable.

- Variables:
  - Controllable
  - Action-relevant
  - Action-irrelevant
- Determining the edge existence by

Theorem 3.1. *Assuming A1 - A5, we define the conditioning set* $\{a_t, s_t \setminus s_t^i\} = \{a_t, s_t^1, \ldots, s_t^{i-1}, s_t^{i+1}, \ldots, s_t^{d_S}\}$. *Then, for any two state variables at indices* i *and* j, *if* $s_t^i \not\perp\!\!\!\perp s_{t+1}^j | \{a_t, s_t \setminus s_t^i\}$, *then* $s_t^i \to s_{t+1}^j$. *Similarly, if* $a_t \not\perp\!\!\!\perp s_{t+1}^j | s_t$, *then* $a_t \to s_{t+1}^j$.

Causality

- Part 1: A historical tour

  How do pioneers understand causality from different perspectives?

- Part 2: Cognitive phenomena

  How people understand and perceive causation?

- Part 3: Computational models

  How to describe the causal cognition in computational tradition?

- Part 4: Frontier topics

  What are researchers doing nowadays?

Thanks for your listening and sleeping